PACNET Meeting Tokyo

Feb 14 2008



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Agenda

- IP NGN Architecture for Tier-1 Providers
- L3 Design Considerations
  - Scalability
  - Operations Efficiency
  - Traffic Engineering
  - Service Creation
- Optical and Control Plane Development
- OAM Development

Summary

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IP NGN Architecture for Tier 1 Service Providers



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Web 2.0 – Evolution Scenario

"The Web As The Platform", "You Control Your Own Data"

Web 1.0	Web 2.0
DoubleClick	Google AdSense
Ofoto	Flickr
Akamai	Bittorrent
Mp3.com	iTunes
Britannica Online	wikipedia
Personal websites	Blogging
Evite	Upcoming.org and EVDB
Domain name speculation	Search engine optimization
Page views	Cost per Click
Screen Scraping	Web Services
Publishing	Participation
Content Management Systems	wikis
Directories (Taxonomy)	Tagging ("folksonomy")
Stickiness	Syndication

Source http://www.oreillynet.com/pub/a/oreilly/tim/news/2005/09/30/what-is-web-20.html

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The Impact of Web 2.0 Is All Around



"You are no longer in control. The consumer has the power."

Peter Weedfald,
Senior VP
Samsung Consumer Electronics

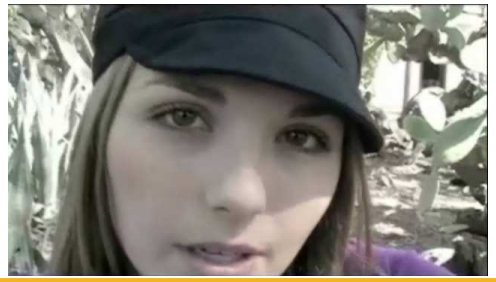
Source: *Time*, January 2007

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New Creators & Consumers of Video Entertainment



Consumer



Prosumer



Professional

NETWORK AS THE PLATFORM



PC



Home TV



Smartphone

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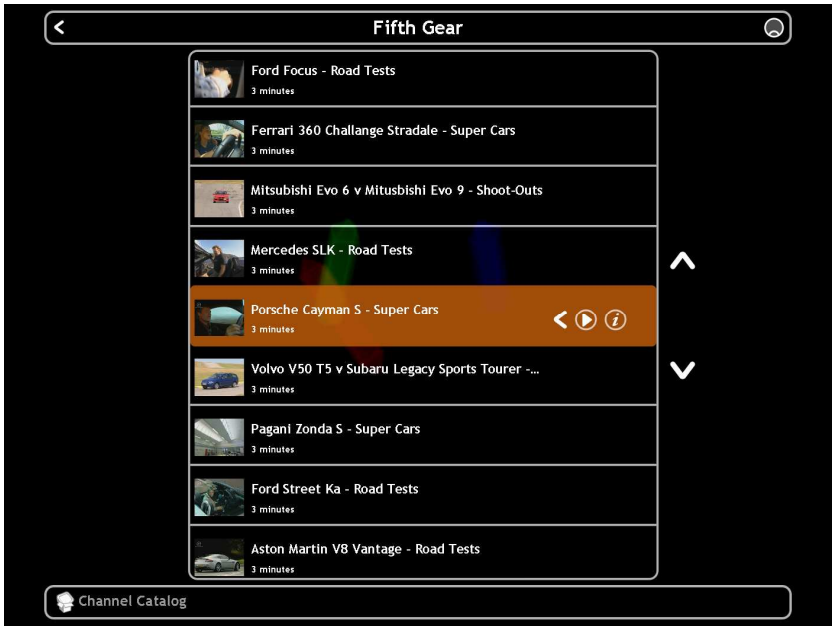
Joost *(from the creators of Skype/Kazaa)*

January 19, 2007

Two poster dudes for file-sharing and VoIP innovation have finally branded their foray into Web TV. Niklas Zennström and Janus Friis, the founders of Kazaa and Skype, unveiled **Joost** this week as the official name of the 'Net video experiment up to now referred to as "The Venice Project."

Currently in beta testing, Joost is being pushed as a service that combines "the best of TV and the best of the Internet." Unlike YouTube, Joost will not offer user-generated content, but streaming content from established providers.

- **Free of charge to Users, Ad sponsored.**
- **Content from: Nat Geo, Viacom, JumpTV, CBS, WCSN, ...**
- **Advertising partners include CocaCola, HP, Intel, Microsoft, Nike, Nokia, Vodafone, P&G, Nestle, Unilever, ...**
- **Streaming at 700Kbit/sec download, 0,32GB/hour & 220Kbit/sec upload, 0,105GB/hour)**
- **1000's of Channels planned.**
- **Rich Search, Navigation... Chat, Rate**
- **P2P runs at deficit (download > upload), Joost will make up for the missing capacity with distributed data centres.**



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Channel Extensions
BBC, Linear TV and VoD

Seven day TV catch-up over the internet

4. The proposed on-demand catch-up facility will give audiences access to the BBC's linear television schedule where rights costs allow. It will be principally organised around a seven day window from the point of original broadcast.

Simulcast TV over the internet

9. This will provide UK audiences with the ability to watch the BBC's existing television channels. Simulcast TV over the internet will be provided via multicast streaming

Non-DRM downloads over the internet

10. The provision of audio programmes and material related to audio programming (with the exception of full track commercial music) without Digital Rights Management (DRM) protection alongside the existing seven day on-demand audio service, where the rights holders agree that this form of distribution is appropriate. Removing DRM restrictions allows a broader range of devices to access the material, extending audience choice around where and when they consume BBC programming.

11. The three internet-based offerings will be combined into a direct-to-consumer offering, BBC iPlayer version 1.0[4], which will incorporate all the existing BBC players and consoles under a consistent user interface and brand[5]. In summary, BBC iPlayer version 1.0 is designed to unify the BBC's provision of audio and video content on-demand and to make it available to UK audiences over the internet.



- **BBC is now a Global ISP**

  They PEER rather than PAY for Internet Access (~500 Peers in UK, NL, DE, US...)

- **BBC iPlayer is based on P2P**

  Seven day TV catch-up and BBC archive are distributed using P2P Technology.

- **BBC Simulcast requires Multicast**

  Only ISP's who provide Multicast Peering to BBC are eligible for Internet Simulcast.

- **Significant Traffic Surge**

  +3GB/User/Month => 400+M£ cost for ISP's (OFcom)

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VoD Streaming, moving to “HD”

based on HTTP/Quantum streaming from Move Networks and VP7 codec

ABC.COM MAKES WATCHING TV AT WORK BETTER

By **David Carnoy** – April 2, 2007, 1:35 PM PDT

Among the networks, ABC.com has been one of the most aggressive in terms of streaming full versions of its shows online after they've aired on TV. During those quieter moments at work--some call them smoking breaks without the smoking--you can watch episodes of *Lost*, *Grey's Anatomy*, *Ugly Betty*, and more, right on your computer screen. Until a few days ago, however, the picture was rather small. Now ABC.com has launched a new full-screen "HD-quality" video player, and the "broadcast" looks shockingly good.

Meanwhile, NBC also has spruced up its online video player, and it, too, has added a full-screen option. The big difference is that NBC's video player uses Adobe's Flash Video, while ABC's is built on technology from upstart Move Networks.

During a "break," CNET editors John Falcone and Matthew Moskovciak huddled around my 19-inch Sony monitor in my office for a quick look at a recent *Lost* episode and were duly impressed. The picture may not quite measure up to true hi-def, but step a few feet back, and you're looking at a very detailed, sharp image. It's the kind of moment that makes you think that someday you'll be able to ditch your cable or satellite provider and get your content from cheaper sources that also happen to be legal. What a concept.



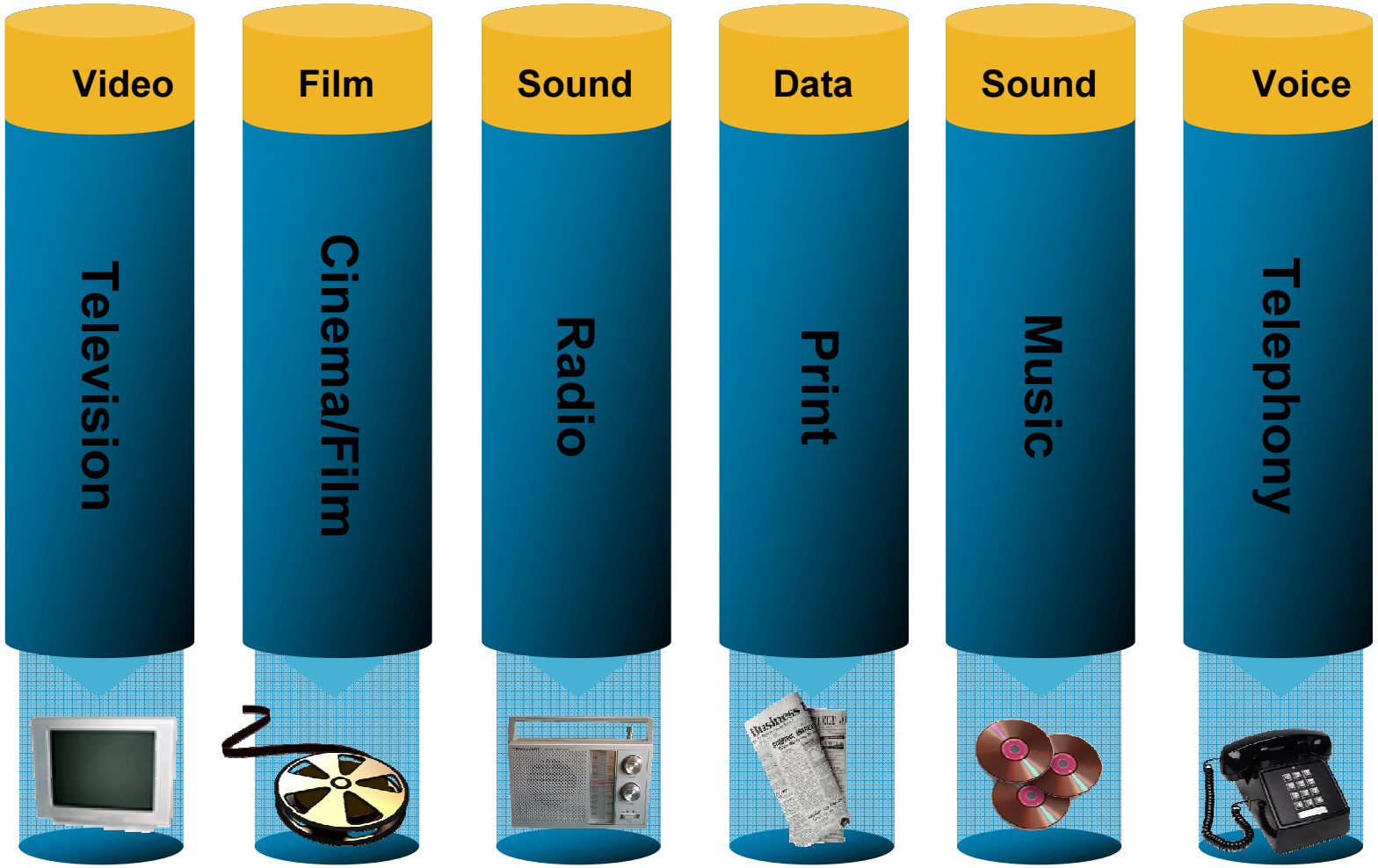
Lost looking good online.

- Applet in Browser.
- HTTP, Quantum streaming from standard Web Server.
- Many parallel TCP sessions for efficiency
- Free-of-charge CDN, Video/Web pages cached by many ISPs..
- Also cached on Client PC (eases replay).
- VP7 codec only requires 2/3rd of MPEG4 bandwidth.
- Can deliver, 1280*720p, 24fps resolution between 0,85-2Mbit/sec.
- Dynamically adapts playout resolution to bandwidth availability.
- Provides very detailed viewing reports based on Client Software (advertisers)

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The Digital Revolution in Entertainment

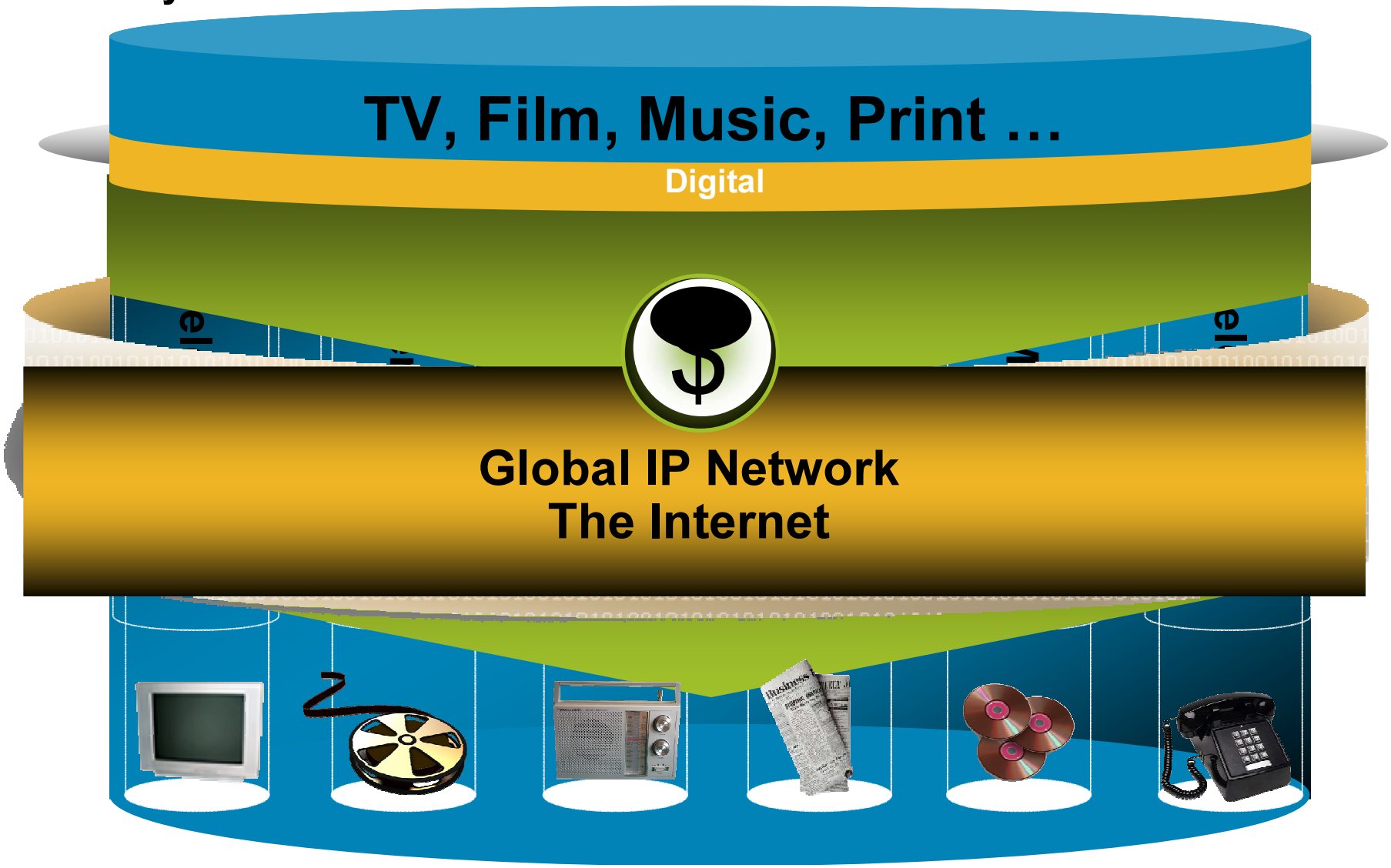
Yesterday



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The Digital Revolution in Entertainment

Today



The Revolution is Causing a Shift in Perspective: Copernicus Was Right



The Entertainment Model Must Evolve

The Implications of Moving from Analog to Digital Distribution

Digital

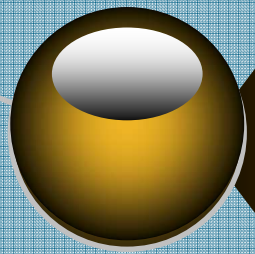
- Personal distribution
- Channel fragmentation
- **More** content
- **More** devices
- The venue of **your** choice



Consumers

Analog

- Mass distribution
- Control the content supply
- Limit the devices/venues



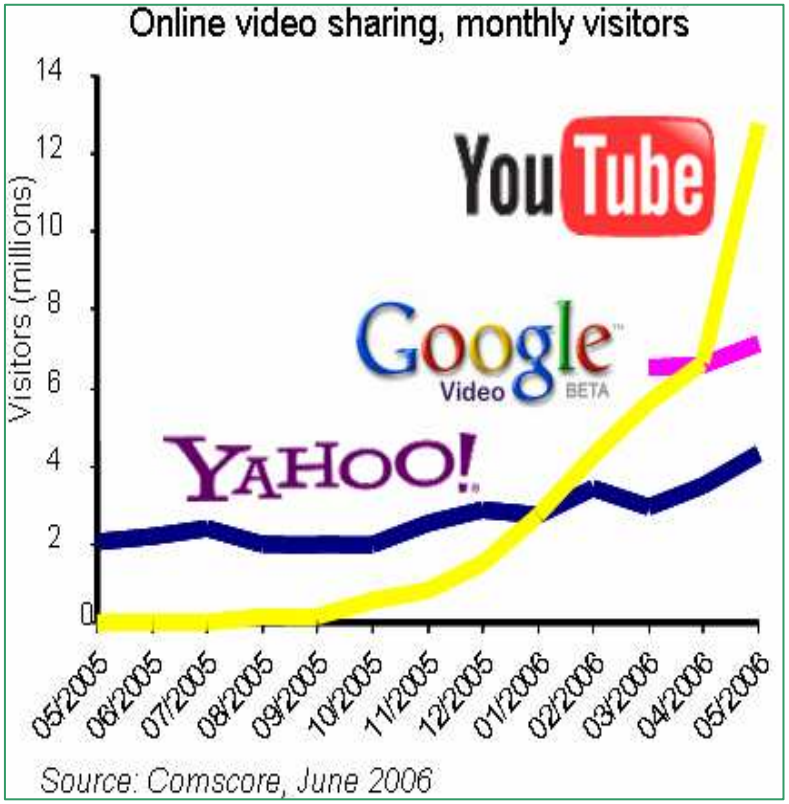
Studios

Social Trends

The Urge to Connect and Converse

10

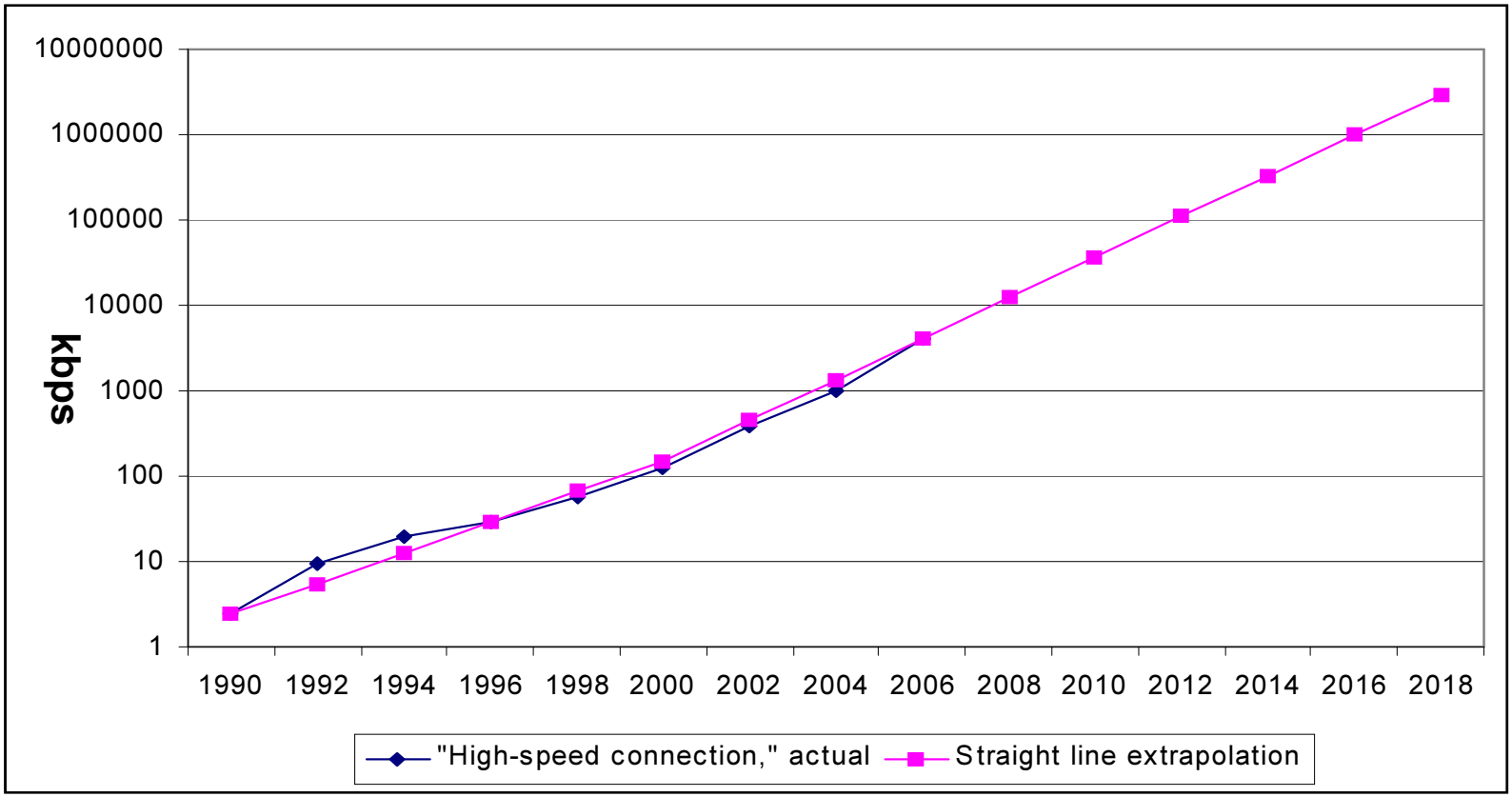
- "Grass roots" is important theme in new business models (e.g., Open source, Youtube, Myspace, Blogs)
- Web sites such as myspace, Youtube show value is in the consumer created data store and value grows with users and usage
- Consumerization will be the most significant trend affecting IT during the next 10 years (0.8 probability, according to Gartner)
- But it comes with a price: fraud, IP issues, theft, spam, poor quality



Increasing bandwidth has shifted power to consumers in value chain

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The Growth in Bandwidth Demand



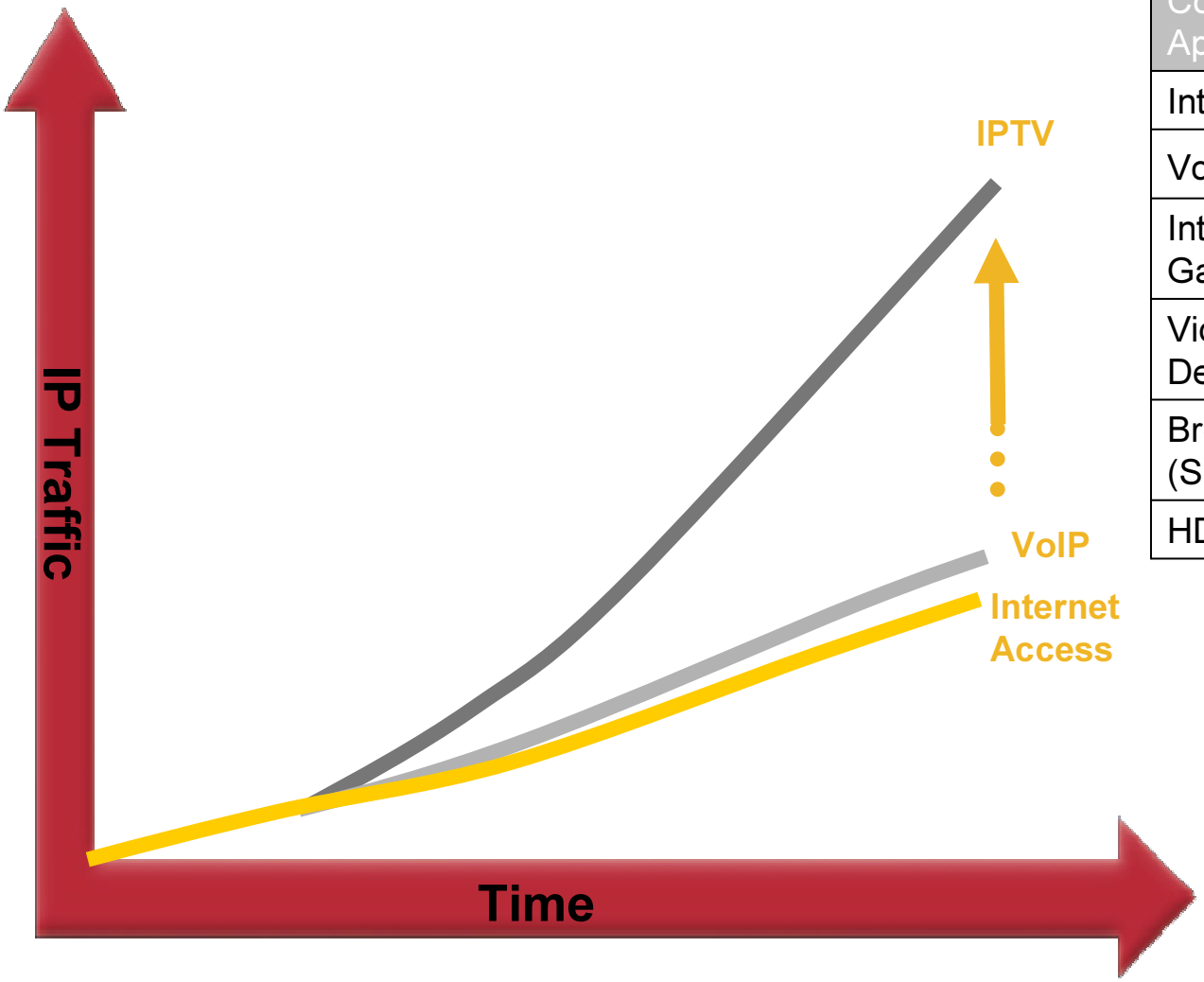
If history is a good guide, 10 Mbit/s will be a standard high-speed connection by 2007, and 100 Mbit/s by 2011

Source: Light Reading 2005

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Global Traffic Growth

Video and IP Rich Media Drive Growth

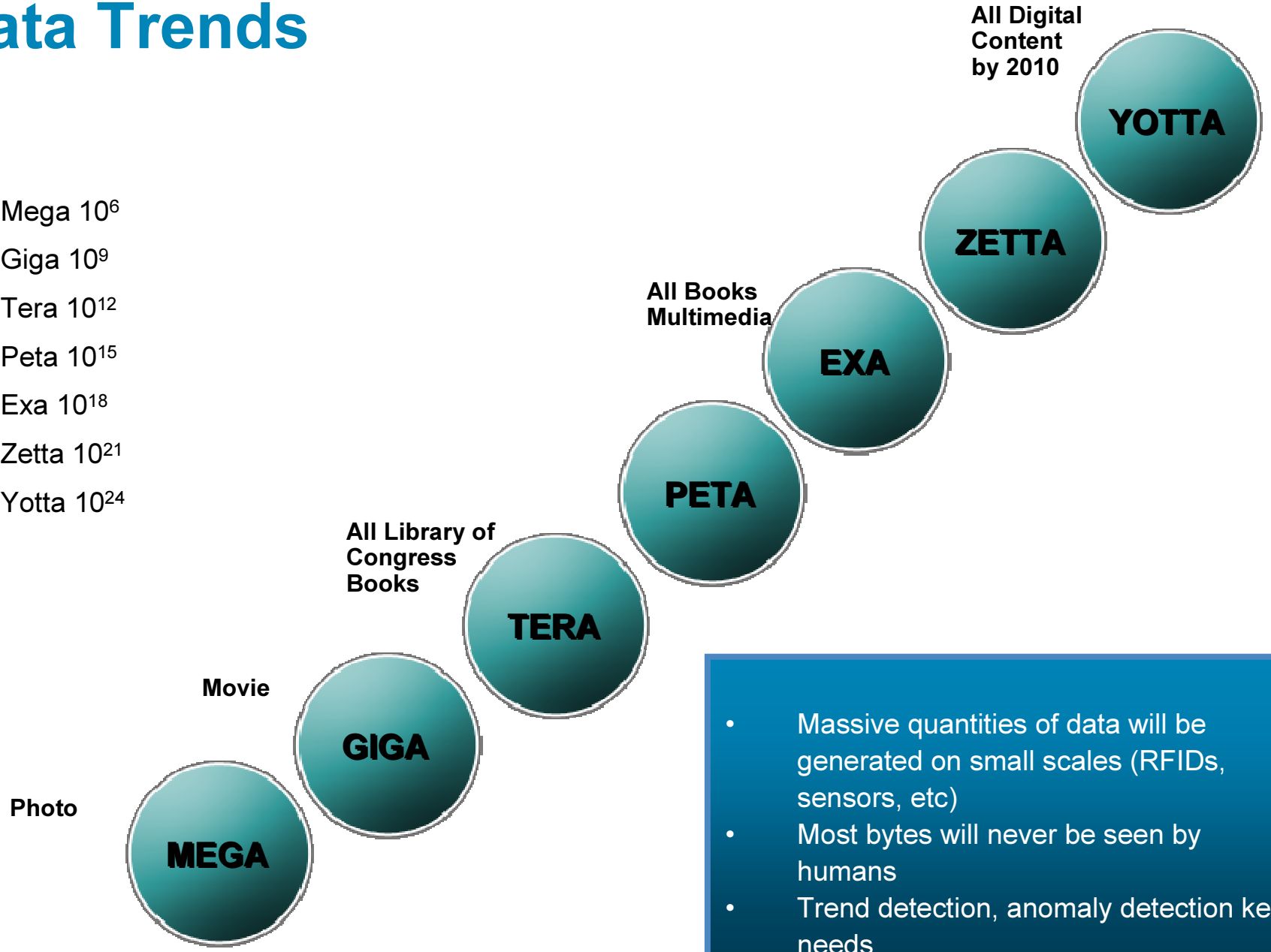


Consumer Applications	Bandwidth Required
Internet	.500 - 1.5 Mbps
VoIP	30Kbps-100 Kbps
Interactive Gaming	128k - 6.0 Mbps
Video on Demand	3.0 - 6.0 Mbps
Broadcast TV (SD-TV)	3.0 – 5.0 Mbps
HDTV MPEG-4	6.0 – 7.0 Mbps

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Data Trends

- Mega 10^6
- Giga 10^9
- Tera 10^{12}
- Peta 10^{15}
- Exa 10^{18}
- Zetta 10^{21}
- Yotta 10^{24}



- Massive quantities of data will be generated on small scales (RFIDs, sensors, etc)
- Most bytes will never be seen by humans
- Trend detection, anomaly detection key needs

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Data Trends

Emergence of Rich Media

Emergence of Rich Media

Commercial

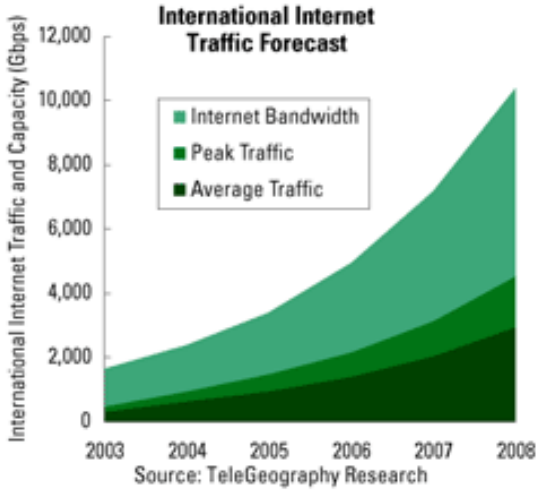
- **4500 motion pictures -> 9,000 hours/year (4.5 TB)**
- **33,000 TV stations x 4 hrs/day -> 48,000,000 hrs/yr (24,000 TB)**
- **44,000 radio stations x 4 hrs/day -> 65,500,000 hrs/yr (3,275 TB)**

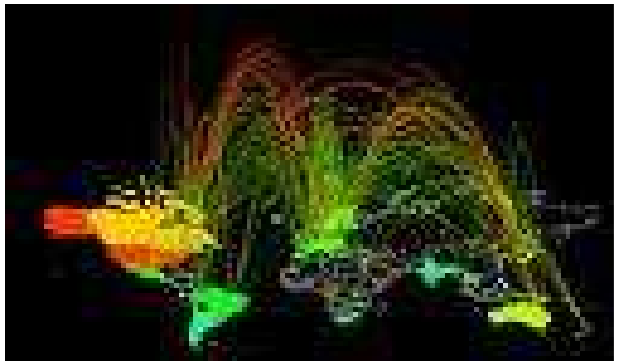
Personal

- **Photographs: 80 billion images -> 410,000 TB/yr**
- **Home videos: 1.4 billion tapes -> 300,000 TB/yr**
- **X-rays: 2 billion -> 17,000 TB/yr**

Surveillance

- **Airports: 14,000 terminals x 140 cameras x 24 hrs/day -> 48 M hrs/day**





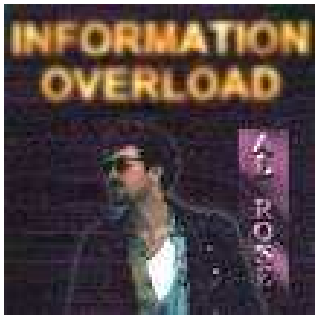
Technology to index, search, and recognize images will be key.

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Data Trends

Information Obesity

1. More information choice but ability for humans to consume it is static. One solution is to deal with information asynchronously
2. Our future consumption of bits will be very conversational, characterized by bursts
3. Consumers want to streamline, time shift
4. Social behavior will also become more asynchronous, with all of us moving in much less lockstep. The net result is fragmentation: the loss of mass shared experience
5. Moreover, the dominant user of the Net in the future will not be people at all. It will be machines talking to one another.
6. Increasingly, these bits will arrive wirelessly
7. " the value of any product or service increases exponentially with mobility. …"





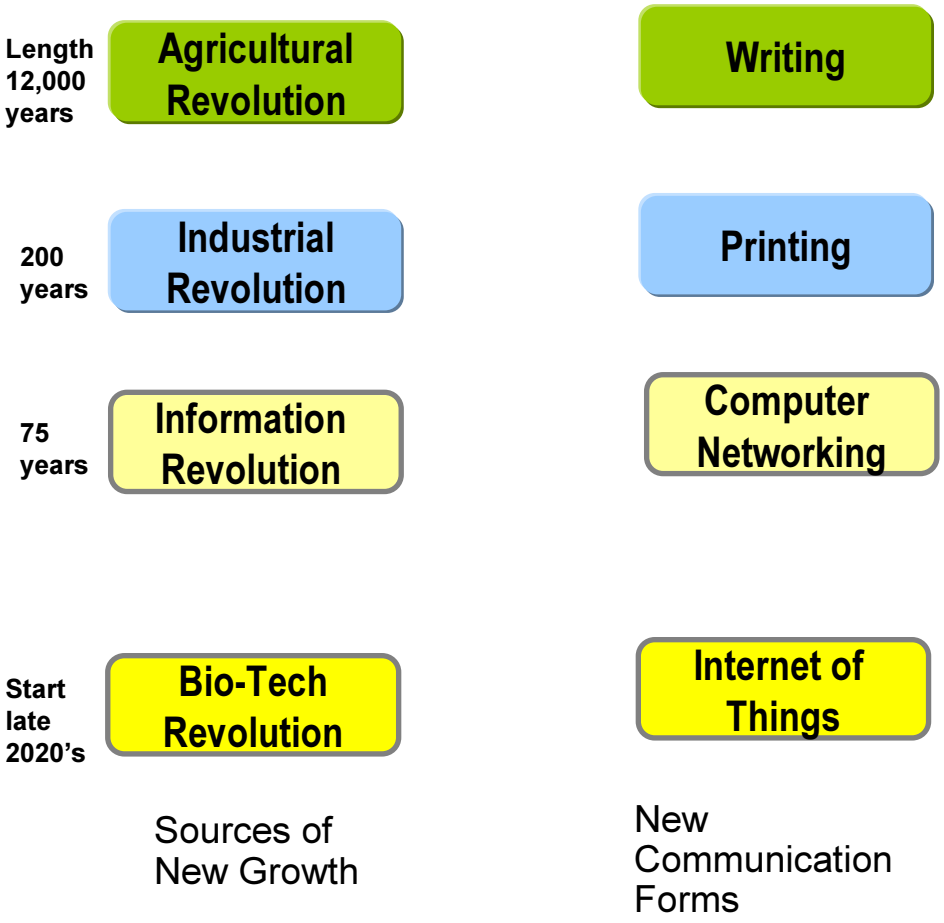
"Everywhere you look, every time you listen, someone is trying their very best to snag your attention… Every week sees another new magazine, supplement, cable channel or radio station. Then there are e-mails, websites, text messages and those DVDs with special extra bits… We drown in data."

Info-besity epidemic by John Naish

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Economic Trends

Major Phases of Development



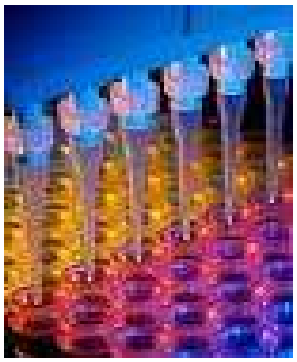
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Economic Trends

Bio-tech Revolution

- We're halfway through the information economy. BioEconomy will take off during the 2020s.
- The BioEconomy started in 1953, when Crick and Watson identified the DNA helix, then the human genome was mapped.
- During the overlap of infotech and biotech many biological processes will be digitized.
- Aging of population in many countries is a biotech driver.





Each era has its dark side.

The industrial age concern is pollution.

Information age concern is privacy.

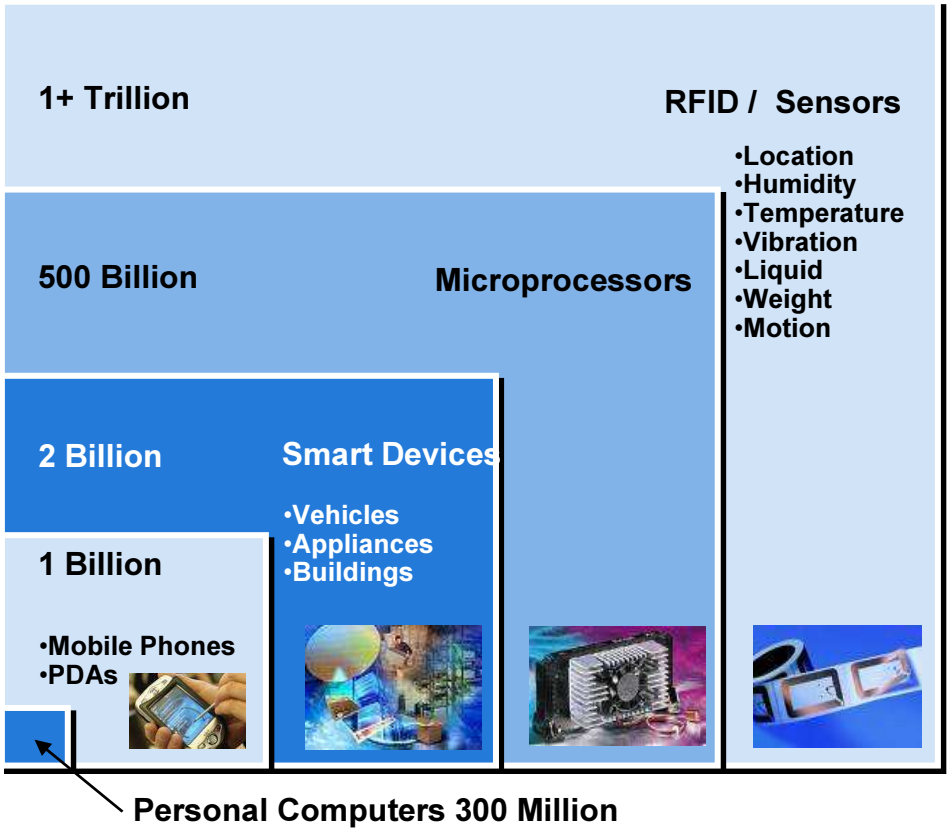
BioEconomy, the issue is ethics - cloning, stem-cells, etc.

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Technology Themes

The Largest Network on the Planet

- So many appliances, vehicles and buildings will be online by 2020 that there will be more things connected than people
- Internet of Things



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Network Model?

- **Everything in RFID is dependent on the network**
  - The value of RFID in inherently about linking across enterprises, thus integrating the business network or supply chain
  - The applications on this network are increasingly mobile/wireless

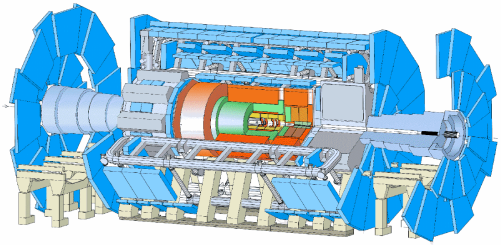
- By **2009** significant share of traffic on our networks will be RFID related
  - So, leverage my network assets, converging all application and frequencies on one platform (data, voice, video, RF, GPS)

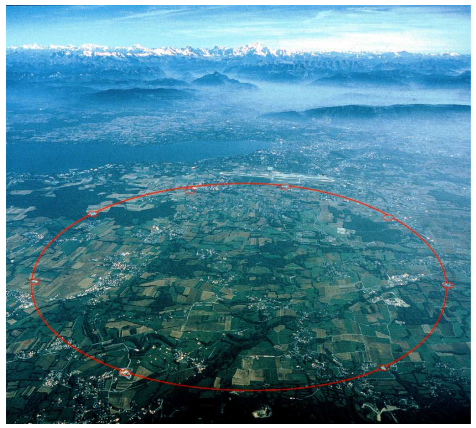
- By **2014** reader populations may approach 300 million
  - Help me preserve my bandwidth by making decisions as close to the edge as possible
  - Help with the chaos to manage my heterogeneous devices

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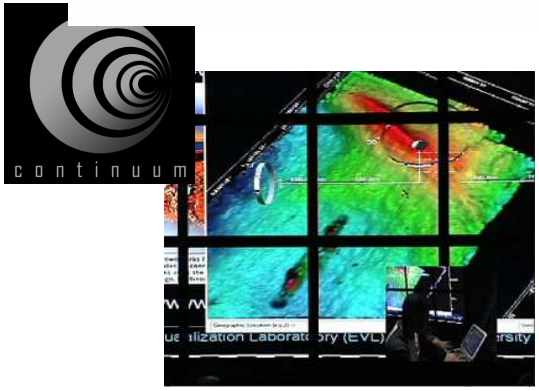
Grid Applications

- High Energy Physics (HEP)
  - Today 1 PetaByte per sec
  - Tens of PetaByte 2008
  - 1 ExaByte 2015





- Distributed Visualization
  - http://www.evl.uic.edu/cavern/optiputer/
  - http://www.evl.uic.edu/cavern/continuum/indexmain.html
- 3D visualization tools are used
- Key tools needed to process & analyze approximately 64 Tbyte of data by 2008

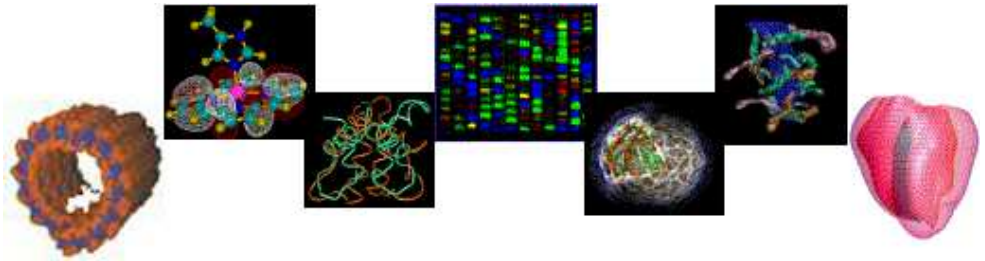


Continuum - Enhanced Distributed Collaboration
http://www.evl.uic.edu/cavern/continuum/indexmain.html



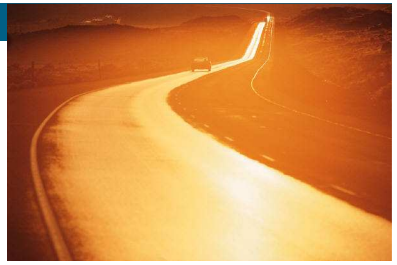
GeoWall2 (NSF) - GeoScience Advanced Visualization
http://www.evl.uic.edu/cavern/optiputer/geowall2.html

- Remote screening - Mammography
  - Digitized image results 75MB
  - Radiologist performs 100 patient readings per day (1 image every 30sec)
  - 16 images per patient results in 16 * 75MByte = 1.2GByte
  - 100 patients screened remotely means 1.2 Gbyte data every 30 sec



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Impact



- Change form processor centric to BW dominated computing
  - http://www.calit2.net/news/2002/9-25-optiputer.html
  - Around 2010 Grid applications will require an International Distributed Cyber Infrastructure based on
  - Petascale computing, exabyte storage, and terabit networks
- Terabit challenge
  - http://www.cmf.nrl.navy.mil/CCS/
  - Terabit global Large Data SOA
  - Integrate federated, distributed computational grids, realtime sensors, and digital historical information
  - Scalable to support exponentially increasing data
  - Privacy, authenticity and security demands: InfoAssured
  - Affordable … highly available … E2E QoS/QoP flows
  - Legacy and rapidly evolving technology integration
  - Perf, NetOps, Information Assurance tools/sensors

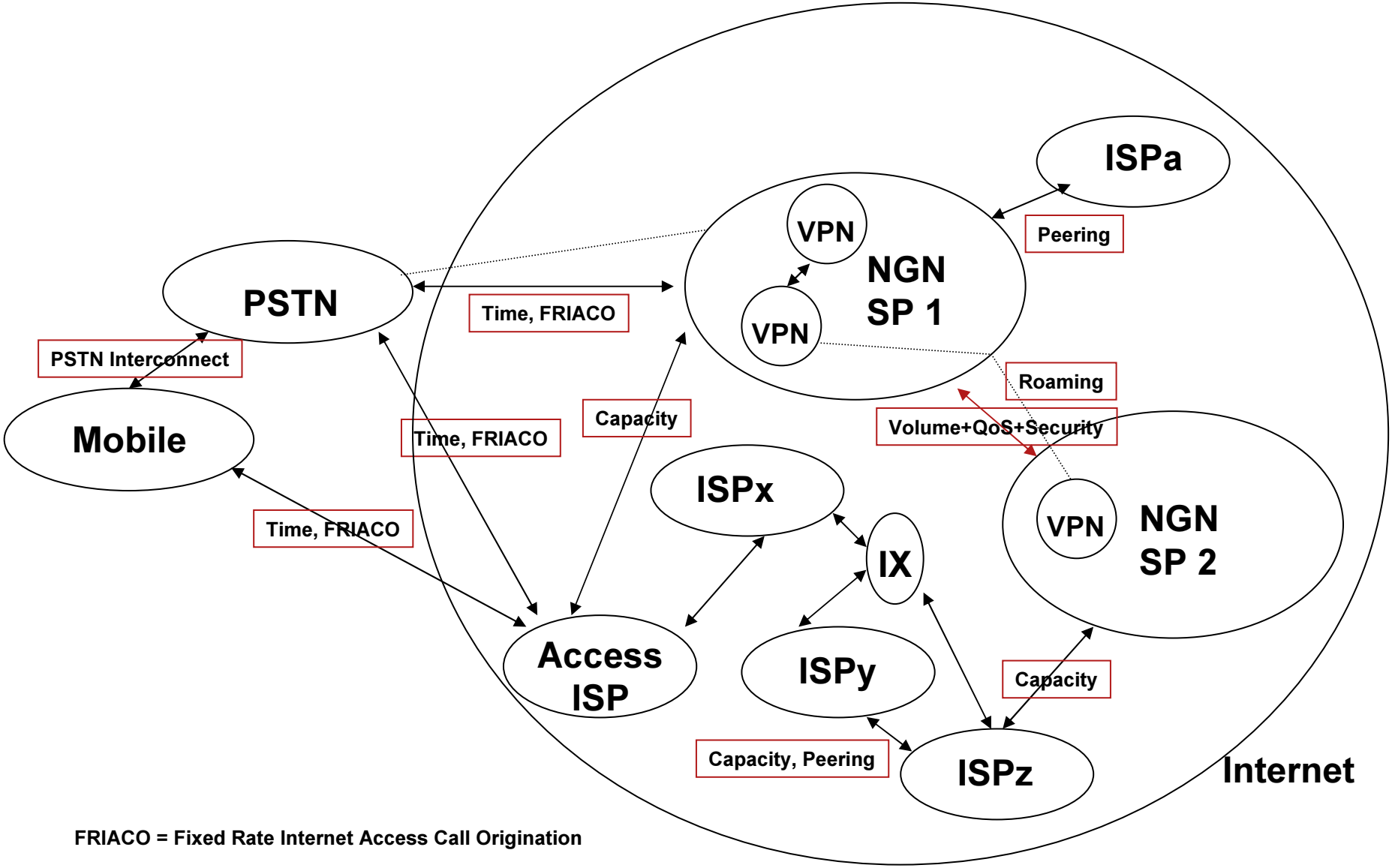
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Network Scaling

	2005	0 - 2 Years	3 - 5 Years	5 - 15 Years
Optical Streams	1 - 10 Gbps	10 - 40 Gbps	120 - 640 Gbps	1 - 10 Tbps
Optical Ctrl Plane	STATIC Provisioned	DYNAMIC GMPLS	DYNAMIC Burst/JIT GMPLS	DYNAMIC Burst/Flow GMPLS
Control Plane	STATIC Tunnel	DYNAMIC SIP	SIP QoS / QoP	
LAN / WAN Technology	IPv4 10GE OC12 4xSDR IB	IPv6 10GE 4x/12x SDR/DDR IB	IPv6 100GE 12xQDR IB 64-128 IB	All Optical System Interconnect
Security Devices	1.0G IPv4 FW, K5, 3DES, CBs, KGs, NTAM	10G KGs HAIPEs, CAC, FEON, PKI, NTAM	40G HAIPE Scalable GFP Encrypter	640G HAIPE GFP Encrypter

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Interconnect?

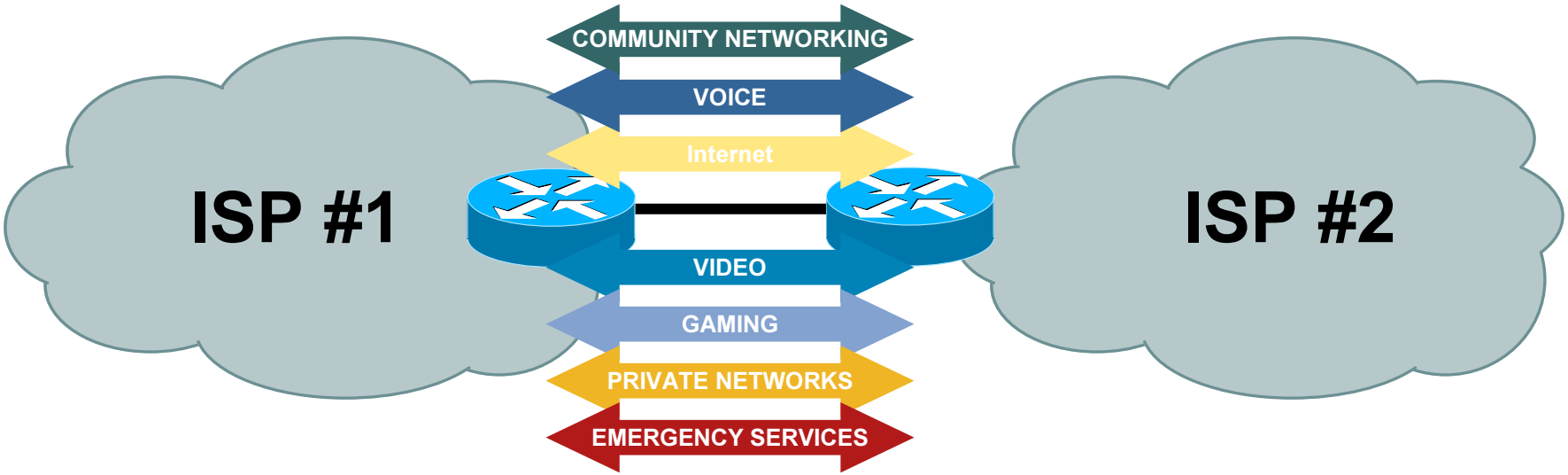


FRIACO = Fixed Rate Internet Access Call Origination

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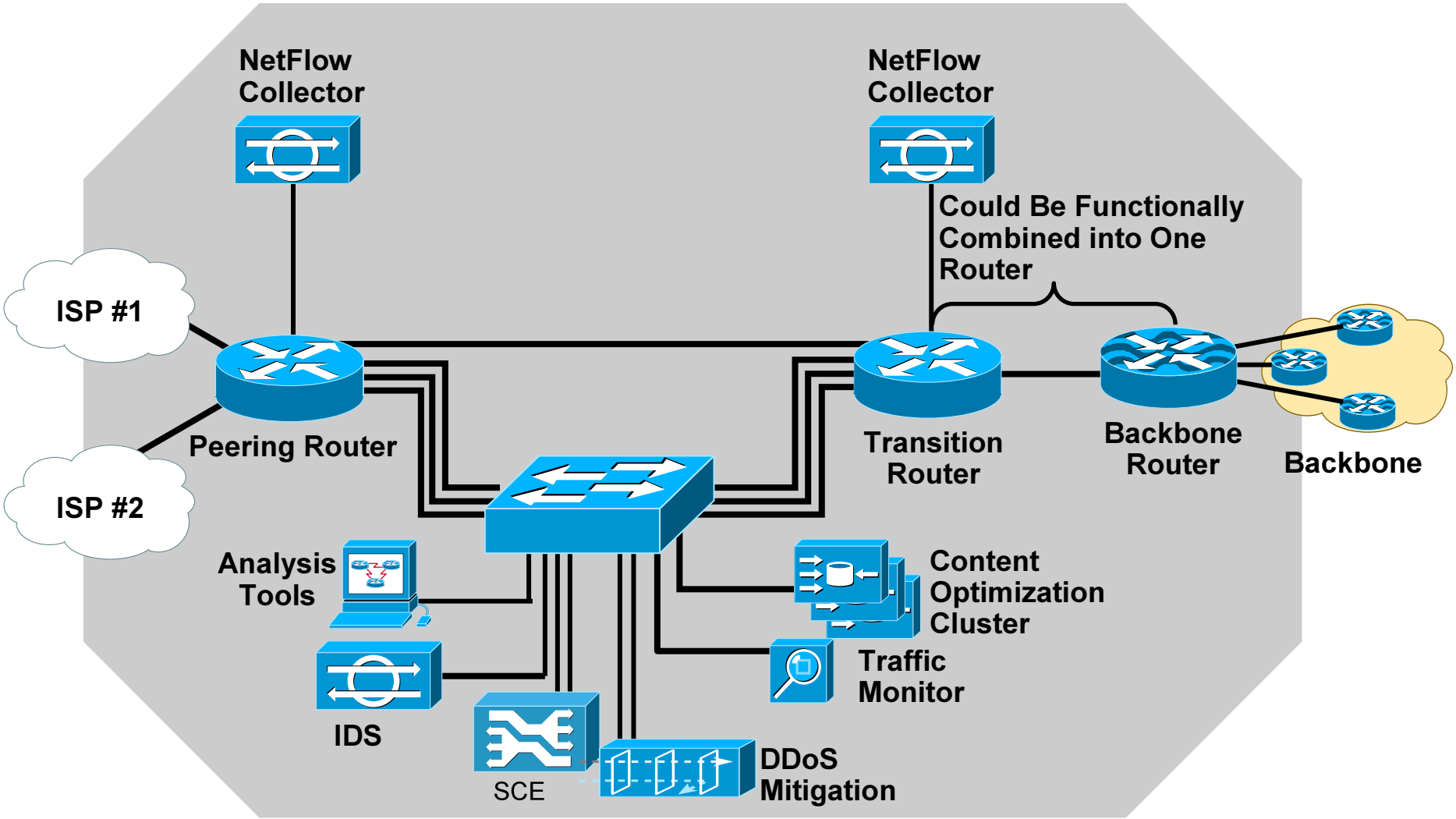
Is Convergence Realistic?

- No longer simply best effort Internet traffic
- Network Convergence and Business Relevance are mandating that new unprecedented network controls be implemented to provide consistent, predictable behavior



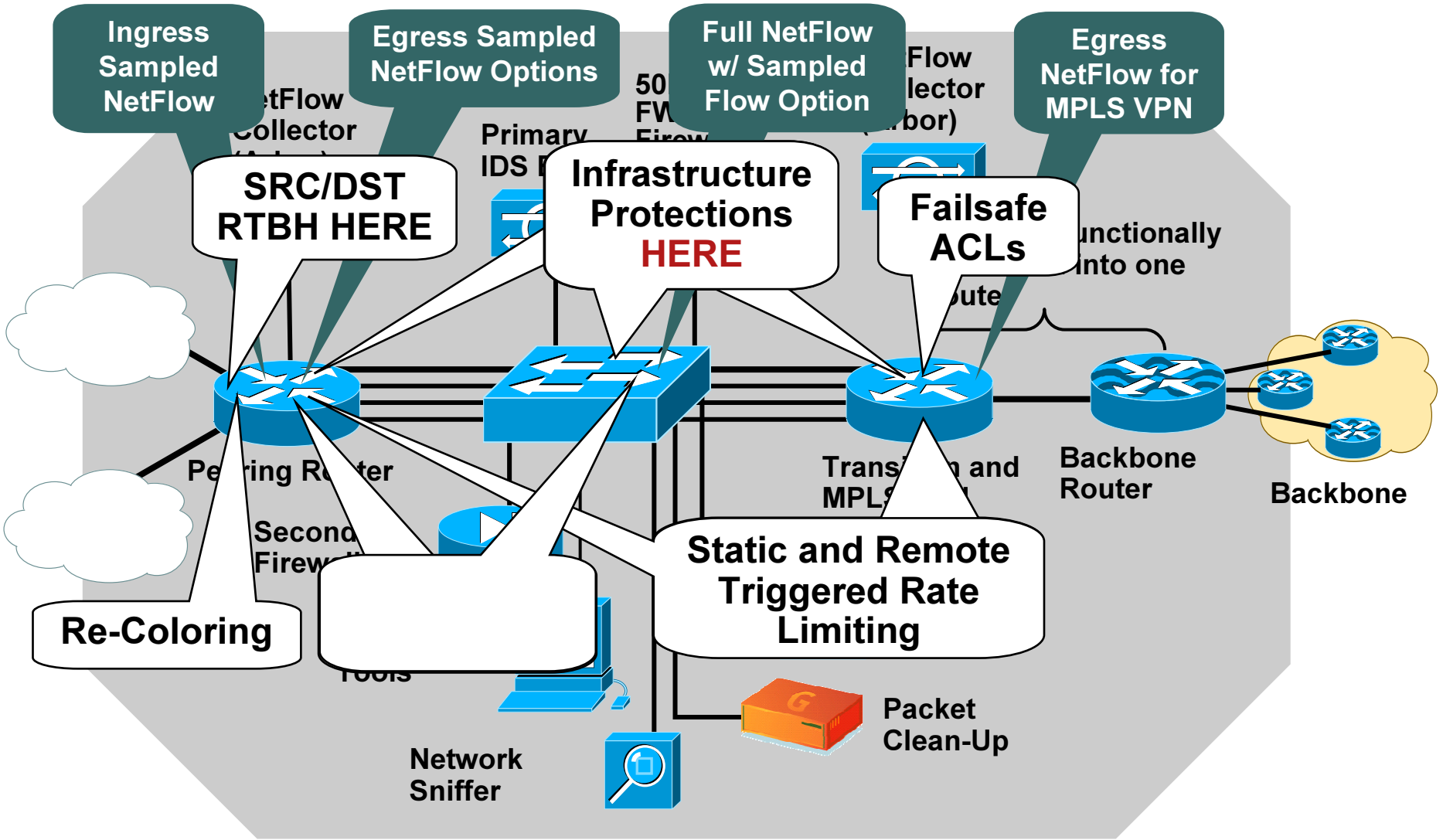
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Wholesale IP Peering Pressures



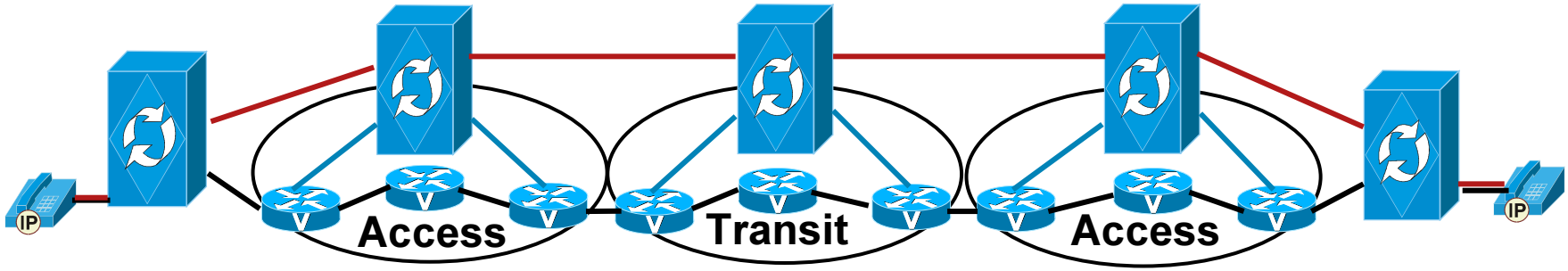
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Layered SP Peering Module Designed Around Security and Compartmentization



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SP Interconnection - Multimedia

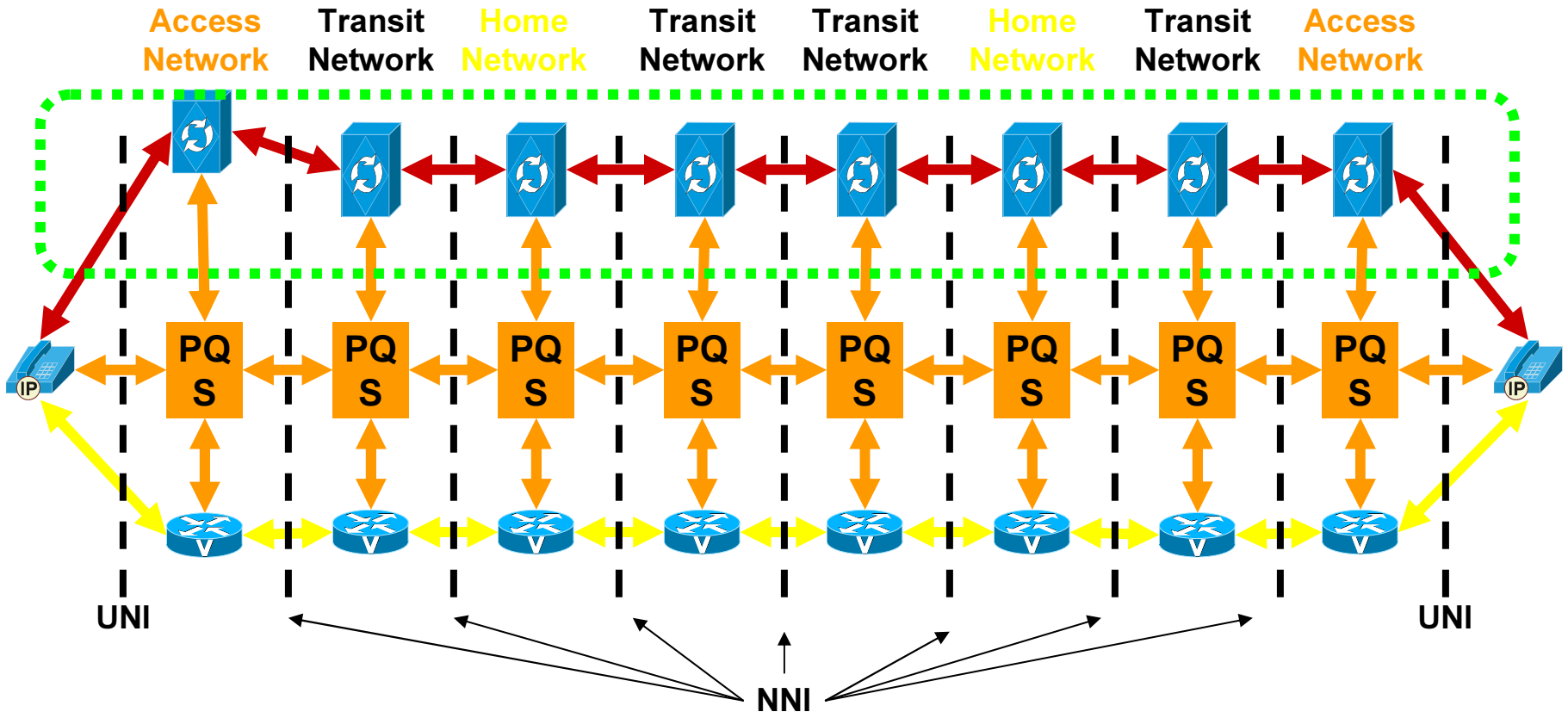


- SIP-based call/session control (end-to-end)
- Session border control

- Session control functions:
  - Protocol interworking
  - Call admission control
  - Address translation
  - Routing
  - Billing
  - Lawful interception
- Border element functions:
  - Firewall
  - NAT
  - NAPT
  - QoS
  - QoS monitoring
  - Transcoding
  - Encryption / decryption
  - Mid-call codec change
  - Media transformation
  - Lawful interception
  - Usage metering

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SP Interconnect – VoIP Viewpoint 1: Hop-by-hop (Logical view)



- Signaling (e.g. SIP/SDP)
- Control (e.g. Policy, QoS, Security)
- Media (e.g. RTP)

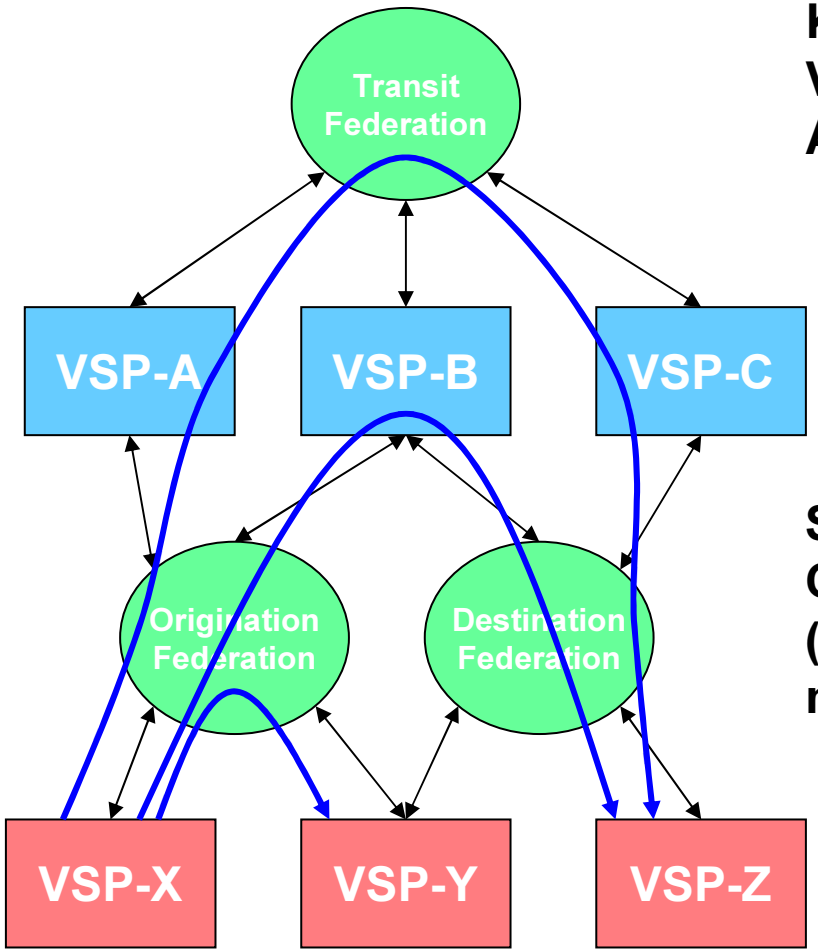
What are the odds that all VoIP SP agree on 381 RFC/ID's?

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SPEERMINT Federation Concept

Federation = Multilateral Rule Set including common Certificate Authority

Federation could be organized and operated by a Service Provider

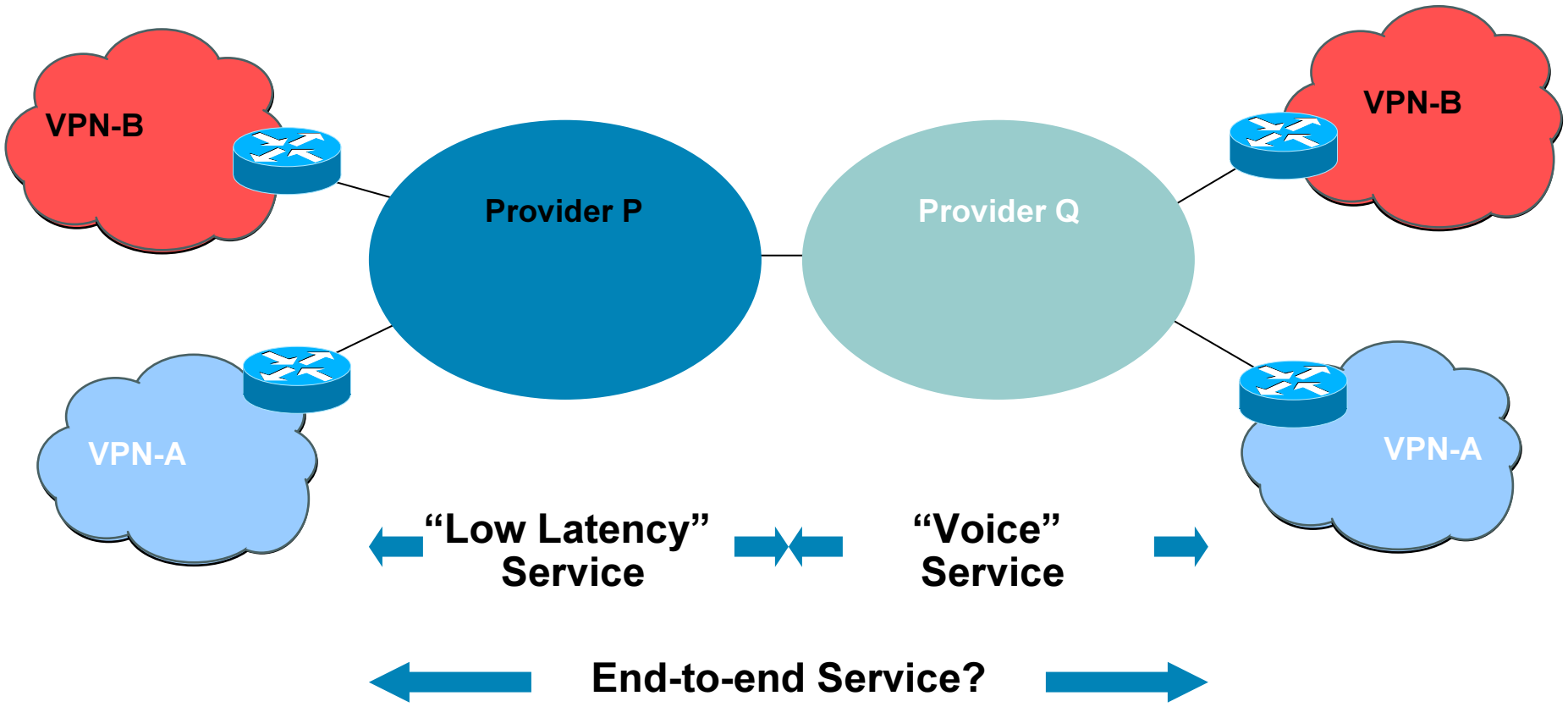


Key Information = VSP to Federation Association in DNS

Stumbling Block = Closed ENUM (Imagine if DNS was not public)

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The Challenge of Inter-provider QoS



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Main Interprovider QoS Challenges

- Service concatenation
- Metric consistency
  - Jitter definitions particularly variable across providers
- Service definition consistency
  - Consistency is a side-effect more than a goal
- Impairment allocation
- Measurement & Reporting
- Troubleshooting
- Routing

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Metric Specification Example: Jitter

- RFC 3393 defines IPDV (IP delay variation) but allows many parameters to be varied, such as:
  - Frequency of measurement
  - Duration of test & number of probes
  - “Selector function” - which 2 points on the delay distribution do you compare?
- CFP doc recommends:
  - 200ms average inter-probe interval, Poisson process
  - 5 minute test (1500 probes per test)
  - Report $IPDV_{99}$ = 99th percentile delay - min delay

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Impairment Allocation

- Suppose:
  - 5 providers are in the path between two end-user sites
  - QoS targets are 100ms delay, 50ms IPDV, 10^{-3} loss
- Question: how much delay, loss, jitter can each provider introduce?
- Jitter is particularly troublesome - not additive
  - For example, if $IPDV_{99}$ is 25ms for one network and 25ms for a second network, then $IPDV_{98}$ is 50ms for both in series

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Implications of two-point promise

- Providers have a common definition of IPDV
- Providers commit to the same “low” and “high” IPDV targets and probability of meeting them
- Reporting can be done on exception basis - report only those intervals where IPDV was “high” or “extreme”
- Delivery of end-to-end SLA considerably simplified
- Providers can collect & report own measurements, but a provider may choose to verify by probing through another provider
- Loss and delay
  - Can be handled with “one-point promise”
  - Again, report only when promise is not met

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A View From One Tier 1 Provider

- Network spans 13 Countries
  - Europe
  - Asia
  - US
- IP Transit services
  - IPv6 unicast
  - IPv4 unicast, multicast

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Network Makeup (2006-11-13)

- 220 Private Peering Ports
- 31 “Fee” Peering Ports
- 794 Backbone Cust/BU facing ifcs
- 5143 “Other” customer/internal ports
- 26 L2VPN ports
- STM-1 to 10G (OC192/10GE) interfaces

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Customer Link Speeds

- 726 100m FE
- 211 GE
- 27 10g
- 4 40g (4x10GE)
- 2 20g (2x10GE)

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Peering Protocols

- IPv4
  - Unicast
  - multicast
- IPv6
  - unicast
- **no “NGN” peering (SBC/VOIP)**

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Overall (projection) Peering

- 2006 - 124g
- 2007 - 223g
- 2008 - 401g
- 2009 - 723g
- 2010 - 1.3tb

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Tier 1 Provider's Personal Opinions

- Google/Yahoo will negotiate private peering with large "eyeball" networks or their upstreams
  - May or may not be settlement free
- 100G+ interfaces are engineering challenge but solutions needed in this space "soon"
- Core network interconnect costs will drive multihoming for tier-2 networks to new levels
- Global and Regional politics (eg: China) will play more of a role in peering agreements
- Parallel networks will emerge to solve capacity and customer specific problems (niche networks)
- Eventual death of public exchange points due to cost (in major markets)

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L3 Design Considerations



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Service Elements
Flexible Service and application Edge

- Why a flexible edge
  - Not all services are the same
- Where the service is created
- Rich set of service functionality based on IP/MPLS control plane
  - L1, L2, L3 VPNs with QoS
- Service Interfaces
  - Ethernet Virtual Connection :- Flexible transport mapping
  - Internet Services Gateway :- Subscriber control and policy interface
- Service modules, blades and appliances
  - VPN aware service functionality
  - Network orientated
  - Application orientated :- Video monitoring, Session Border controllers

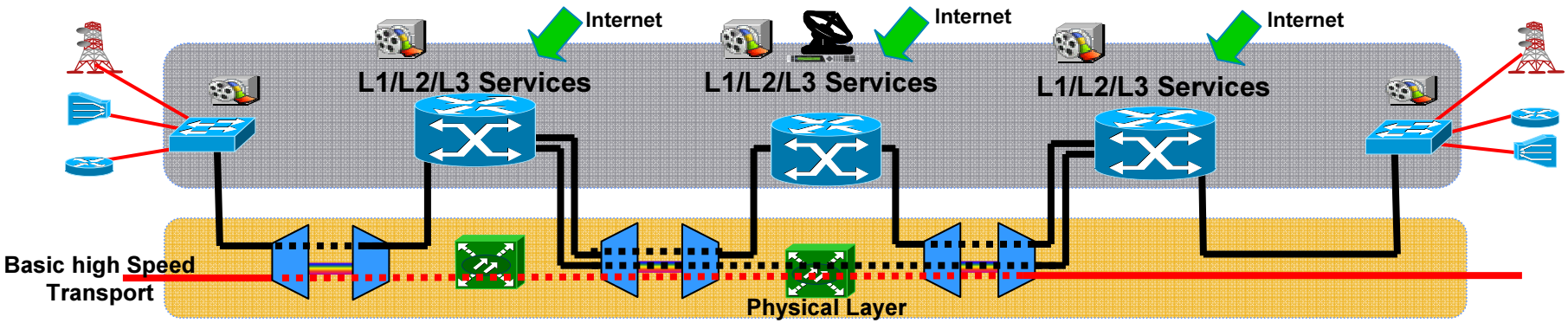
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Service Requirements in NGN infrastructures

Consumer	Business	Wholesale	Evolved services
Internet Voice / Video / data Voice PSTN Multimedia Video Walled Garden Mobility Fixed / Mobile	TDM/ATM→Ethernet L2 VPNs Pt2Pt Pt2MPt MPt2MPt L3 VPNs Connectionless Value-add Services Based on L3 visibility	TDM/ATM→ Ethernet L2 VPNs Pt2Pt Pt2MPt MPt2MPt L3 VPNs L2TP Connectionless	PSTN Migrate Mobile RAN backhaul IP transition TDM Migrate and evolve to Ethernet ATM Migrate and evolve to Ethernet F/R Evolve to Ethernet

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Network structure



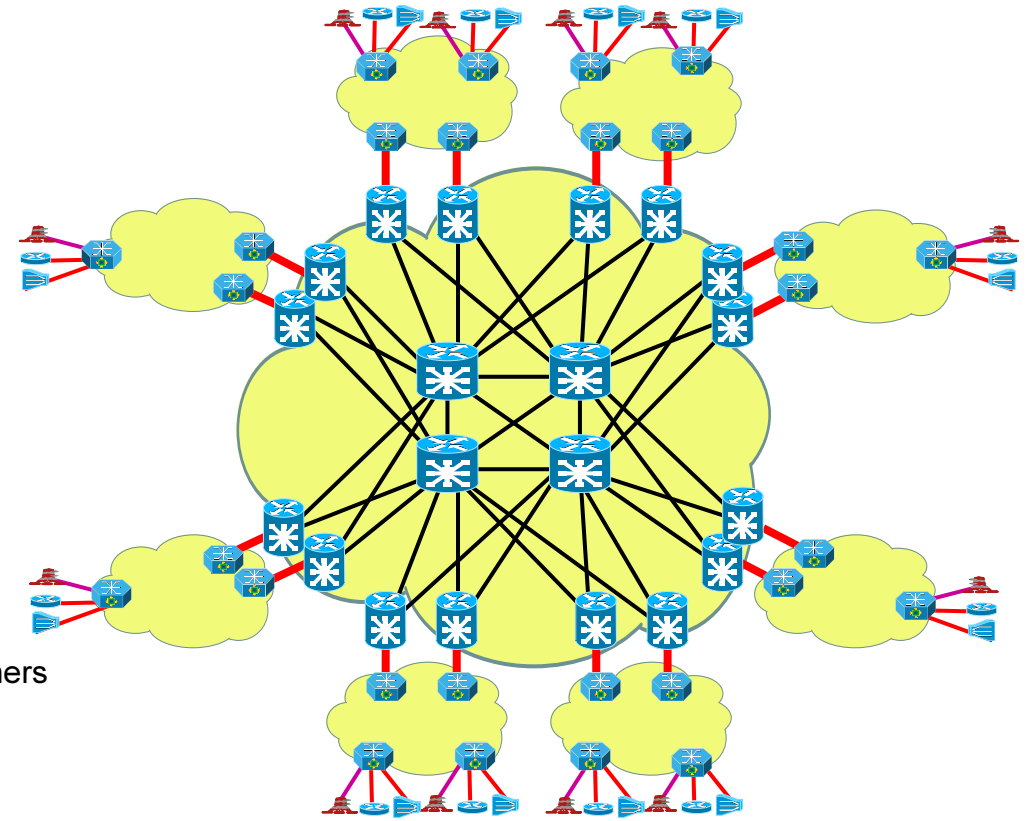
- Optical layer
  - Dark fibre and / or DWDM
  - Basic non-oversubscribed point to point high bandwidth services
  - Under lying transport for IP/MPLS infrastructure
- IP/MPLS
  - End to end IP/MPLS control plane
  - IP/MPLS directly integrating with optical layer – dark fibre or DWDM
  - Concurrent support of L1, L2, L3 services
- Flexible Service Edge
  - Service termination
  - Content injection
- Minimal layering
  - Rapid adaptation
  - CAPEX and OPEX efficiencies

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Network Structure

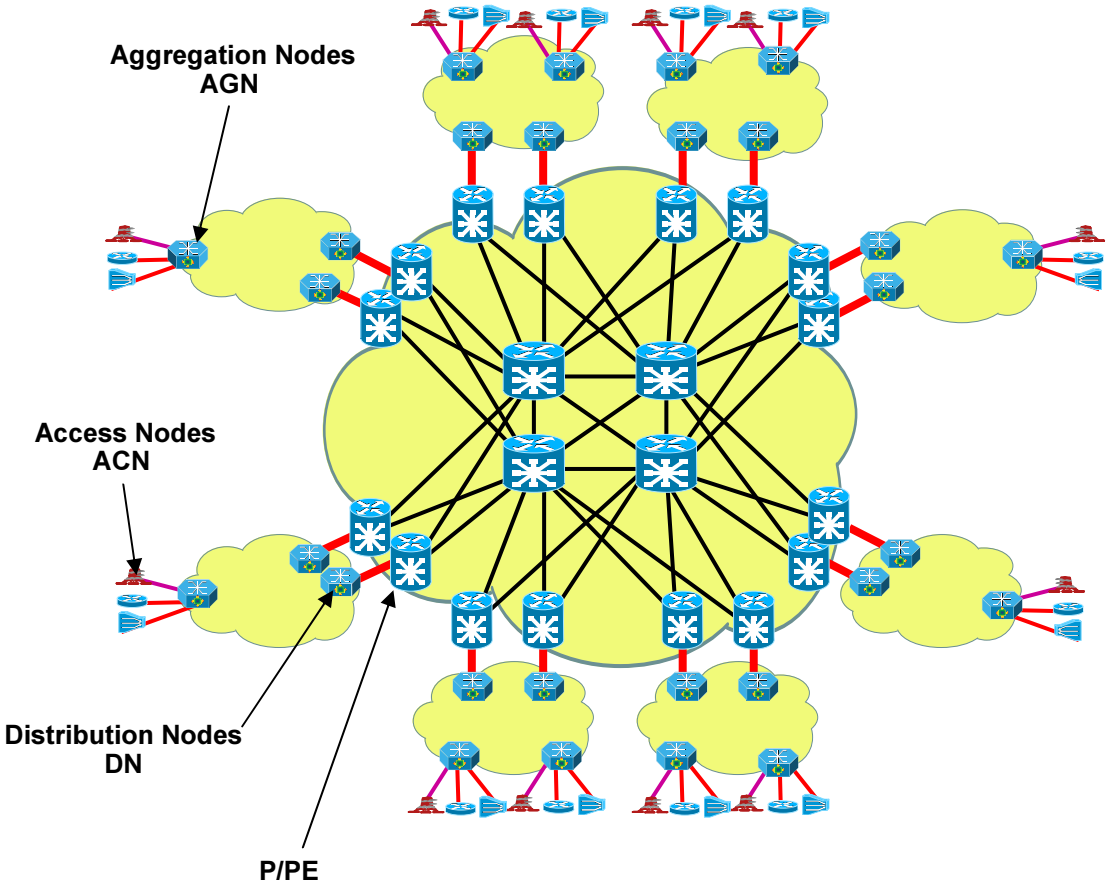
- Core network
  - Partial mesh / full mesh on core
  - Hierarchical structure to IP/PoPs
  - IP/MPLS control plane
  - IPoDWM capabilities
  - Supporting transport of L1, L2 and L3 services
- Metro and city regions
  - Ring / Hub and Spoke
  - Technology options:- IEEE, IP/MPLS, SDH, Others
- Factors determining technology in metro
  - Size of metro and number of customers
  - Physical topology : Ring / hub & spoke
  - Types of services offered in metro:- L1, L2, L3
  - Service offerings :- Pt2Pt, Multipoint
  - Application offerings



Cisco support a variety of technologies in the metro. For an end to end full service infrastructure an IP/MPLS control plane provides the most flexibility

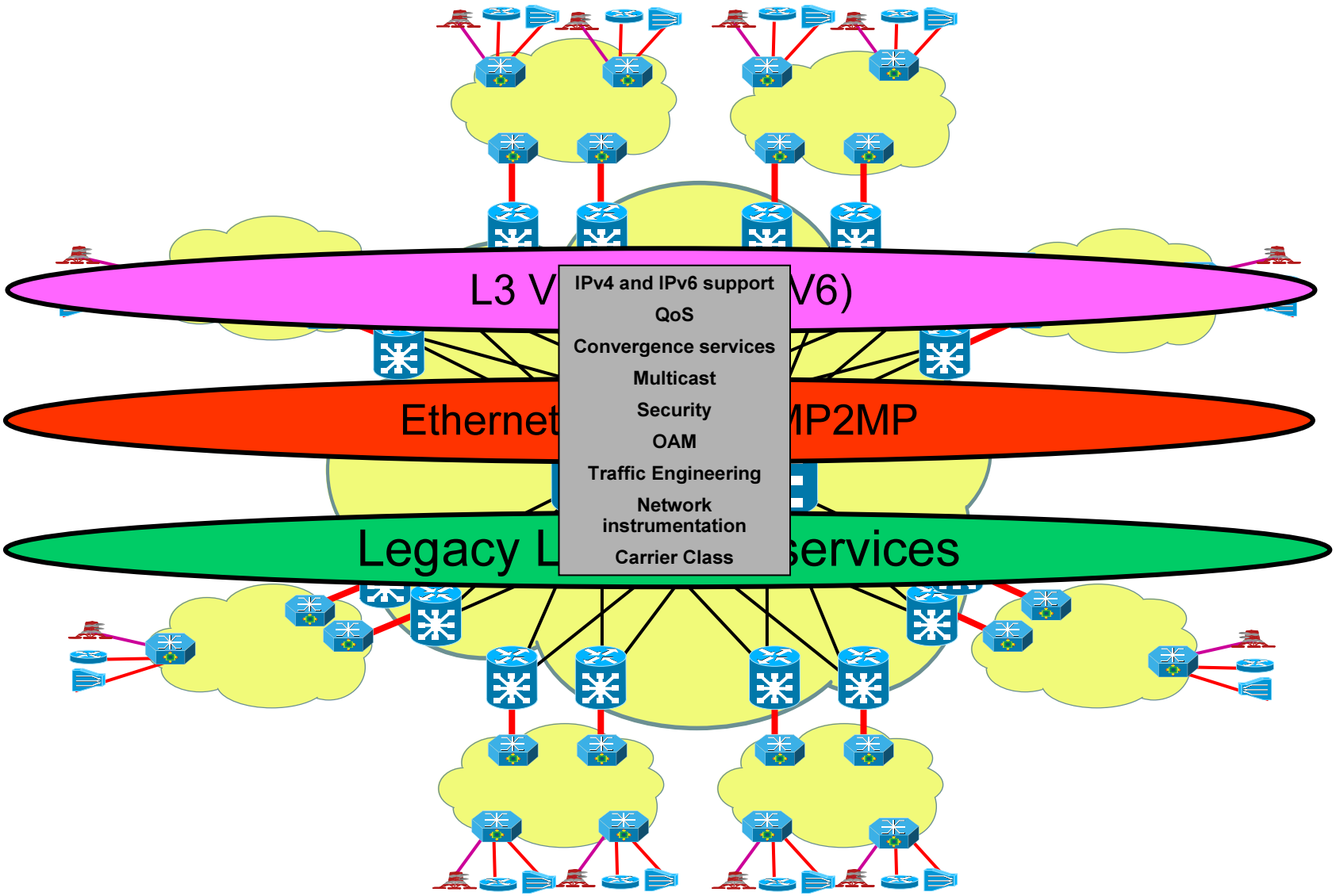
Network Structure
Functional capabilities

- Large scale infrastructure
  - Consolidation of components in small networks
  - Example DN and P/PE node
- Access Node
  - Pure layer 2 device
  - L2+ features :- DHCP opt 82, IGMP snooping
- Aggregation Node
  - IP/MPLS Control Plane
  - Legacy service edge (TDM and ATM)
  - Ethernet layer 2 service edge
  - Video layer 3 service edge
  - Video layer 3 caching
  - Video monitoring and optimisation
- Distribution Node
  - IP/MPLS Control plane
  - Boundary to core environment
  - Legacy service edge (TDM and ATM)
  - Ethernet layer 2 service edge
  - Video layer 3 local content injection
- P/PE
  - IP/MPLS device
  - AS boundary
  - L1, L2, L3 service capabilities



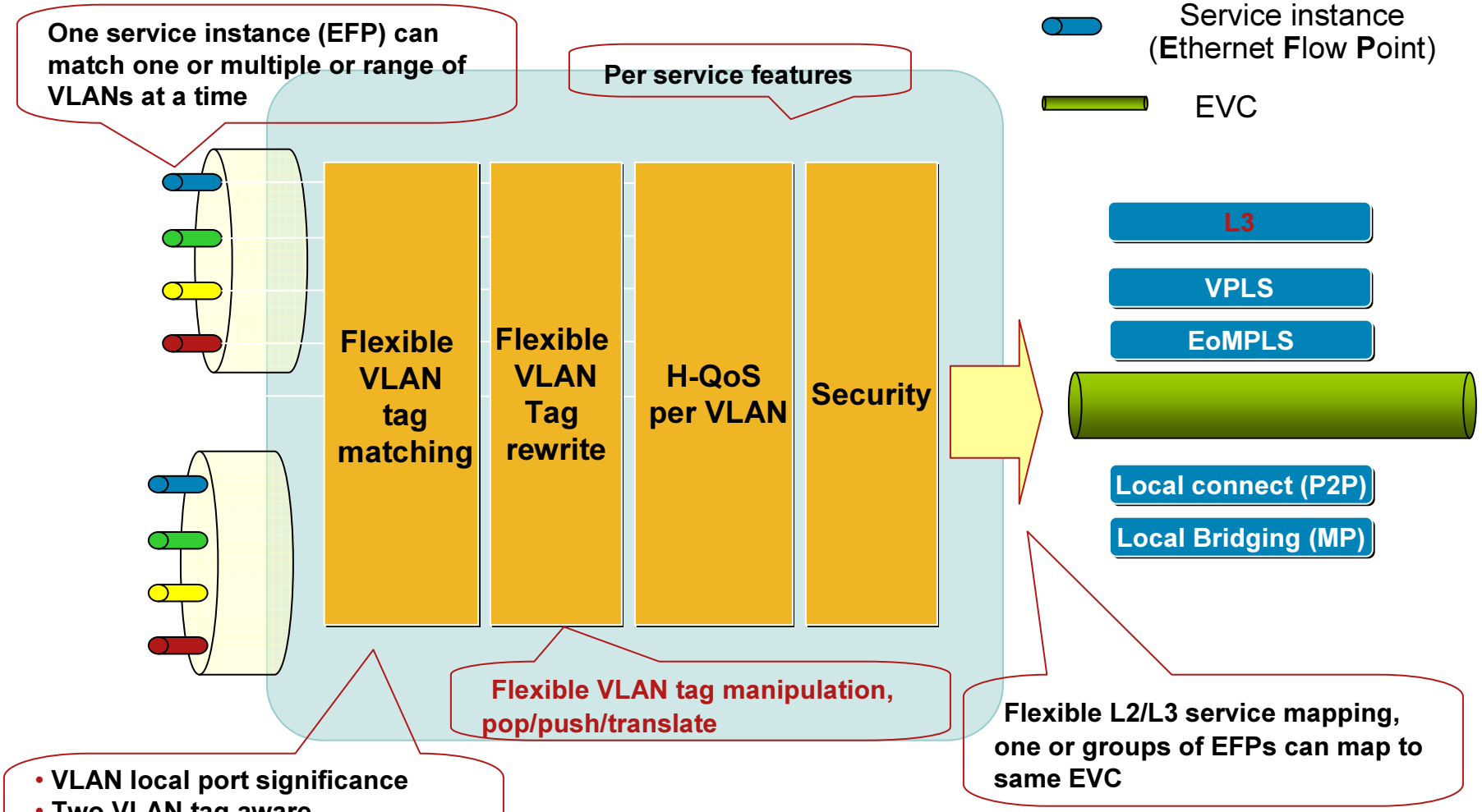
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Overlay architecture



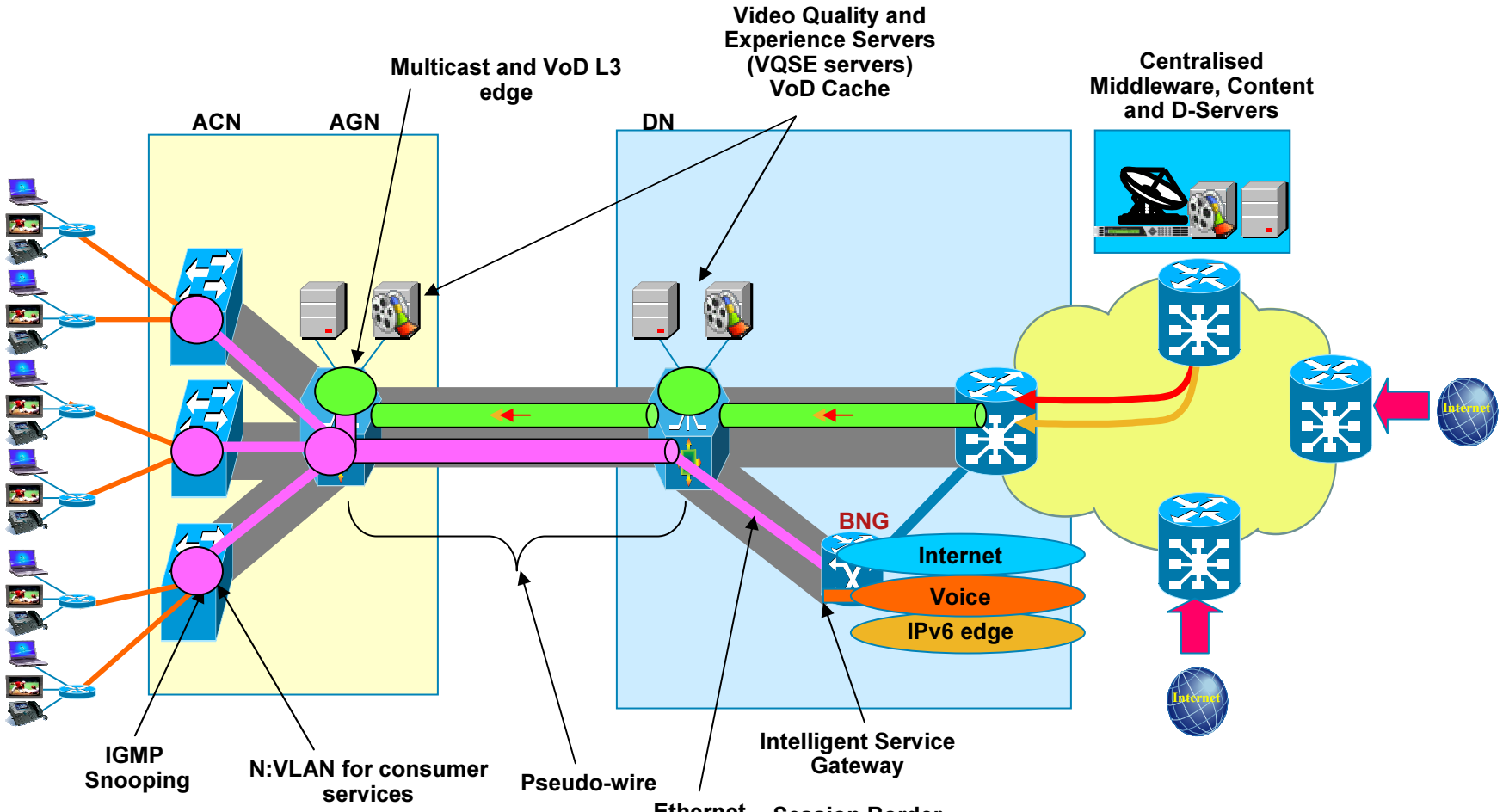
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Service Element
Ethernet Virtual Connection



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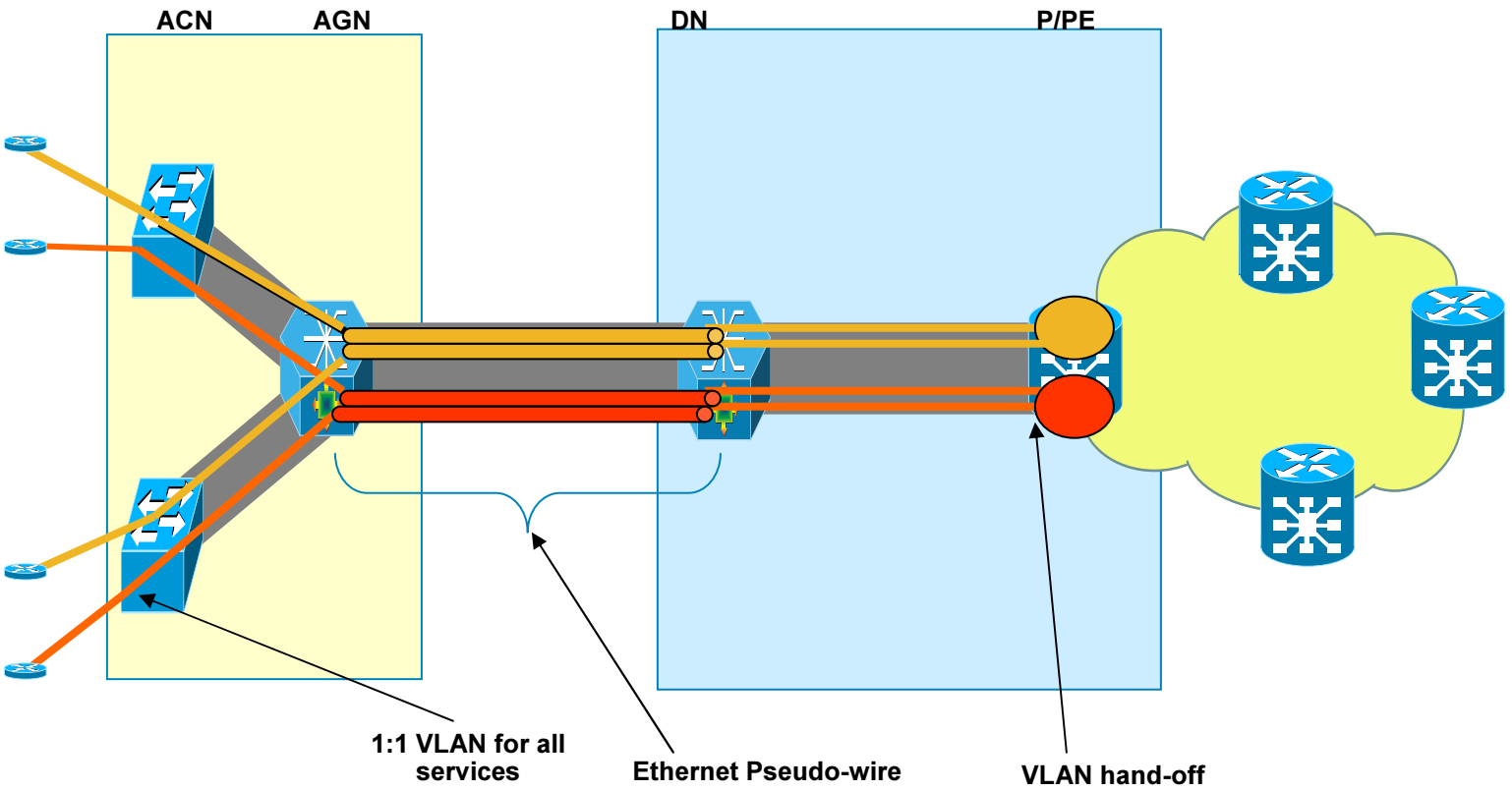
Consumer Services



- Inline video injection (M/C and unicast)
- Out of line HSI (via Ethernet handoff to BNG)
- Unicast and multicast
  - Call admission Control (On Path or Off Path)
- PIM-SSM with PIM fast convergence for TV delivery
- Point to Point Services
  - End to End pseudo-wires

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Business Services L3 VPNs



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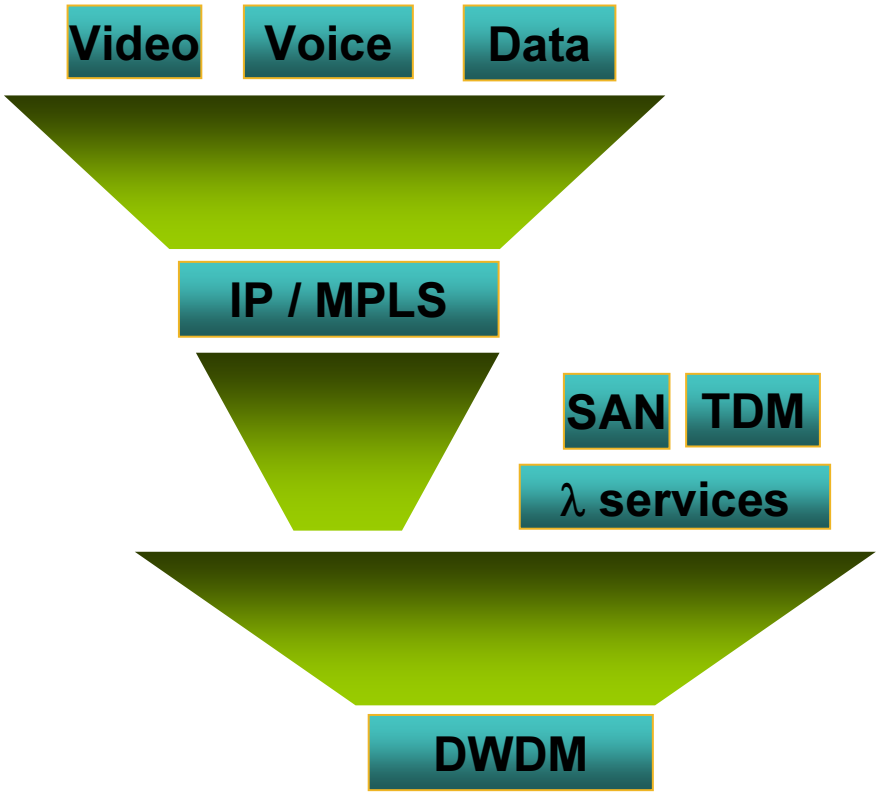
Optical and Control Plane Development



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The Cisco Strategy:
IP-over-WDM Converged NGN Transport

- Packet layer convergence to IP/MPLS has delivered CapEx & OpEx savings in core networks
- WDM layer convergence has enabled CapEx & OpEx savings in multi-service metro networks
- Significant additional CapEx/OpEx saving can be delivered by converging the IP and optical layers!



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Today's Core Network Architecture

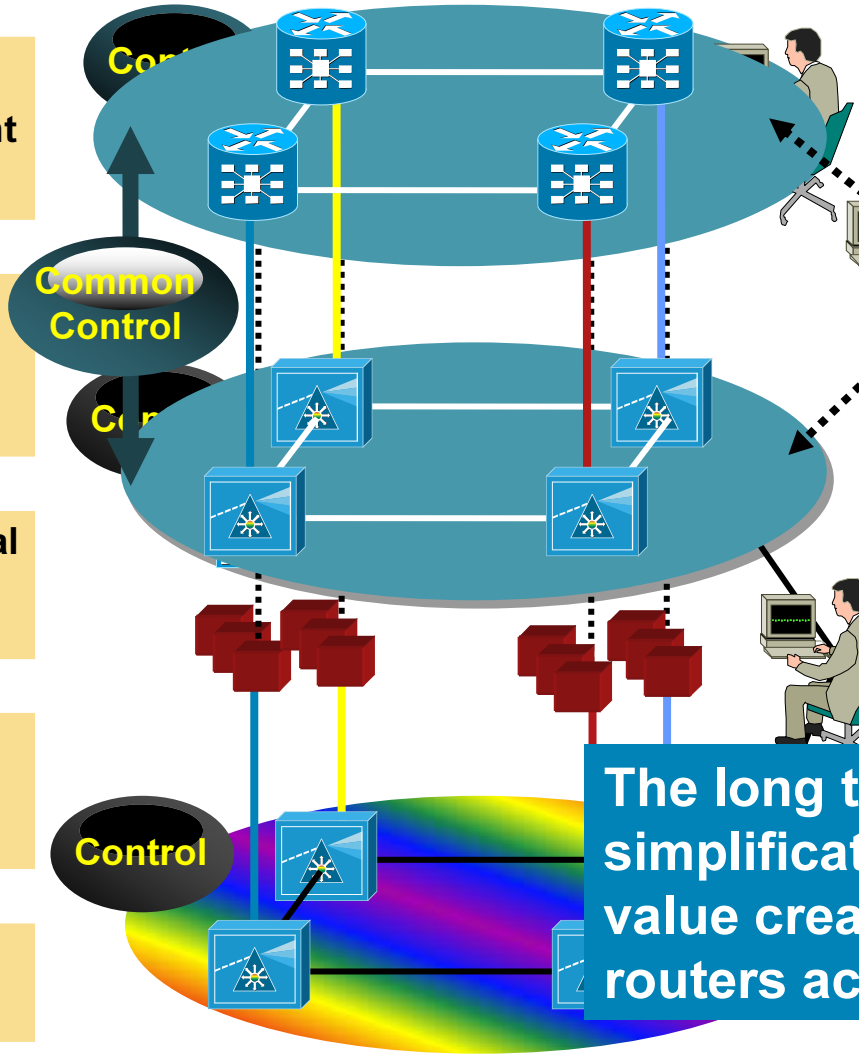
Distinct IP and DWDM Management Planes

Distinct IP and DWDM Control Planes

Expensive Electrical Cross Connects (OEO)

Multiple Transponders per Wavelength (OEO)

Truck Rolls for Reconfiguration



Routers
Aggregation of IP traffic to 10G
Fast restoration at Layer 3
Performance monitoring L2/L3

Cross Connects
Groom low speed circuits
Fast restoration at Layer 1
Performance monitoring L1

Transponders
Convert short reach to color

The long term vision is beyond simplification, it is about new value created by optical + routers acting as one system

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GMPLS Value Proposition

Reduce OpEx

Automated processes
- automatic connection setup and release
- topology auto-discovery
- rapid circuit provisioning Multi-vendor/domain layer integration

Reduce CapEx

Improved network utilization
- Meshed restoration
- Resilience against multiple failures
- Resource management
- Traffic Engineering

Add revenues

Flexible / fast service provisioning
- Bandwidth on demand
- Class of Services at transport layer

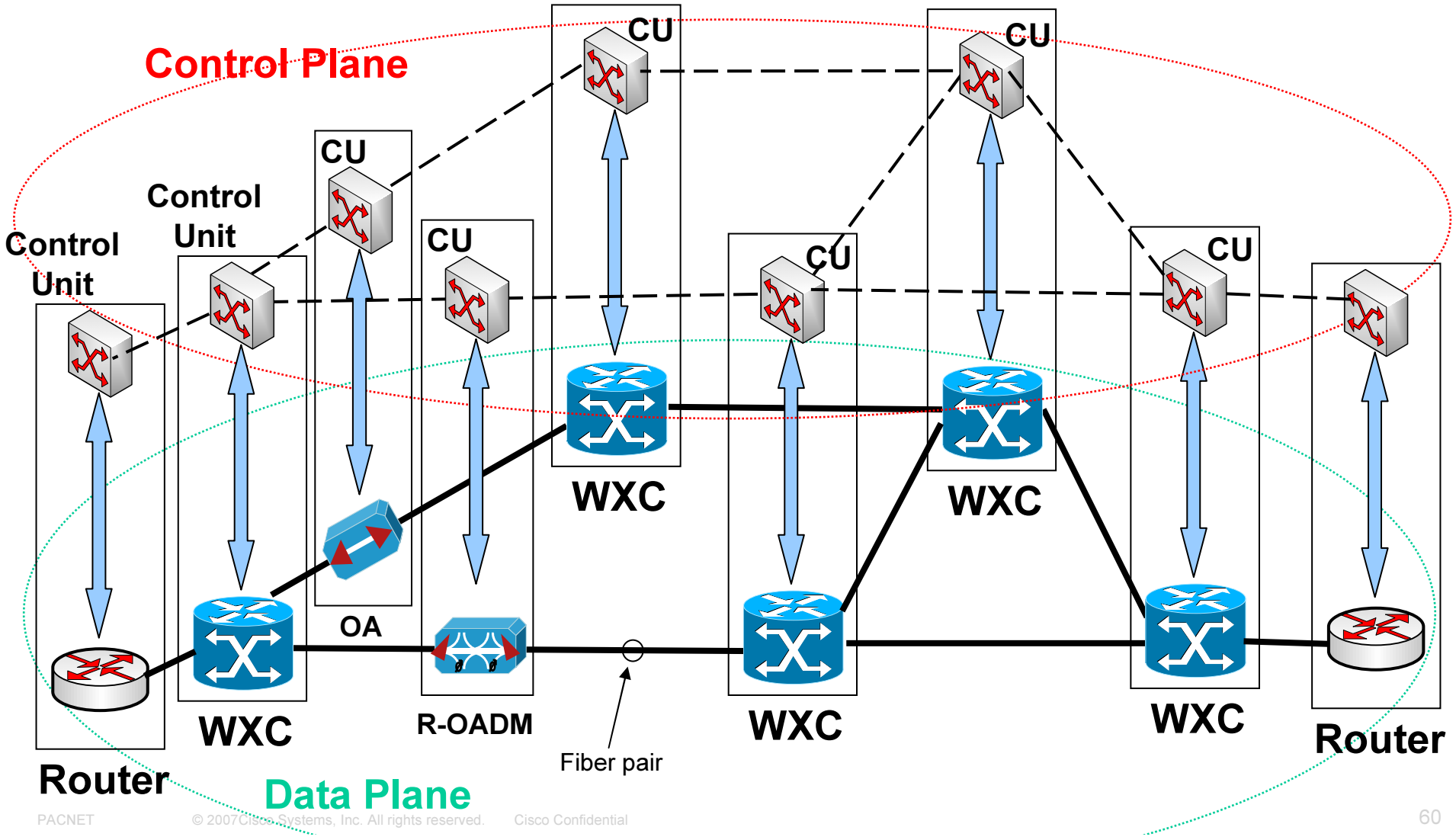
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Main outcomes

- Exploration of Optical Control Plane - OCP architectural options to deal with physical impairments in G-MPLS networks
- Validation of OCP approaches through a Java-based G-MPLS simulator
- Proof-of-concept of OCP through prototype implementation on DCL™ platform
- Prospective contribution to IETF on RSVP extensions for dealing with physical impairments

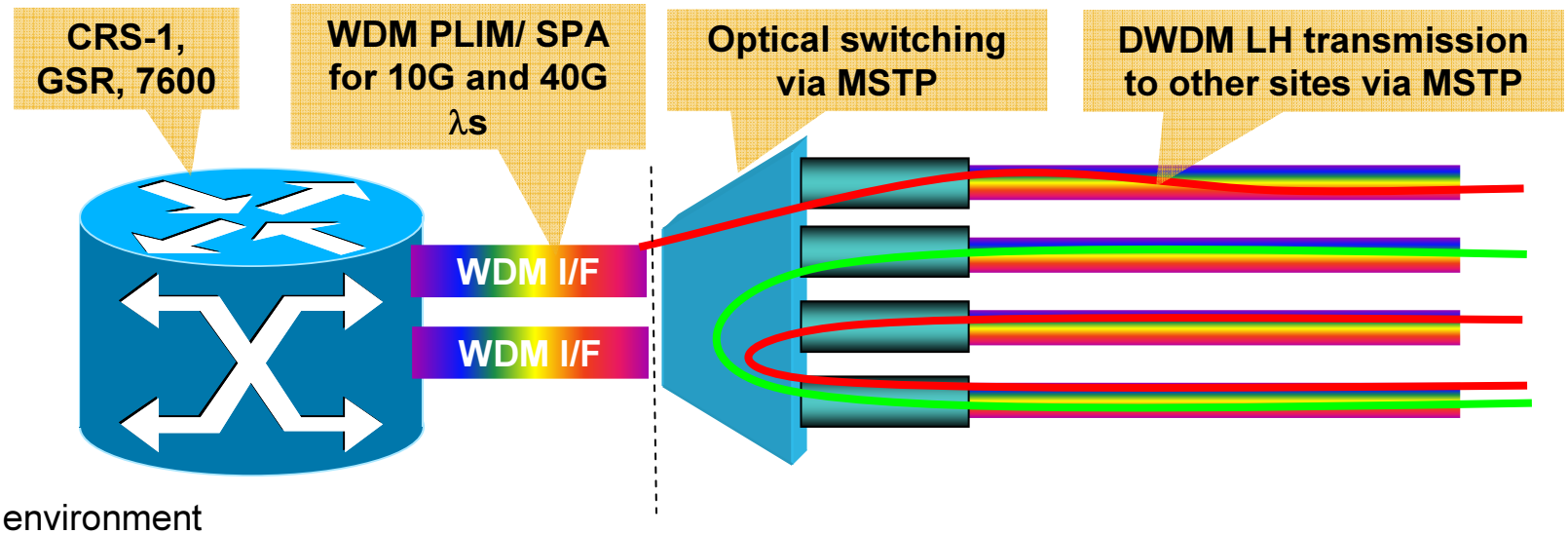
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Network scenario



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IPoDWDM



- Core environment
  - Intelligent tuneable optics with FEC integrated into the router
  - 10gbps GE and POS
  - 40gbps POS using duo-binary technology – onto 10gbps optical systems
  - Router to Router G.709 for enhanced OAM&P plus enhanced capabilities
  - Cisco or non Cisco DWDM layer
- Metro environment
  - Drive distances less, DWDM often not deployed
  - Less complex solutions required
  - Fixed optics, no FEC

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Goal

- The initial values of IPoDWDM are around CAPEX and OPEX savings due to the elimination of the transponder
- However, IP and DWDM integration offers synergies that provide more value than the 2 separately

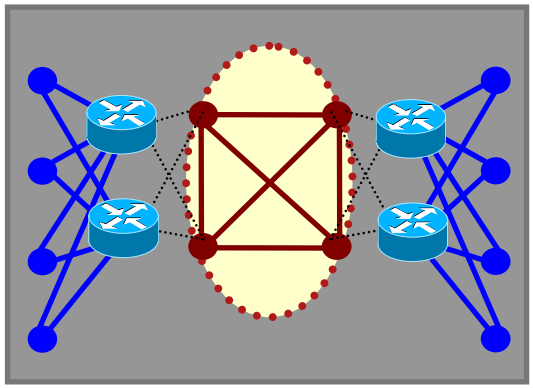
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IPoDWDM Synergy Features
Near term

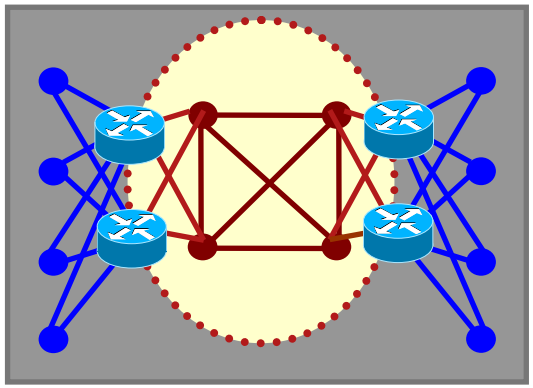
1. Segmented GMPLS infrastructure
2. Proactive protection based on pre-FEC errors

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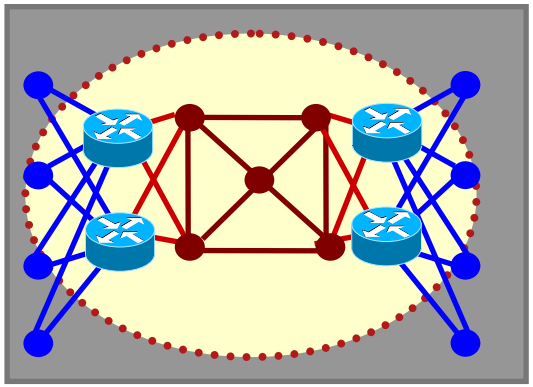
Segmentation Model of GMPLS (S-GMPLS)



Overlay Model
(OIF/ASON)



S-GMPLS Model*



Peer GMPLS Model
(IETF)

Hide Network Details from Other Depts. and Customers

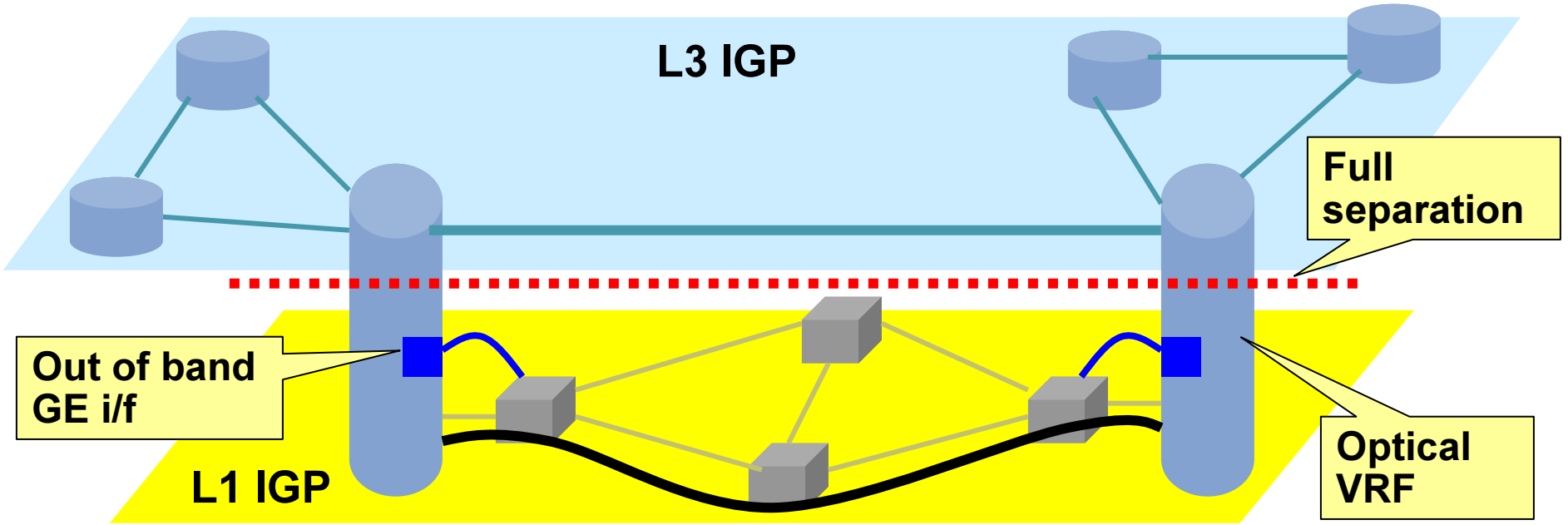
Only Segmentation Model Provides Both Benefits

IP Control Plane with intelligence and Lowest OpEx

Optical Control Domain

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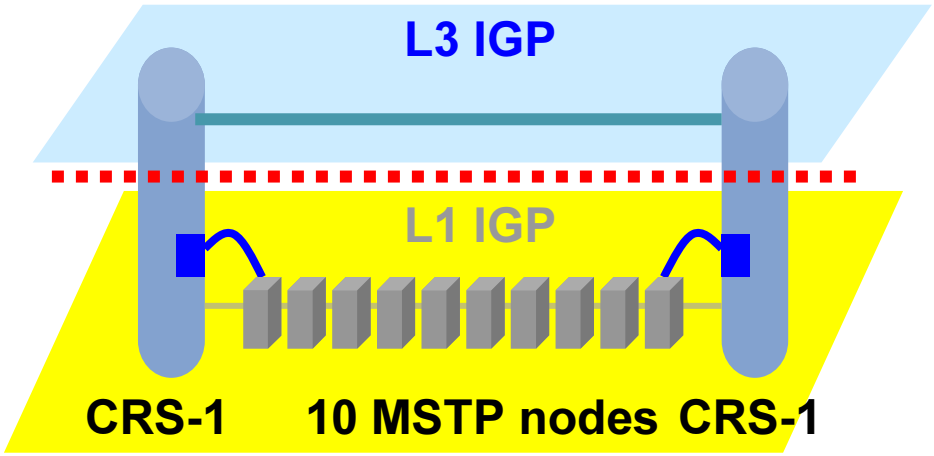
S-GMPLS Infrastructure
Implementation



1. MSTP nodes run OSPF and discover the optical layer topology
2. GE port on CRS connected to MSTP controller via out of band cable
3. OSPF enabled on CRS port and on MSTP port
4. CRS port added into a special “optical layer” virtual routing/ forwarding table (VRF)
5. CRS learns optical layer topology and stores it in the VRF
6. Users outside the router cannot see or access nodes inside the optical VRF

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S-GMPLS Infrastructure
Proof points



L1 topology visibility (OSPF database)

```
RP/0/RP0/CPU0:MiniRex#sh ospf vrf optical database

        OSPF Router with ID (10.10.10.10) (Process ID 1)

                Summary ASB Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
10.85.85.200    10.85.85.229    182         0x80000006 0x00027e
10.85.85.200    10.85.85.238    177         0x8000000b 0x002840
10.85.85.201    10.85.85.229    182         0x80000006 0x0049c7
10.85.85.201    10.85.85.238    177         0x8000000b 0x00cb0a
10.85.85.230    10.85.85.229    182         0x80000006 0x004f6d
10.85.85.230    10.85.85.238    177         0x8000000b 0x007f70
10.85.85.235    10.85.85.229    182         0x80000006 0x00b60b
10.85.85.235    10.85.85.238    177         0x8000000b 0x00b32d
10.85.85.236    10.85.85.229    182         0x80000006 0x00c163
10.85.85.236    10.85.85.238    177         0x8000000b 0x0095e5
10.85.85.237    10.85.85.229    182         0x80000006 0x00cbbc
10.85.85.237    10.85.85.238    177         0x8000000b 0x00779e
10.85.85.239    10.85.85.229    182         0x80000006 0x00658f
10.85.85.239    10.85.85.238    177         0x8000000b 0x00b4f0
10.85.85.244    10.85.85.229    182         0x80000006 0x00ae9b
10.85.85.244    10.85.85.238    177         0x8000000b 0x00083e
```

Control plane debugging (Ping)

```
RP/0/RP0/CPU0:MiniRex#ping vrf optical 10.85.85.200
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.85.85.200,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 1/1/4 ms
```

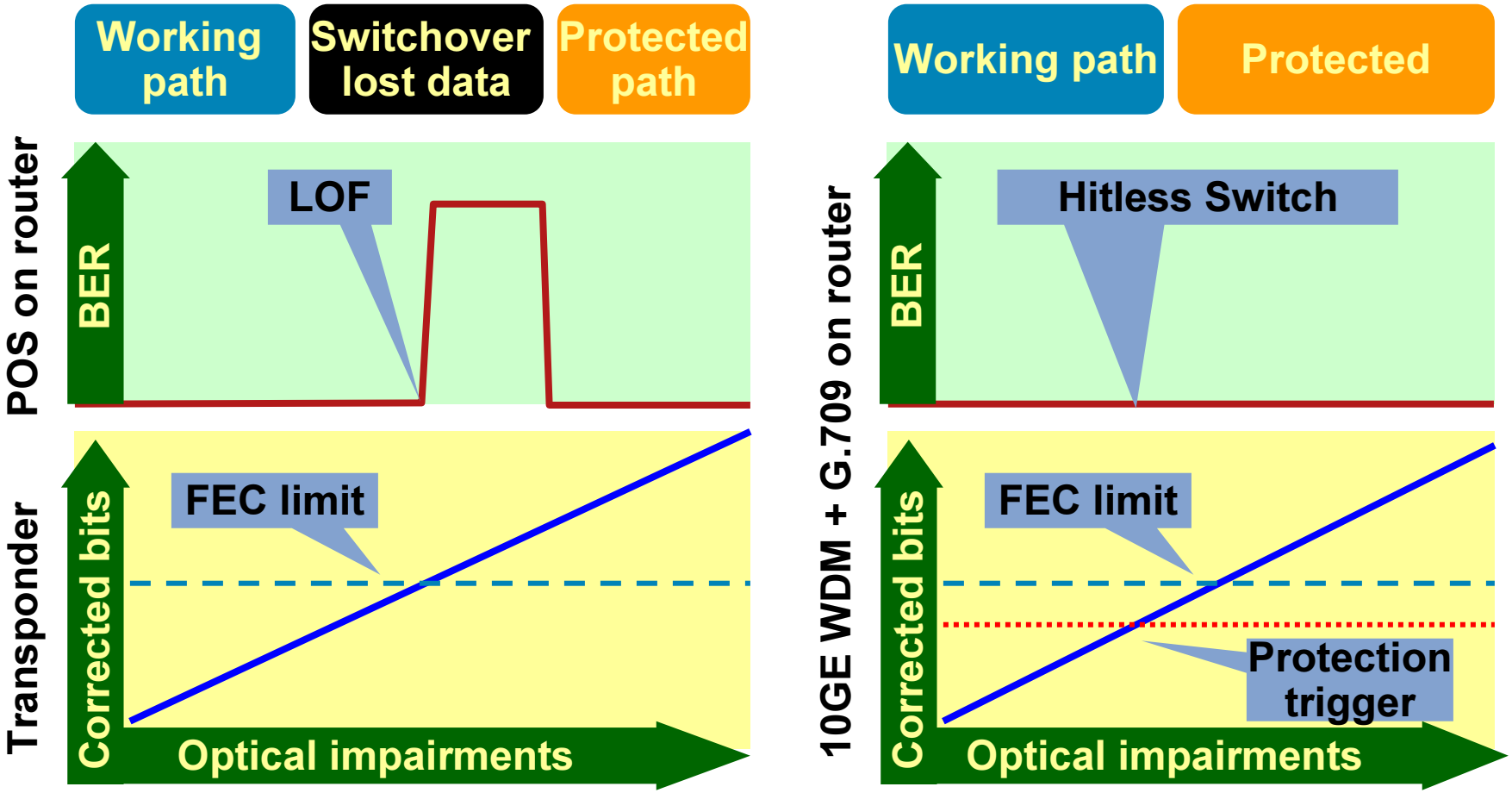

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S-GMPLS Infrastructure
Why is this important?

- We validated that the S-GMPLS model works and provides sufficient isolation of the optical layer from the L3 topology, yet allows routers to have sufficient visibility into the optical layer
- Now that a control plane is established between the two layers, the following capabilities are easy to introduce:
  1. Sharing of SRLGs
  2. Sharing of routing metrics between the layers

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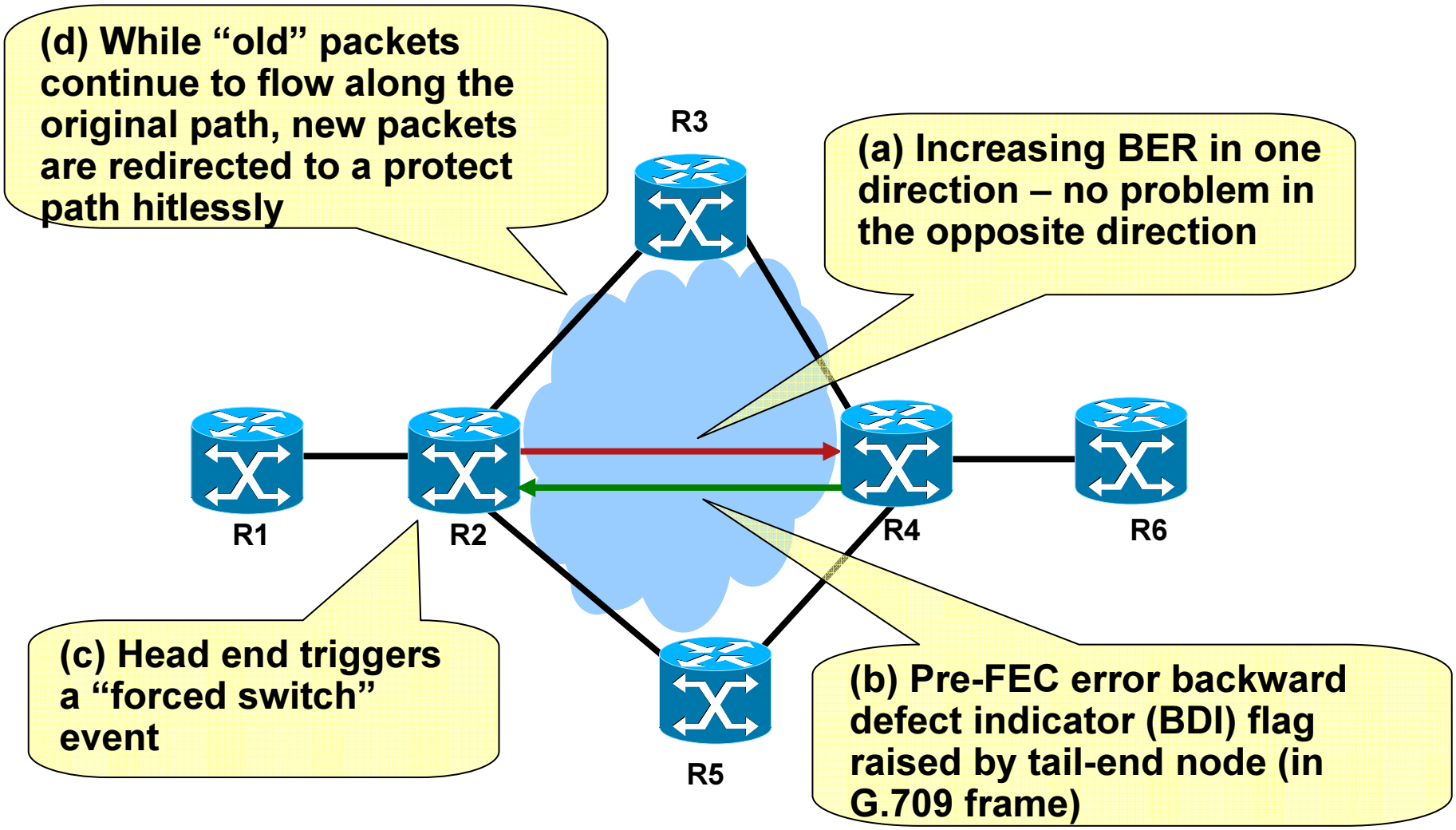
Proactive Protection based on pre-FEC errors
Introduction



Superior protection compared to transponder based networks

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Proactive Protection based on pre-FEC errors
How it works



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Proactive Protection based on pre-FEC errors
Covered Failure Modes

- Slow signal degradation due to:
  - Aging of fiber plant
  - Pinched patch cord
  - High PMD – assuming it is statistically compensated for but not in the worst case
- Even fast events may be covered:
  - Cable cuts by backhoe take 100's of ms – this has not been tested but cutting a patch cord w scissors seems to take 100ms
  - Human error: pulling a connector takes 10's ms – may or may not apply

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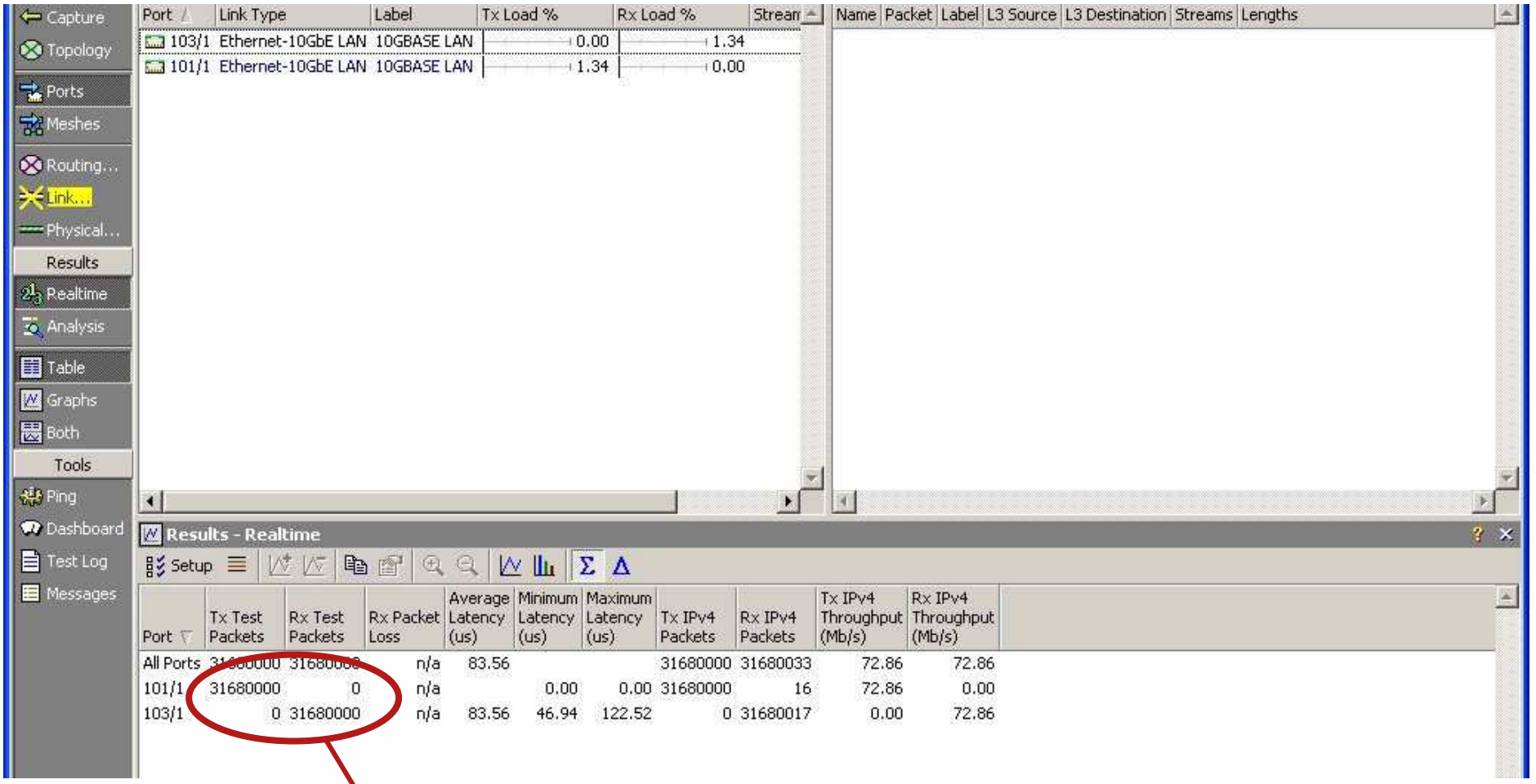
Proactive Protection based on pre-FEC errors
Which Protection Schemes are Supported and When?

- The scheme is implemented in a generic way that will work for whatever convergence mechanism is used for the interface. This could be:
  1. MPLS Fast Reroute
  2. IP Fast Reroute
  3. IGP/BGP Fast Convergence. The solution will be hitless if failure mode takes more than ~300 msec.s, due to the “make before break” approach in XR
- Timeframe:
  - XR 3.6: Implemented and tested for MPLS FRR using 10G WDM i/fs on CRS
  - XR 3.7: extended to 40G WDM
  - Testing for IP FRR and IGP convergence based on customer demand

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Proactive Protection based on pre-FEC errors

Proof point: screenshot from Agilent router tester



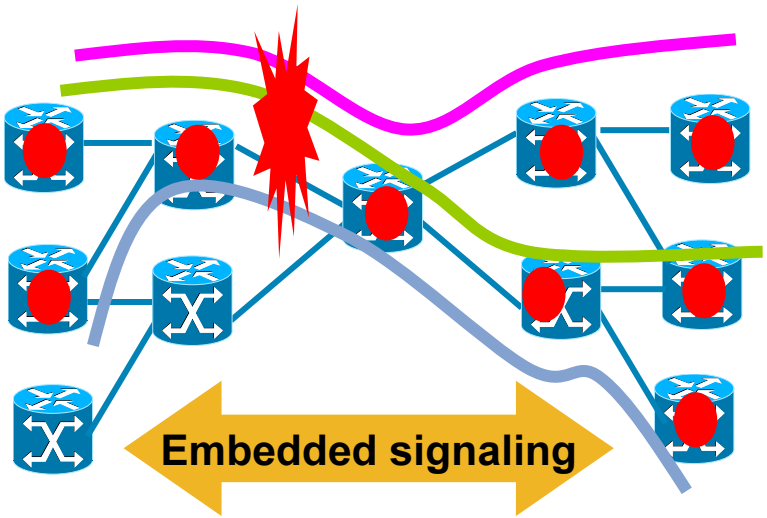
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IPoDWDM Synergy Features Mid-term

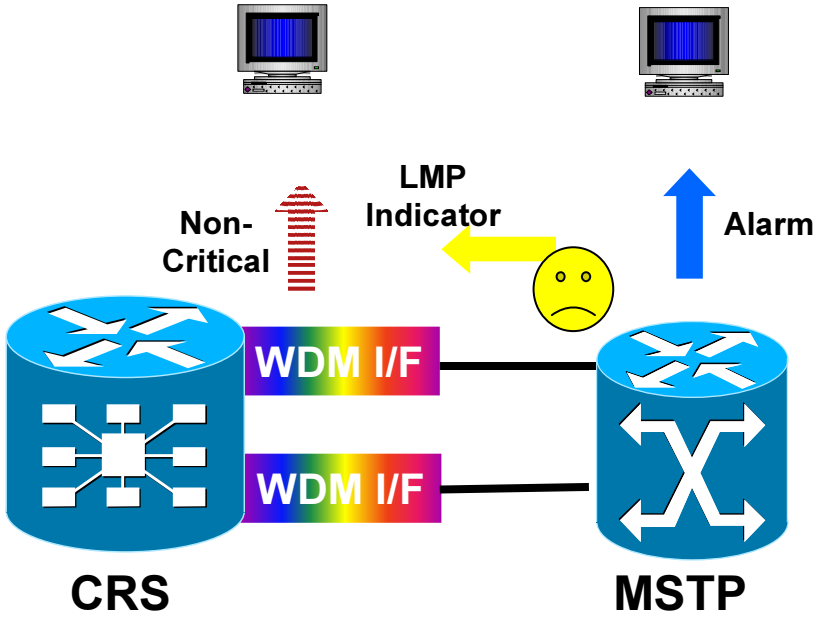
1. Extension of alarm correlation to router
2. Propagate SRLGs from optical layer to router for protection
3. Auto-discover router--DWDM connectivity
4. Auto-tune wavelength based on ROADM port
5. Optical 1+1 splitter protection, either integrated into the PLIM or triggered by the router I/F
6. Switch traffic away from span/node (planned maint activity)

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Extension of alarm correlation to router



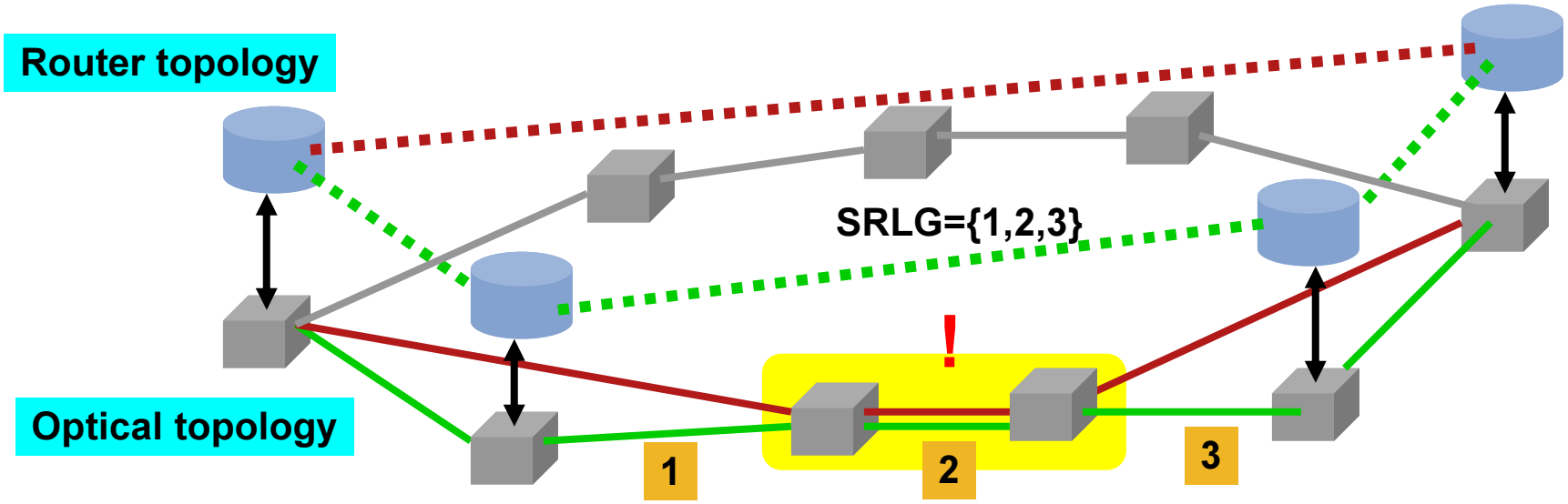
Step 1: correlation within DWDM layer



Step 2: correlation including router

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Propagate SRLGs From Optical Layer To Router For Diverse Routing Decision for MPLS/IP FRR

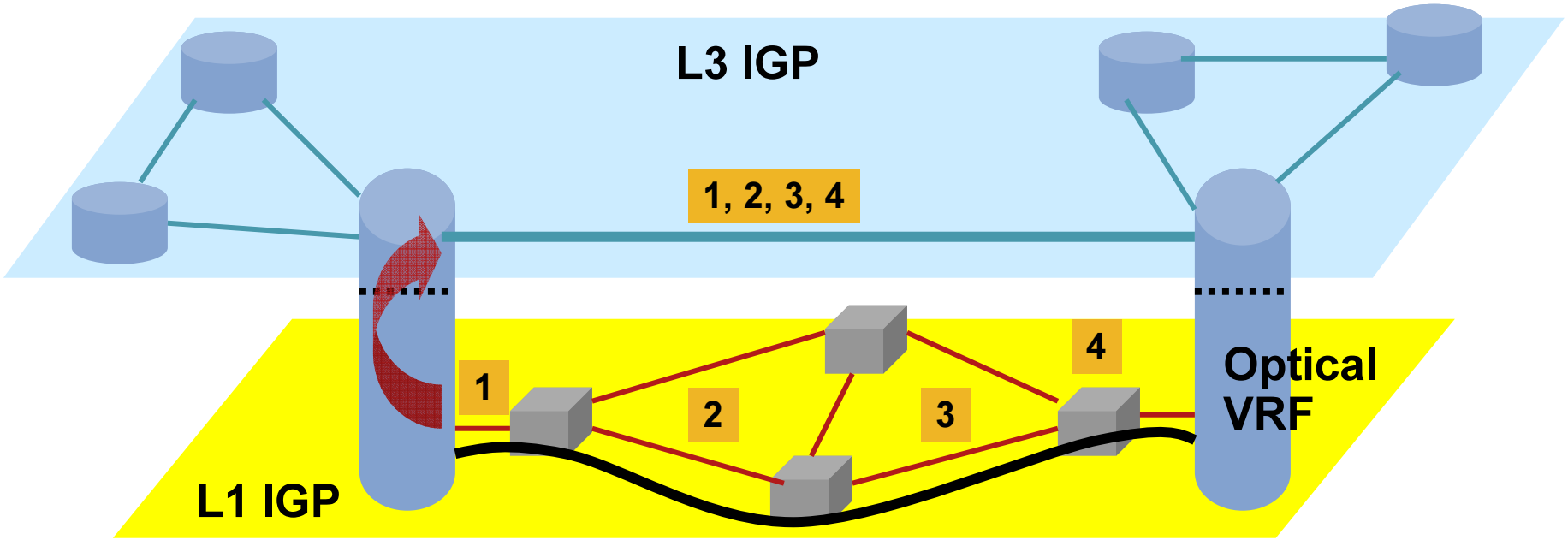


- MPLS can route based on SRLGs (Shared Risk Link Groups) but require manual setup – typically not available (not in IOS-XR 3.5 ?)
- Also needed for IP FRR (3.5?)
- W/out SRLGs: what appears diverse in L2/L3 may not be diverse in L1
- SRLGs can be mined from the OTN layer and fed to MPLS/IP layer
  - Need to add duct risk groups into DWDM nodes and CP

Enhance network resilience w/o error-prone manual work

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Propagate SRLGs From Optical Layer To Router Details

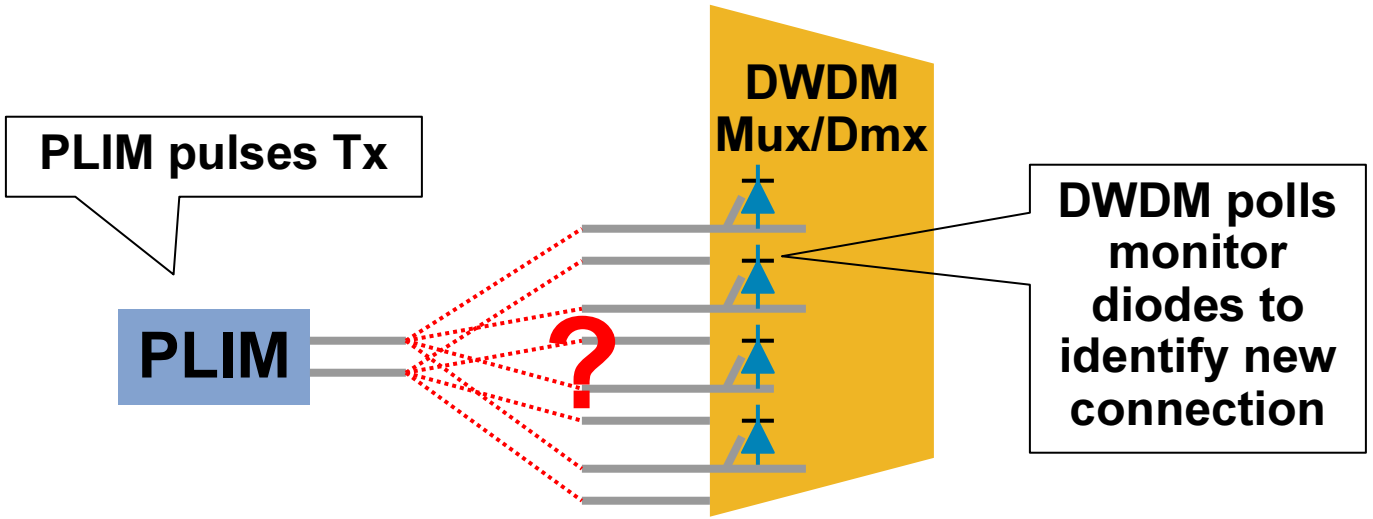


1. Provision L1 SRLGs into DWDM nodes
2. Discover them via L1 IGP
3. Discover route in optical network via RSVP “record path”
4. Collect all SRLGs along the path
5. Auto-provision into L3 IGP
6. Use for MPLS routing decisions for backup tunnel

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Auto-Discover Router To DWDM Connectivity

- Eliminates possibility for misconnections
- Simplifies installation (less training)
- Out of service slow OOK coding of ID detected by PDs on Mux, coordinated through LMP between the layers

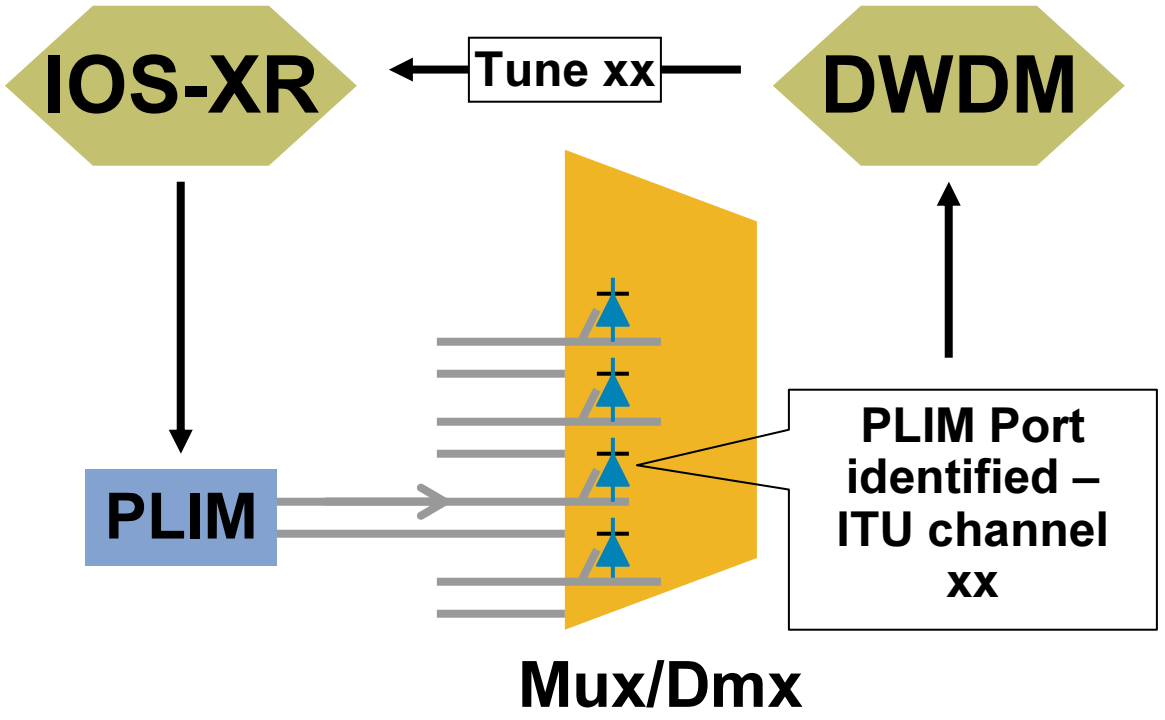


Simplified Operations

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Auto-tune Wavelength Based On ROADM Port

- Once PLIM to Mux Port connectivity established, DWDM can command PLIM to tune to the unique wavelength of that port
- Simplifies channel turn up – one less opportunity for error

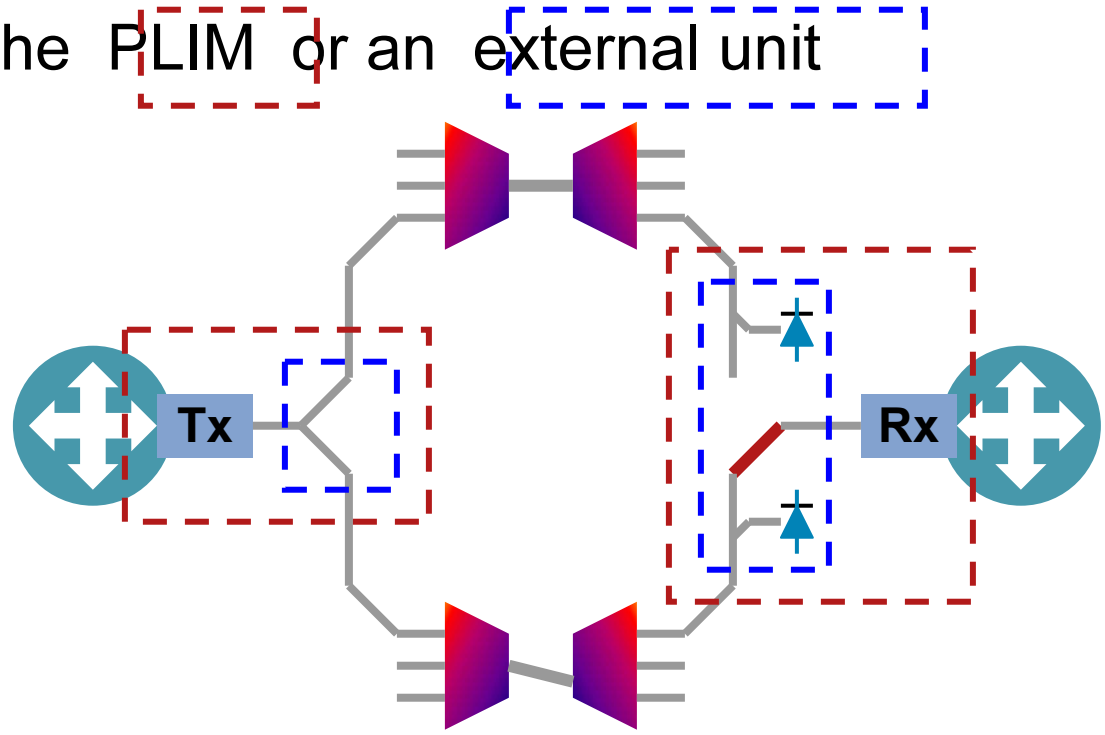


Prevents misconfiguration, simplifies turn-up

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Optical 1+1 Splitter Protection

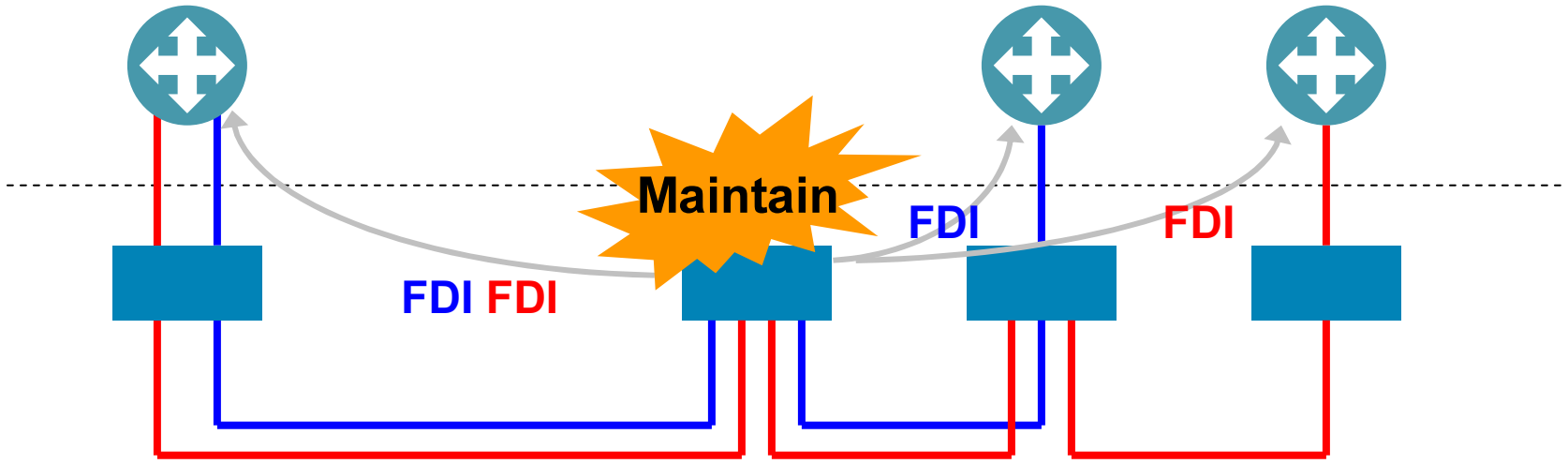
- Low-cost protection not including I/F failures or to deal w uncoordinated maintenance (e.g., VZ)
- Can be integrated on the PLIM or an external unit controlled by the PLIM
- Since controlled by PLIM, can switch on LOF or high BER – not just LOL



Allows low-cost optical layer maintenance in many places simultaneously without compromising protection

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Switch Traffic Away From Span/ROADM (Planned Maintenance Activity)



1. User informs network of a span/NE that needs to be maintained
2. FDI sent to end points of lightpaths traversing optical NE requiring maintenance
3. FDI signal can be propagated to router, invoking hitless FRR for all tunnels using these lightpaths

Eliminates need to coordinate maintenance activity between L1 & L3. Reduces incentive to duplicate protection in L1 and L3

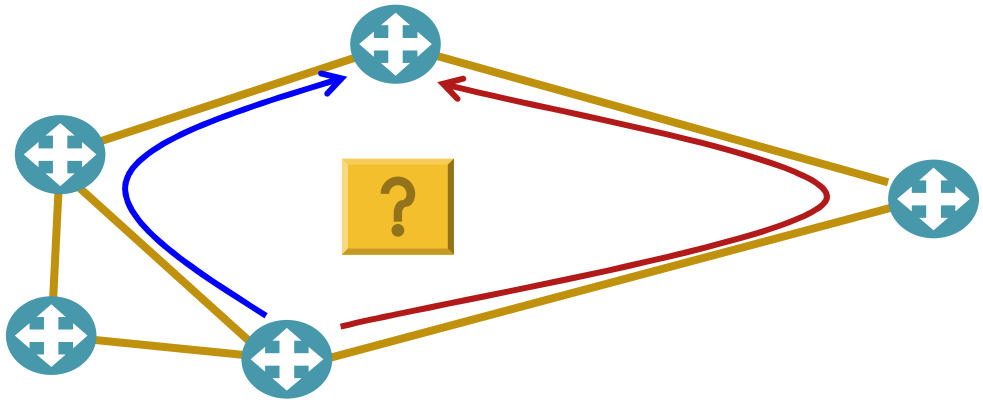
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IPoDWDM Synergy Features
Long-term

1. Influence IP routing decisions based on optical parameters
   - Distance/delay, bandwidth, other metrics?
2. IP user initiated connections
   - User sets up end to end connection via CLI on router – no need to involved transport group
3. Automatic lightpath setup based on router loads (auto bandwidth)
4. Photonic protection/restoration triggered by router

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Automatic setting of router metrics based on optical layer attributes

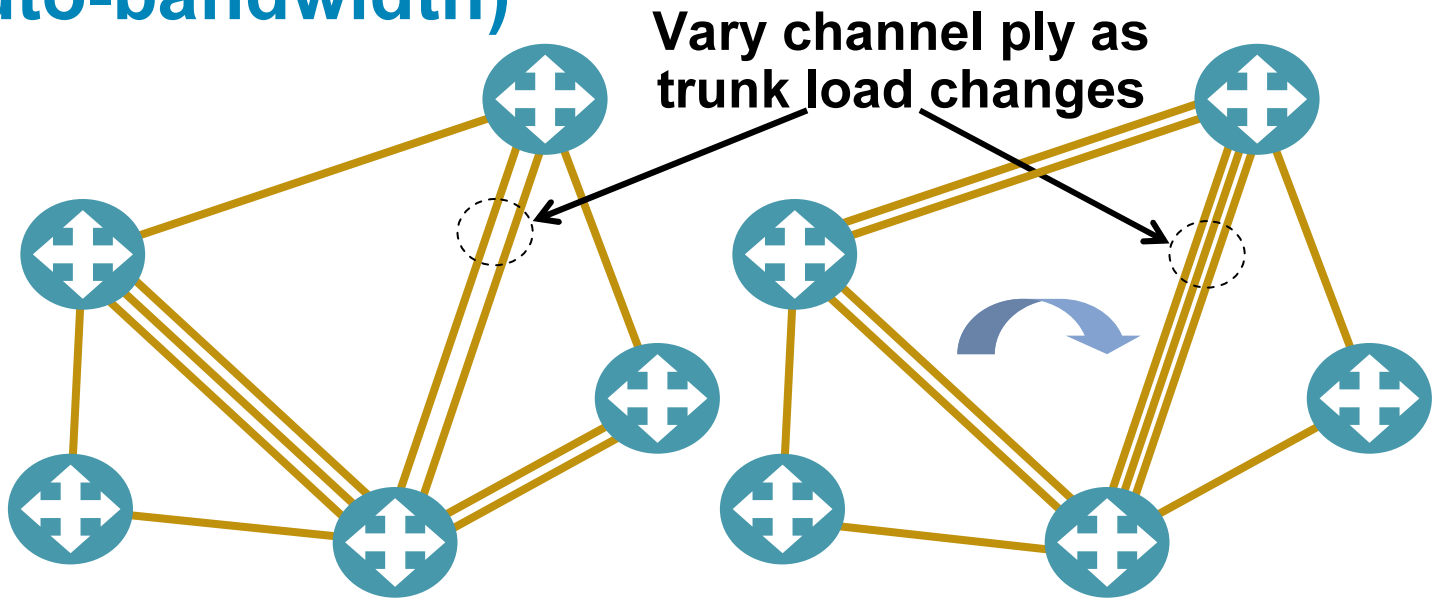


- The distance is not automatically factored into routing metrics and there is no way for the router to directly find it out
- The DWDM layer has the required information as well as the cost of a route in terms of the number of line amps and ROADMs traversed
- Factoring these into the routing will reduce congestion on expensive routes and avoid having to upgrade them
- While this can be done manually, it requires manual coordination between transport and IP groups and is subject to change if the routing of a lightpath changes due to a maintenance activity
- Solution: Use RSVP “record path” to discover the metric - see SRLG slides

Ensure packets go over shortest physical route
W/o manual work for tweaking routing metrics

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Automatic lightpath setup based on router loads (auto-bandwidth)



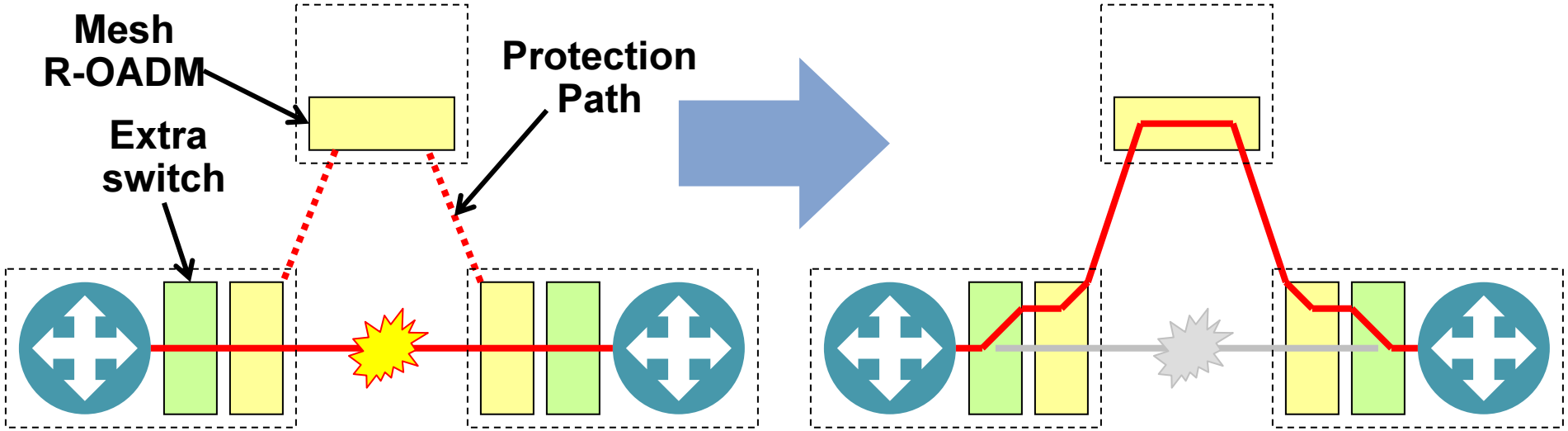
- Any interface can serve any trunk, so less interfaces needed
- Original topology retained to avoid convergence issues – channel add/remove within existing link bundles
- No need to manually measure link loads – the router figures this out
- Need to maintain I/F pool at each router, only need to replenish pools upon alarm

Can run trunks at higher utilization;
Reduce manual work during network growth

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Photonic protection triggered by router

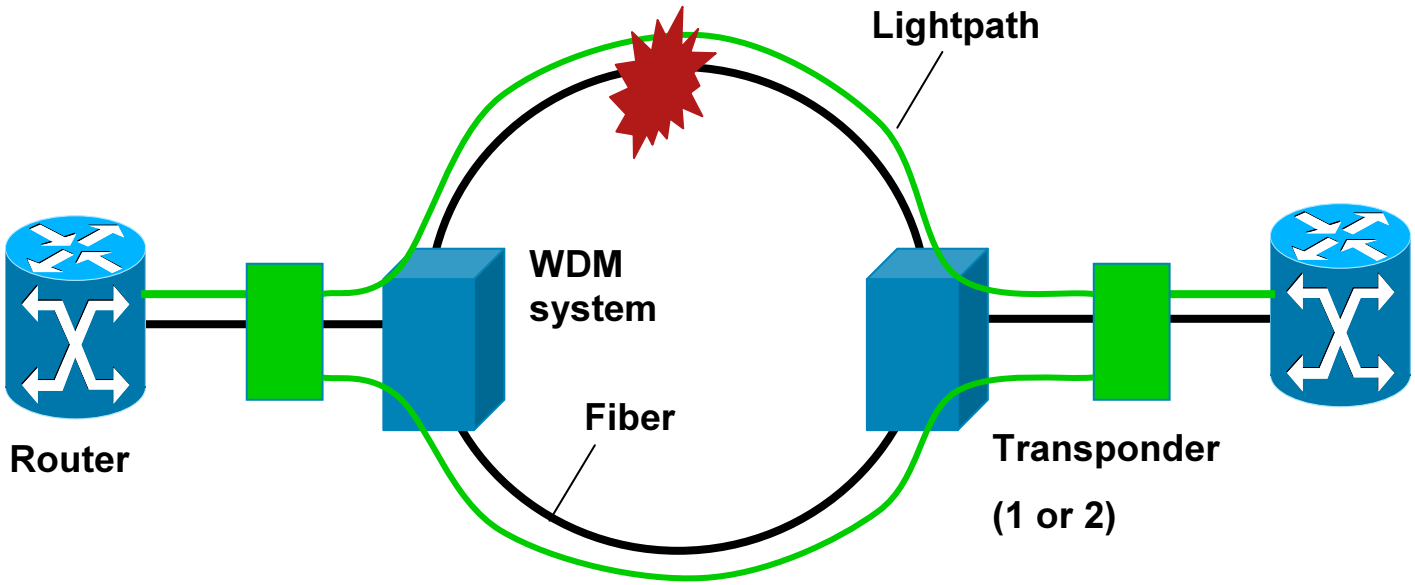
- Protects channel path (interfaces protected by 1:N)
- Switch channel to alternate fiber route – router interfaces unaffected
- Distributed coordination required, as well as fast switching in MSTP
- Only savings vs. 1+1 splitter protection: DWDM bandwidth



Eliminates CAPEX: ~30% less interfaces viz. IP/MPLS protection

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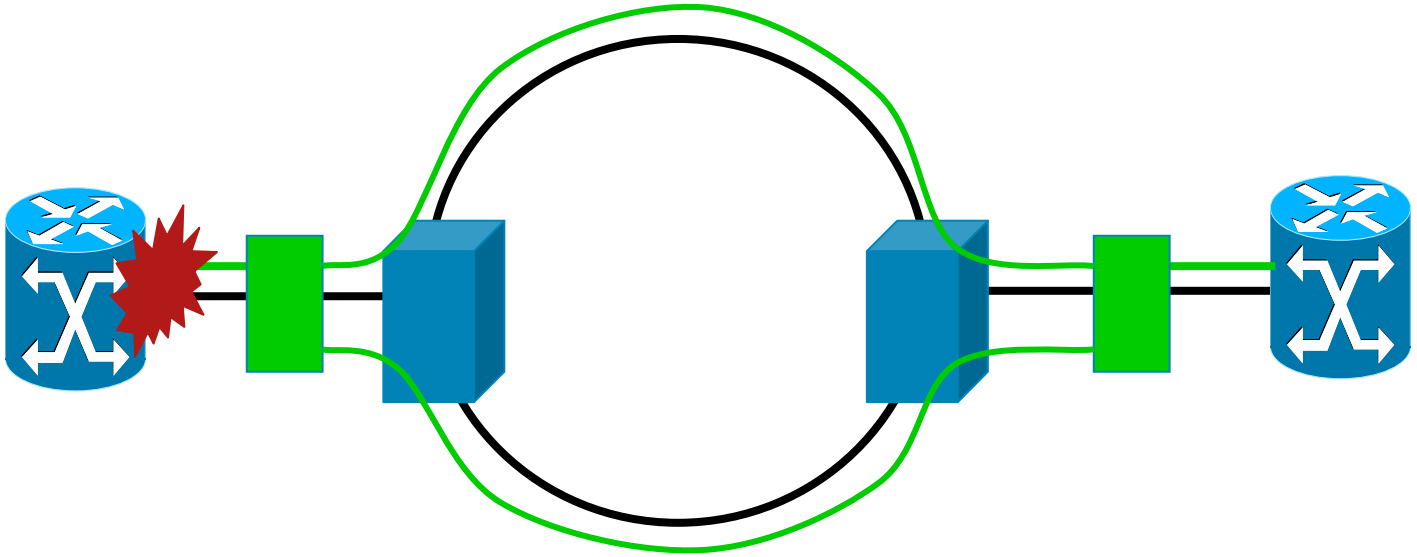
Optical protection protects well against optical failures



- This could be 1+1 optical protection (1 transponder at each end),
- Or 1+1 Y-cable protection (2 transponders at each end),
- Or even dynamic protection assuming the optical layer is capable of it (it's not, but no need to go into it here)
- Let's assume the optical layer could even achieve hitless protection using pre-FEC indications (it can't, but again let's not complicate this discussion)

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But optical Protection is useless if the router interface fails

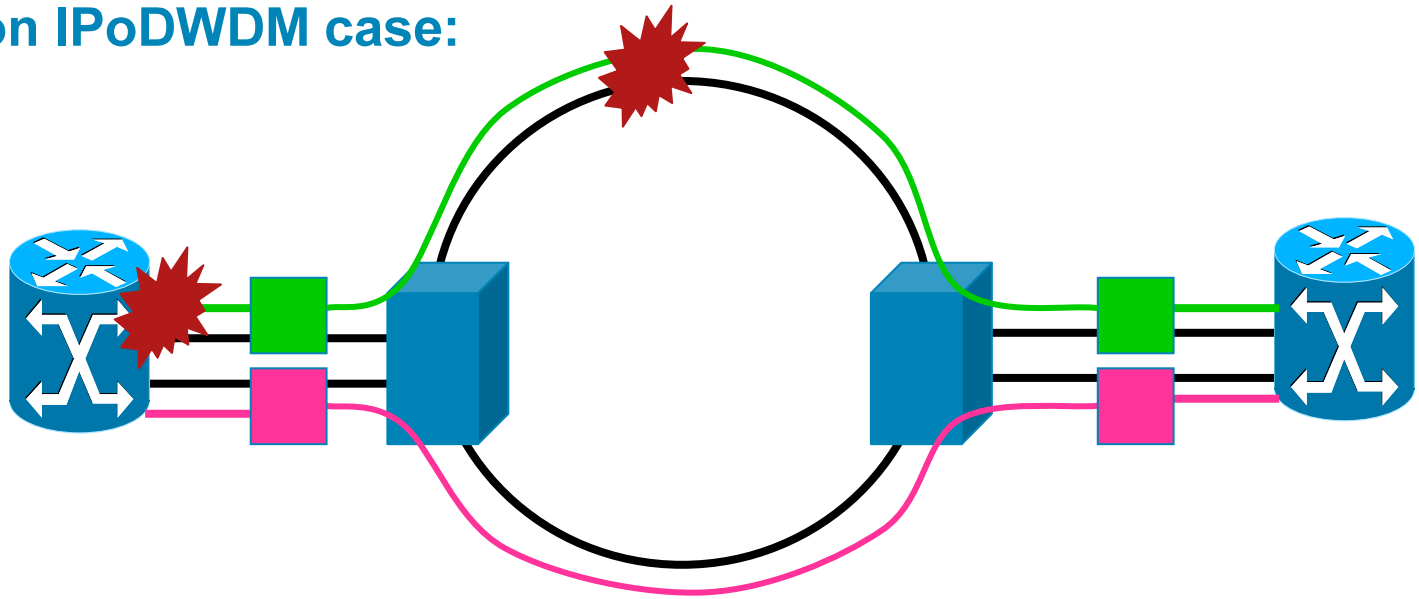


- Or the whole router fails,
- Or the router SW fails,
- Or the cable between the router and the transport layer

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Therefore the only way to protection IP traffic is via L3 protection

Non IPoDWDM case:

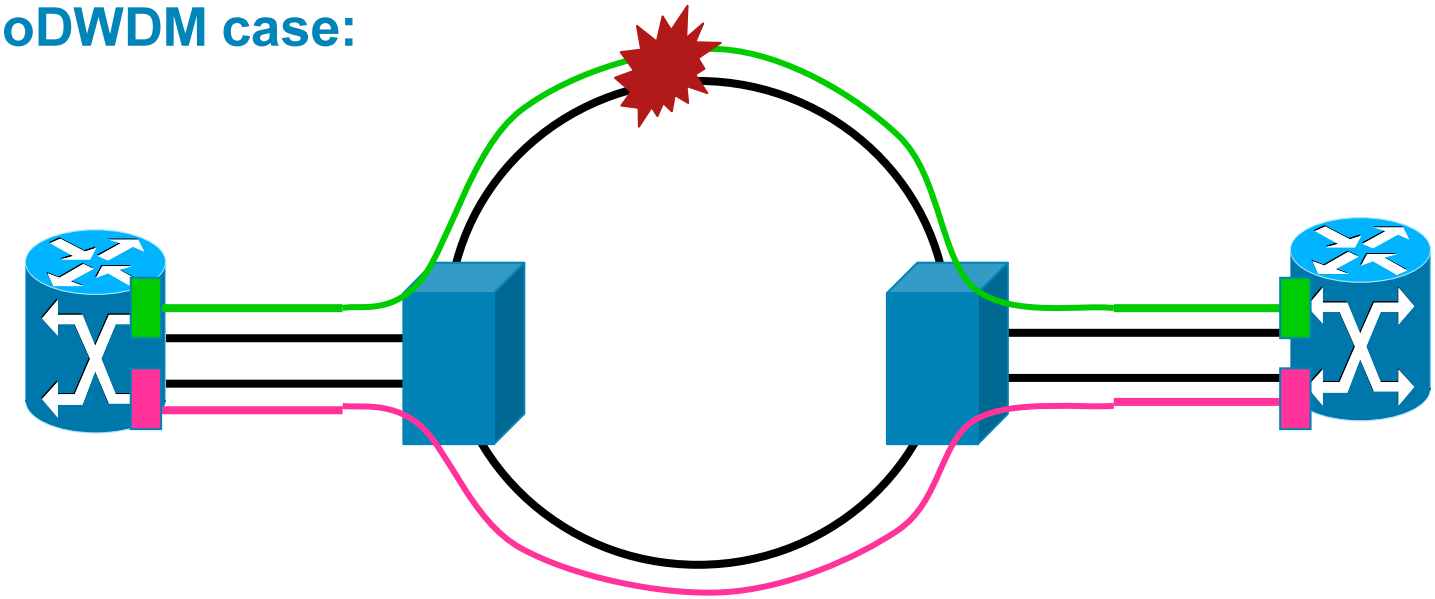


- This covers both L3 and L0 faults
- Quite fast (50-200 ms)
- And it is also more bandwidth efficient since routers can more easily share protection bandwidth, or drop packets that don't need protection

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Therefore the only way to protection IP traffic is via L3 protection

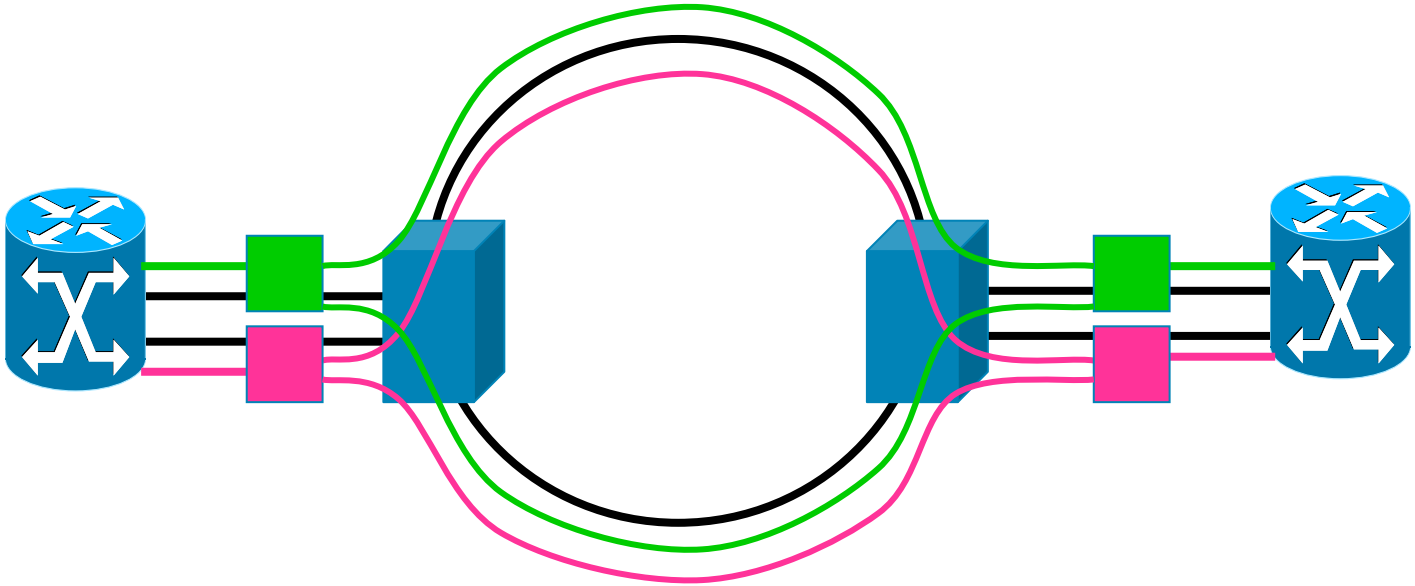
IPoDWDM case:



- Much faster w/ IPoDWDM:
  - Hitless for MPLS FRR and IP FRR under many realistic optical layer failures
  - Almost hitless even for FC (approx. 10ms!)
- And with lower CAPEX/OPEX (separate discussion)

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But what if you want hitless protection and do not want IPoDWDM?



- Then you need both L0 and L3 protection
- Which doubles the cost of the optical layer
- And consumed more than 2x bandwidth (protect is longer)
- And runs into multi-layer protection coordination issues
- 75% of the bits do not generate revenue…

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OAM Development



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MPLS Embedded Management

- MPLS management capabilities integrated into IOS
- IETF standards based + Cisco-specific value adds
- MPLS embedded management feature areas
  - MPLS SNMP MIBs (Draft, RFC-based + Cisco extensions)
  - MPLS OAM (Draft, RFC-based + Cisco-specific automation)
  - MPLS-aware Netflow
- MPLS SNMP MIBs
  - MPLS LSR, LDP, TE, FRR, and L3VPN MIB
  - VRF-aware MIB support
- MPLS OAM
  - LSP Ping, Trace, and Multipath (ECMP) Tree Trace
  - IP SLA – LSP Health Monitor

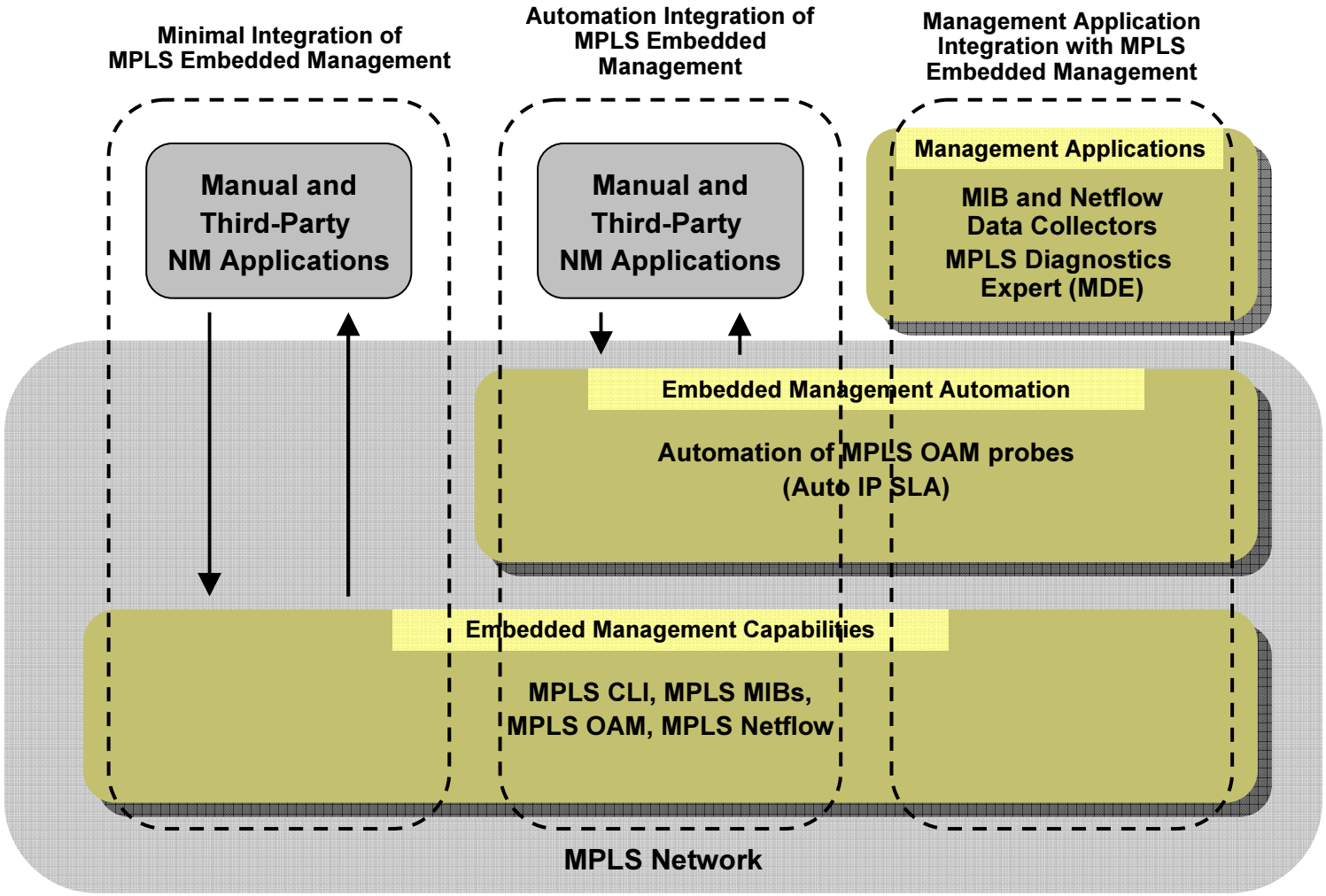
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MPLS MIBs and OAM

Deployment Scope	MIB Module/OAM Feature	Standard Compliance	Cisco Value Add
Base MPLS Core	MPLS-LSR-MIB	RFC3813	VRF-aware MIB capabilities
	MPLS-LDP-MIB	RFC3815	VRF-aware MIB capabilities
	MPLS-FTN-MIB	RFC3814	VRF-aware MIB capabilities
	MPLS LSP Ping/Trace for LDP IPv4 FECs	RFC4379	OAM automation via IP SLA
Advanced MPLS Core	MPLS-TE-MIB	RFC3812	-
	MPLS-FRR-MIB	IETF Draft	-
	MPLS LSP Ping/Trace for RSVP IPv4 FECs	RFC4379	-
	LSP Multipath (ECMP) Tree Trace	RFC4379	OAM automation via IP SLA

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MPLS Management Solution



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Cisco MPLS Diagnostics Expert (MDE)



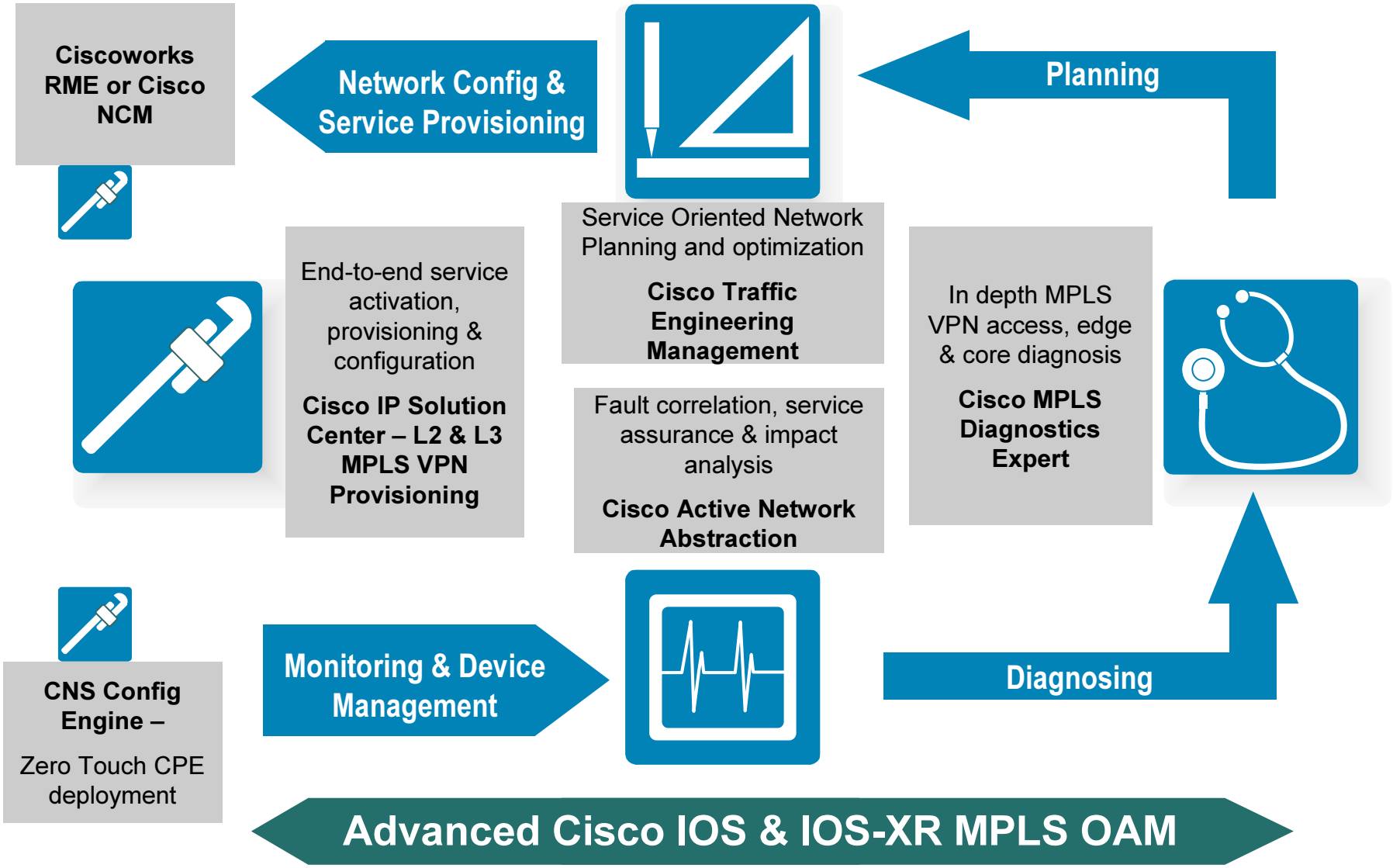
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Agenda: Cisco MPLS Diagnostics Expert

- Cisco IP NGN Network Management
- MPLS VPN Operations - Business and Technical Problems
- The Product - Cisco MPLS Diagnostics Expert
- What Is Troubleshooting & Diagnostics?
- The “Traditional” CLI / EMS approach
- The Cisco MPLS Diagnostics Expert – Demo
- Typical Competitive Approaches

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The Bigger Picture: Cisco IP NGN Management



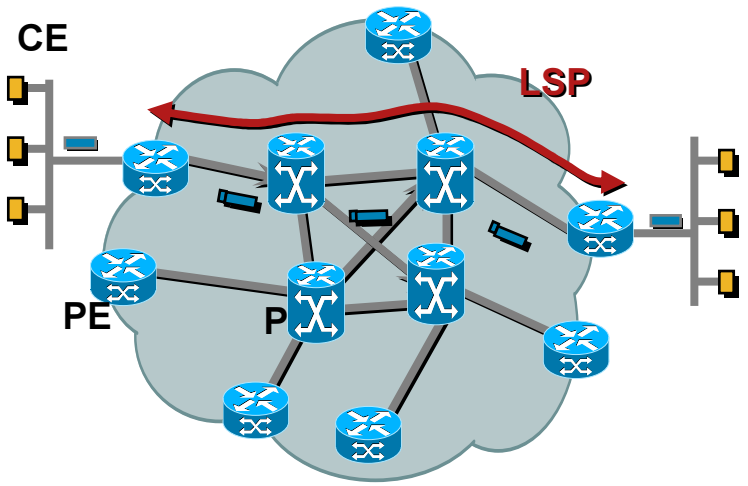
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The Business Problem: Overview

- Outage Detection
- Complexity of IP/MPLS networks e.g. *LSP Black holes* in MPLS core
- Network Configuration changes … typically cause 50% of network outages
- Access circuit issues –typically 90% of customer support calls
- Customer network problems – all too often "MPLS VPN outage" root cause
- Need for MPLS skills on the Helpdesk
- NOC Organizational "hand-overs"

  Front-line helpdesk -> Level 2 support –> Level 3 support/Design Engineering)

Real life customer problem … delete VPNV4 address family on route reflector … took hours to find & diagnose

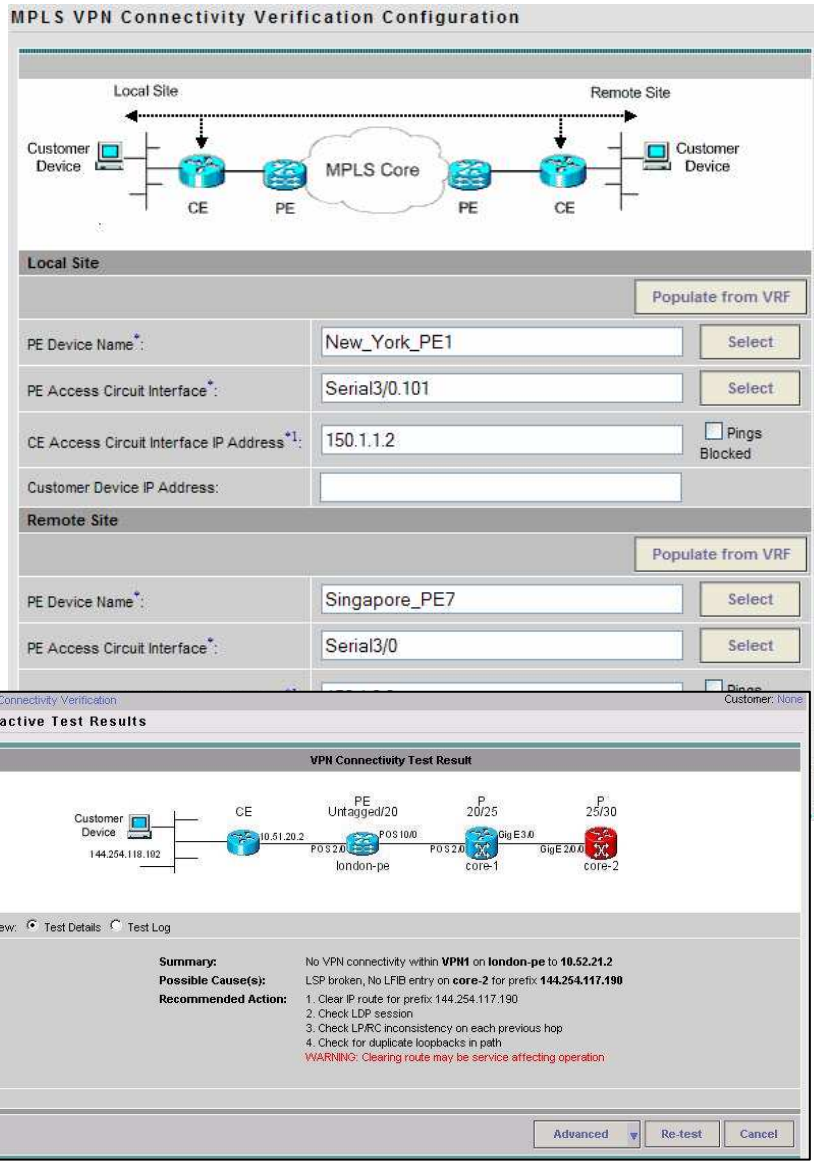


Real life customer problem … Inconsistency between MPLS forwarding table on CPU & Line Card … few hours outage

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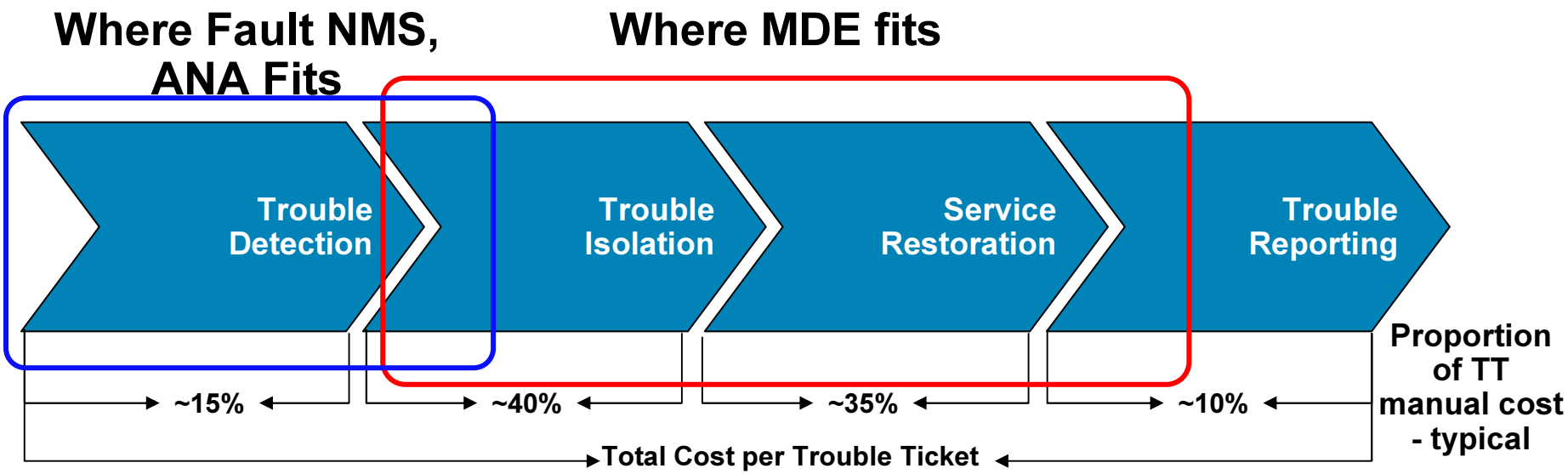
Cisco MPLS Diagnostics Expert (MDE) Product Overview

- New, problem centric, approach to NMS - Automated Troubleshooting and Diagnosis of MPLS VPN problems - 80/20 Rule
- Diagnoses problems in MPLS VPN core, edge and access … to end customer network
- Reduces MPLS outages from hours to minutes using MPLS VPN Failure Knowledge Base developed by NMTG – with TAC & ITD support
- Diagnoses complex MPLS core LSP "blackholes" via Cisco IOS MPLS OAM support – LSP ping & traceroute in mixed IOS/IOS-XR networks
- Usable by, and Empowers, the NOC Helpdesk
- Accelerates NOC Organizational Handover via detailed Test Log- all steps logged
- Unique Cisco capability, complementary to existing, traditional fault NMSs
- Deployable in customers who don't use ISC as well as those who do
- Comprehensive patent filed on MDE algorithms
- 60+ customers since Jan CY2006



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Assurance Process Breakdown
Where MDE Fits







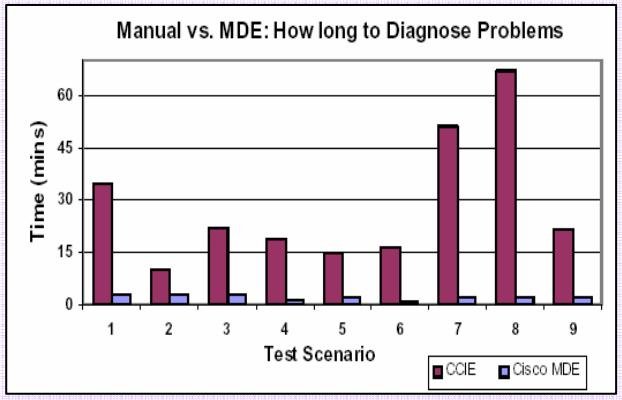


Fault NMS, ANA ← → MDE Focus Areas

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Where MDE Fits … The MDE Value Proposition

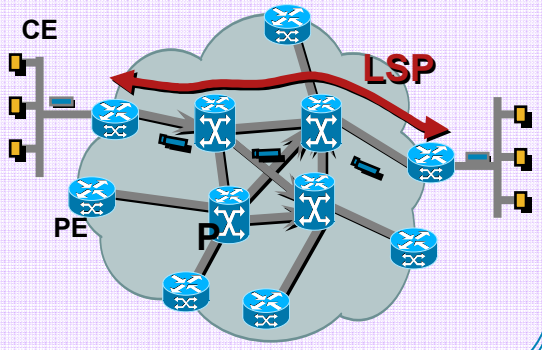
Assurance



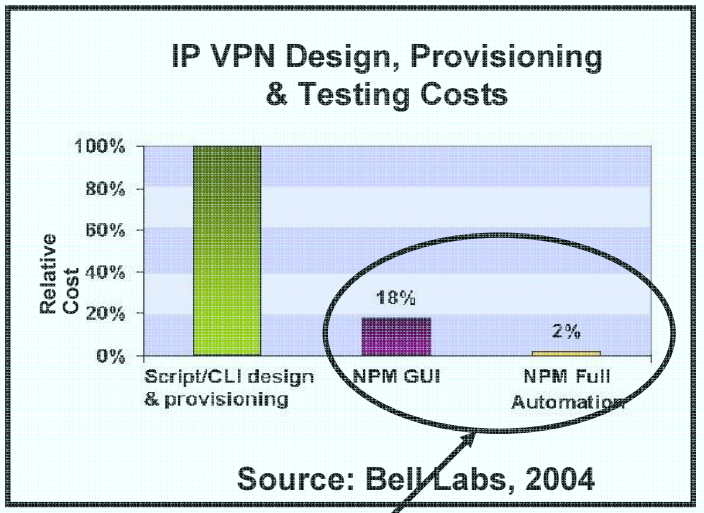
- Day to day troubleshooting
- Operational cost reduction
- Empowering level 1 support

Strategic Network Protection – Black Hole Diagnosis

- Major city PoP out for 10.5 hours …. router labelling issue
- MDE diagnoses in ~ 5 minutes



Provisioning



- MDE has key role in Automated Post-Provisioning Verification
- *"Up to 75% of our provisioning costs are in testing that the provisioning operation has been successful"* – Tier 1 Service Providers

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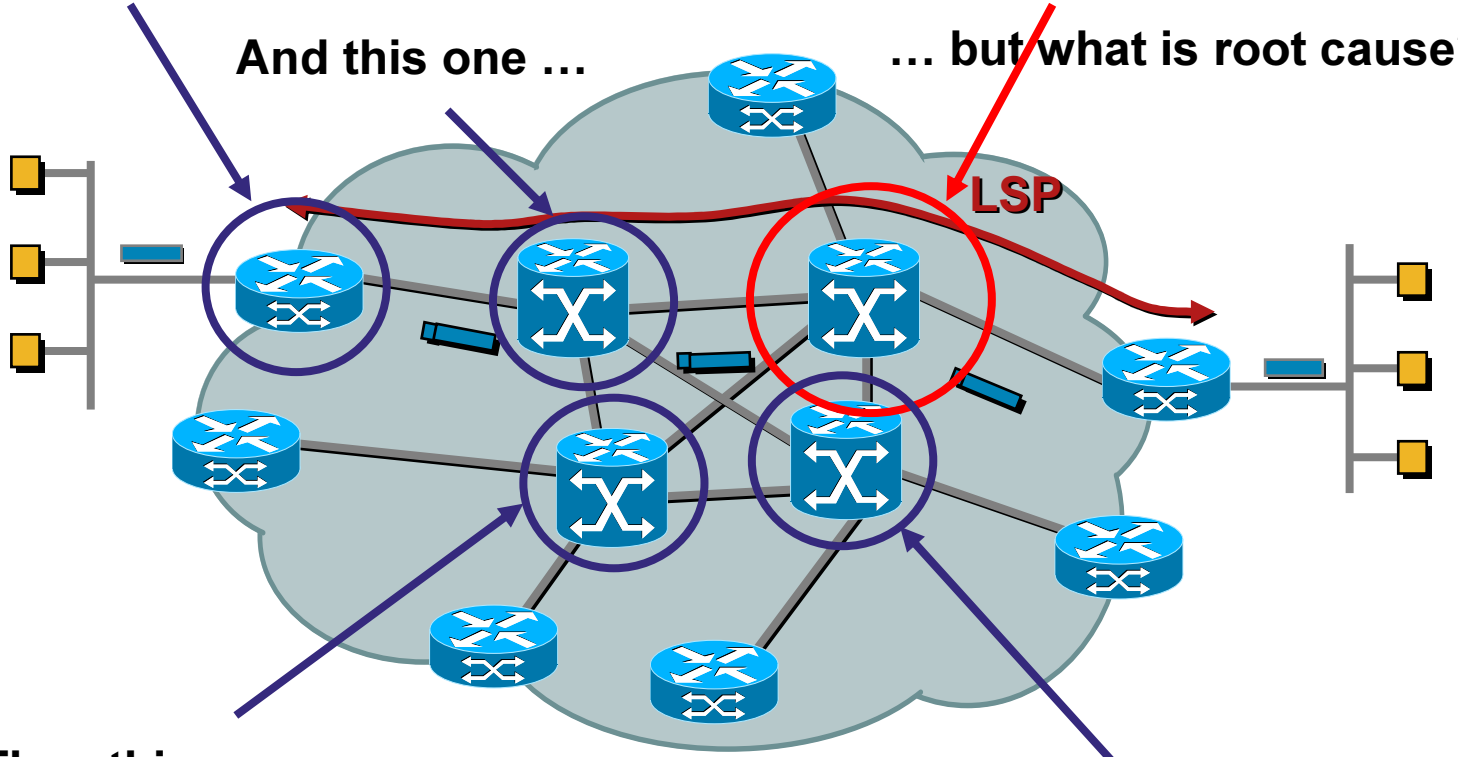
Example troubleshooting scenario: MDE replaces…

Check this node, using MIBs & CLI

The fault is here … but you had to check 70 out of 100 P/PE nodes before you found the location of this "black hole"

And this one …

… but what is root cause?



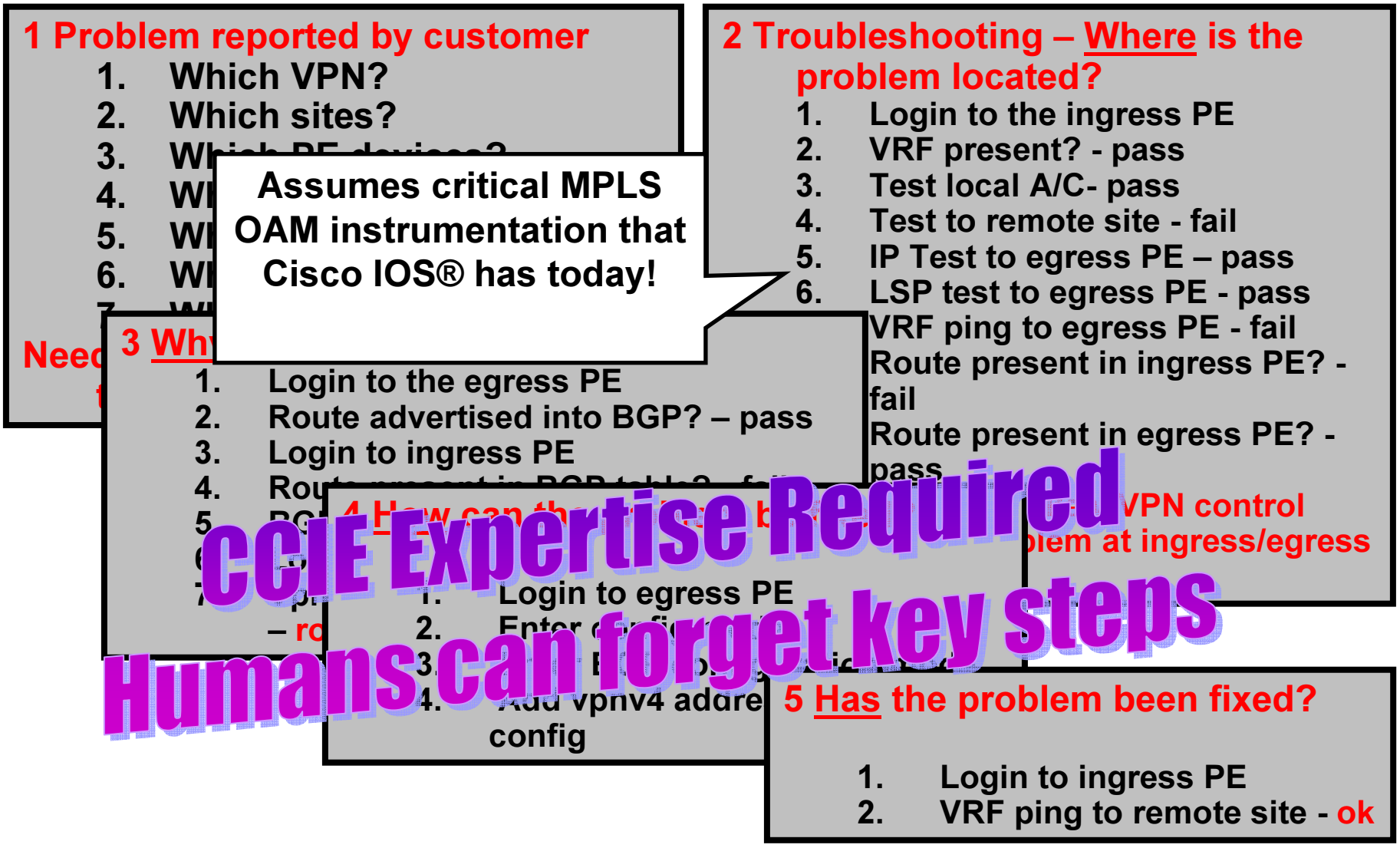
Then this one

And this one

Real Life Examples .. Can take hours (without Cisco IOS and MDE!)

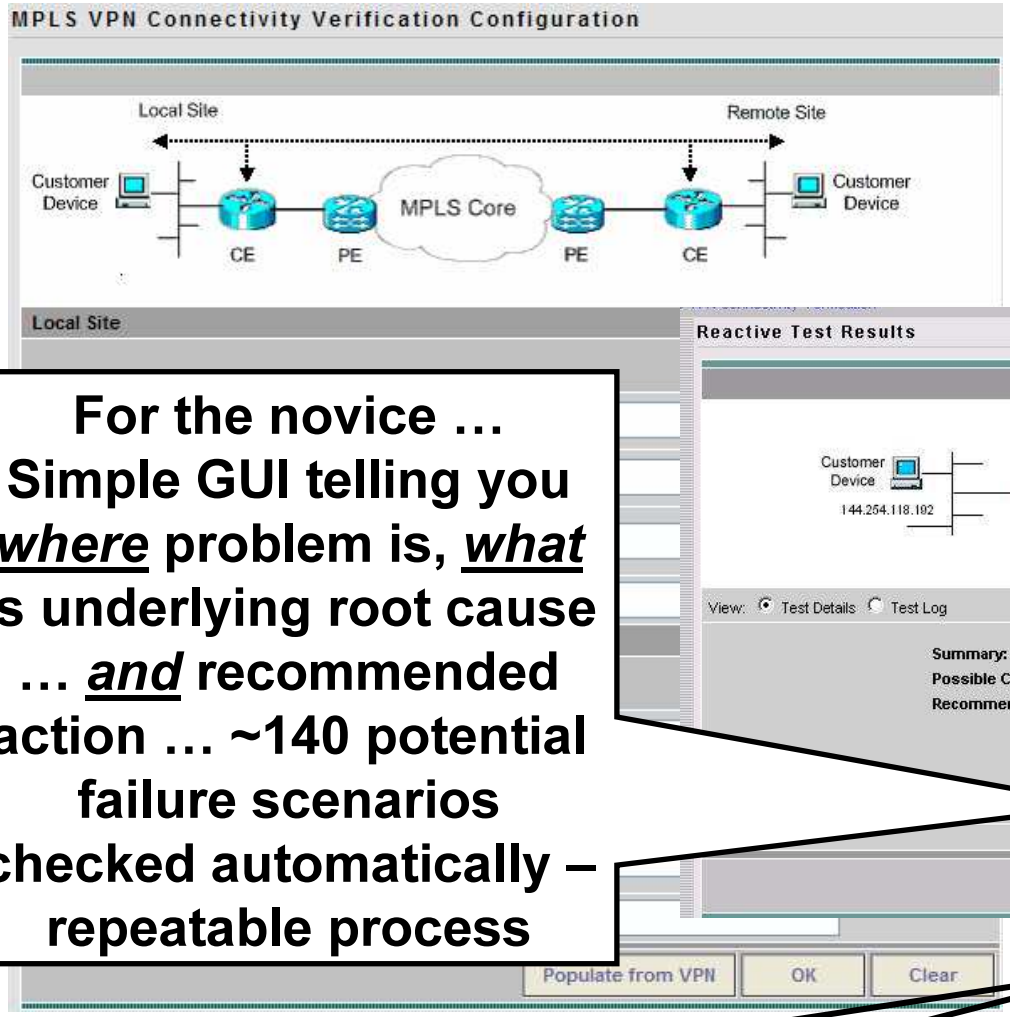
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… and … (Manual CLI Process) …



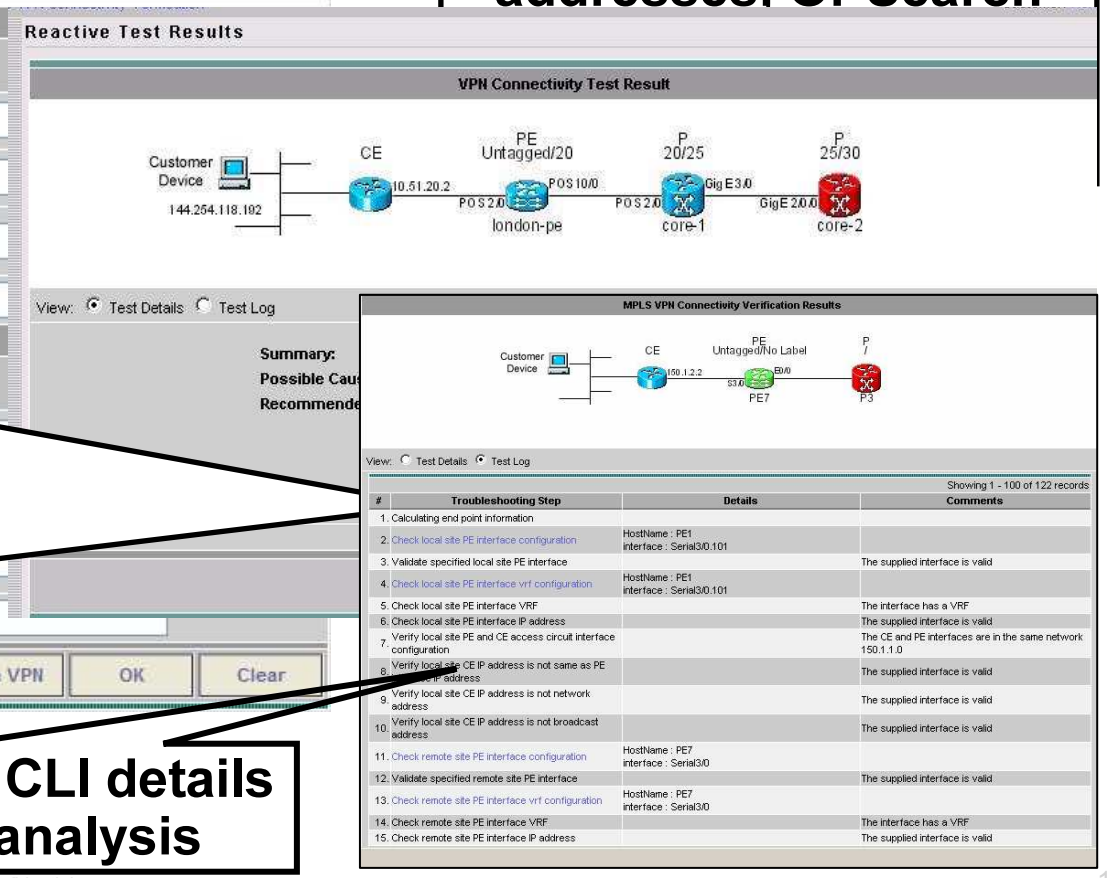
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… With Simplification and Acceleration
Cisco MPLS Diagnostics Expert



Type in simple details e.g. Customer Edge IP addresses. Or Search

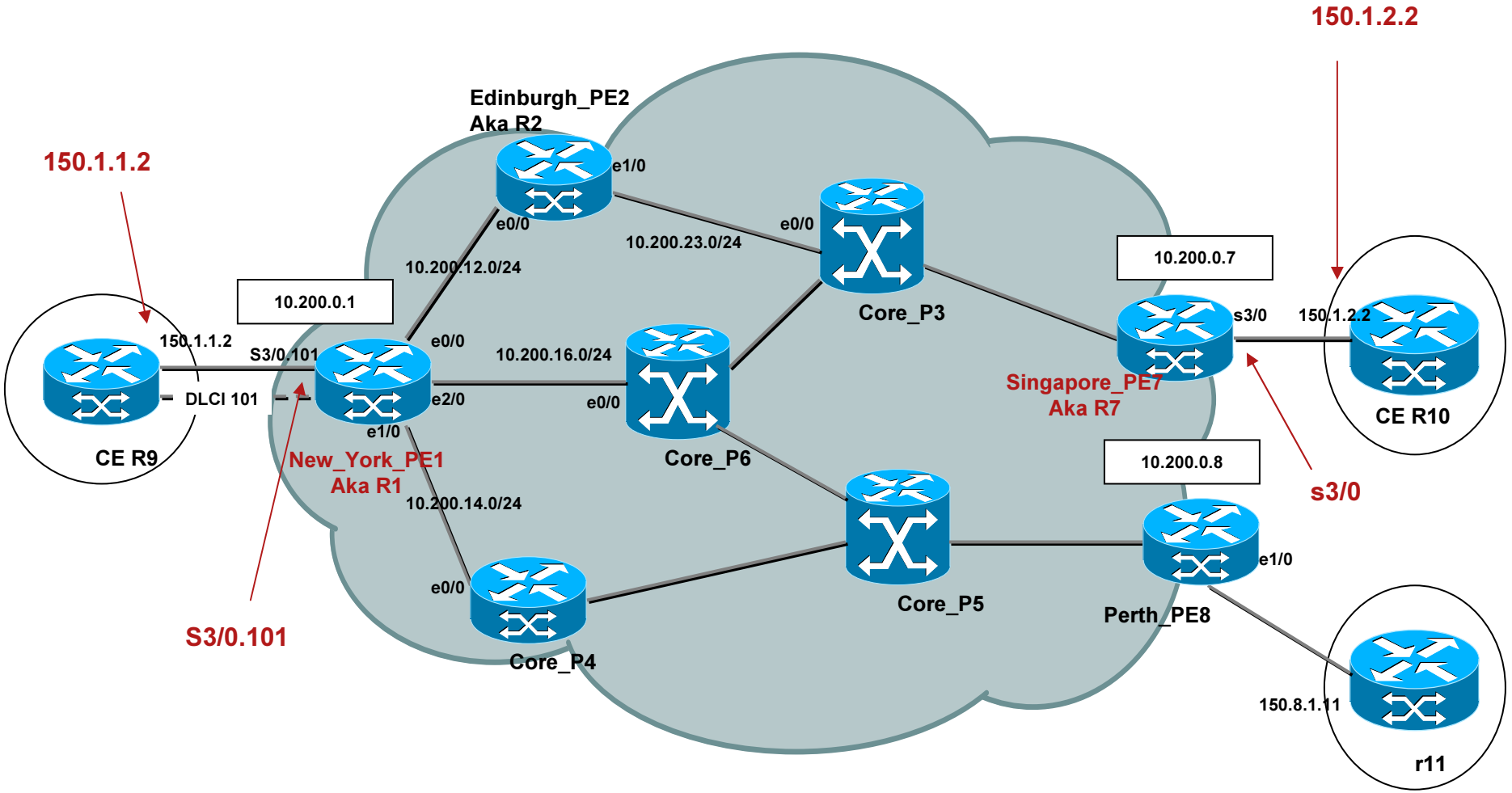
For the novice … Simple GUI telling you *where* problem is, *what* is underlying root cause … *and* recommended action … ~140 potential failure scenarios checked automatically – repeatable process



For the expert - All CLI details for subsequent analysis

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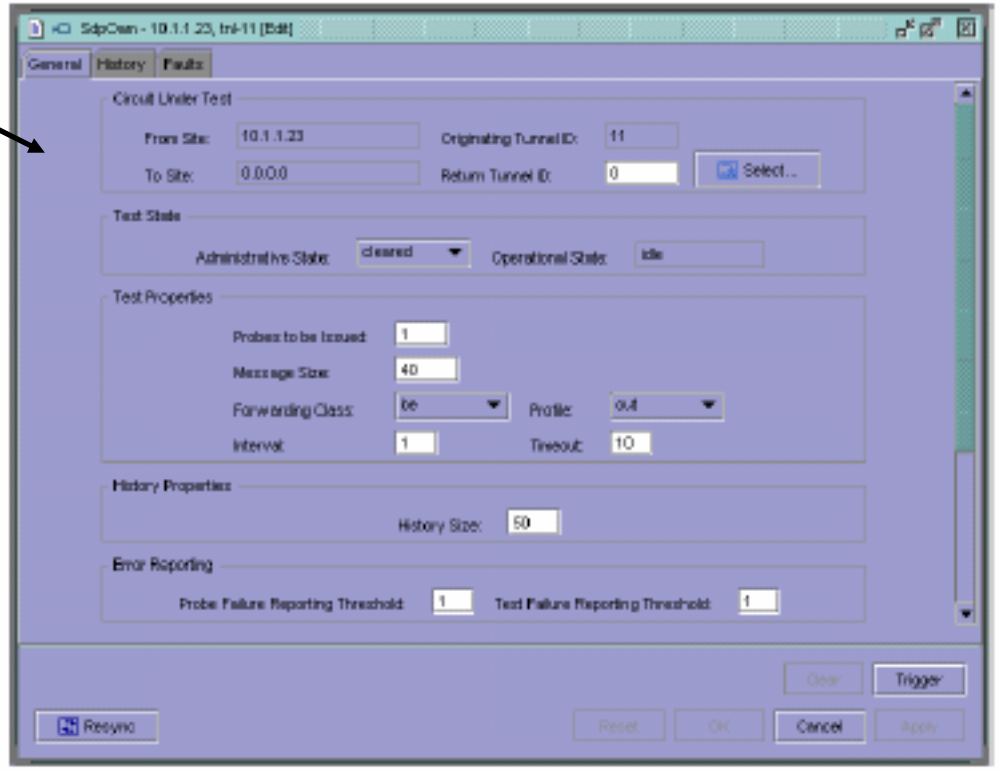
MPLS Diagnostics Expert
Demo Lab – Network Topology



Competitive Approaches: NOC User Complexity

Typical Competitor LSP Ping GUI...

- What this *doesn't* help you with ….
- Do you want to find out *if and why* connectivity is broken OR execute an LSP ping?
- Is LSP Ping the right tool to use *at all*?
- Do you know remote PE address to ping to?
- Do you know subnet mask on remote PE?
- What if issue is in return LSP?
- What about all other device commands we need … does each have a separate GUI? (lack of automation)
- How does (entry) NOC operator work out what loopback addresses, forward and reverse tunnel IDs are required?
- OK, LSP ping has failed, how does this help me find out *why* it failed



With Cisco MPLS Diagnostics Expert …

- You only need to know end test points
- MDE automates all decisions as to what OAM tools to use (ATM OAM, MPLS OAM etc)

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Summary: Why Use Cisco MPLS Diagnostics Expert

- Cisco MPLS Diagnostics Expert – Automated troubleshooting & Diagnostics for MPLS VPN networks – SP + Enterprise
- Reduces troubleshooting times, in cases from hours to minutes
- Ease of use for Helpdesk, Diagnostics Detail for 2nd Line Support
- Repeatable software vs “Humans do forget”
- Encapsulates & Automates Cisco internal experience & intellectual property in troubleshooting MPLS networks in software
- Automated use of advanced Cisco IOS® and IOS XR MPLS OAM
- Works for customers who do or don’t use ISC Provisioning
- Future roadmap to converge on ANA platform
- Strategic advantage for the NOC
- Strategic advantage for Cisco - winning hardware & NMS deals

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Background



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More Information …

- More info ….
- www.cisco.com/go/mde
- www.cisco.com/go/isc
- www.cisco.com/go/ana
- Miercom report: http://www.miercom.com/?url=reports/&v=16&PHPSESSID=aece1a14c87200b43cdd930404341fdd
- http://www.cisco.com/warp/public/732/Tech/mpls/mpls_oam.shtml

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Approved Customer Reference: Thus PLC

"THUS is highly focused on delivering operational excellence to customers of our next generation IP/MPLS network. We see the Cisco MPLS Diagnostics Expert as a strategic tool for our operations and network engineering teams that will deliver not only operational cost savings to THUS but also allow THUS to deliver industry leading responsiveness to our customers. The Cisco MPLS Diagnostics Expert makes automated and intelligent use of the MPLS Operations and Management (OAM) features in Cisco IOS MPLS Embedded Management. Yet the Cisco MPLS Diagnostics Expert is simple enough for our first line customer support operators to use, and empowers them with MPLS VPN troubleshooting skills normally only found in CCIE-level networking experts. And for our network specialists as well as second and third line support, the audit trail of advanced diagnostic tests executed by the Cisco MPLS Diagnostics Expert will accelerate diagnosis not only of rare customer issues but also help testing of new networking solutions. It's not only the operational time savings that are valuable - what's more important are the other, more strategic, value-added tasks our team can focus on as a result of the troubleshooting cycle time improvements delivered by the Cisco MPLS Diagnostics Expert."

THUS (www.thus.net) is one of the leading telecoms providers in the UK, focused on the corporate and SME markets. Thus are already delivering a range of innovative data services based upon a Cisco next-generation network. Thus products and services range from broadband to pioneering and market-leading MPLS and Managed Solutions.

Jason McInally, Network Engineering Manager, Thus PLC

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The Opportunity - Revolutionizing The MPLS VPN Customer Support Process

(Real) Customer Case Study: CEF/BGP Label Mismatch

6:41am:
- EBGP Flap - Customer Calls "I can't connect across your VPN

6:50am:
- Customer data collected into Trouble-ticket;
- Level 1 investigation starts

7:30am:
- Level 1 support escalates to Level 2 support

8:30am:
- Level 2 support escalates to Level 3 support

10:00am:
- Level 3 support still investigating ….

4:30pm:
- Cisco TAC engaged

5:15pm:
- Cisco TAC recommends solution – clear ip route on local PE

With Cisco MPLS Diagnostics Expert

6:41am:
- EBGP Flap - Customer Calls "I can't connect across your VPN. Customer data collected into Trouble-ticket

6:50am:
- Customer data collected into Trouble-ticket
- Level 1 support decides to apply Cisco MPLS Diagnostics Expert – MDE

7:00am:
- MDE Diagnosis complete – passed directly to Level 3 support

7:05am:
- Level 3 support executes MDE recommended action – verifies with MDE – customer problem solved

7:15am:
- Customer informed

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Deployment Services – Key to the Solution

"Harbrook Consultants demonstrated an in-depth knowledge of Cisco's IP Solution Centre (ISC) and MPLS Diagnostics Expert (MDE) and most importantly they knew how to tailor the solution for our business. IPI now have first class MPLS provisioning and diagnostics system that will help us deliver standard services cost effectively and to a high quality."

Pete Edwards, Technical Director,

IP Infrastructures Ltd (UK SP)



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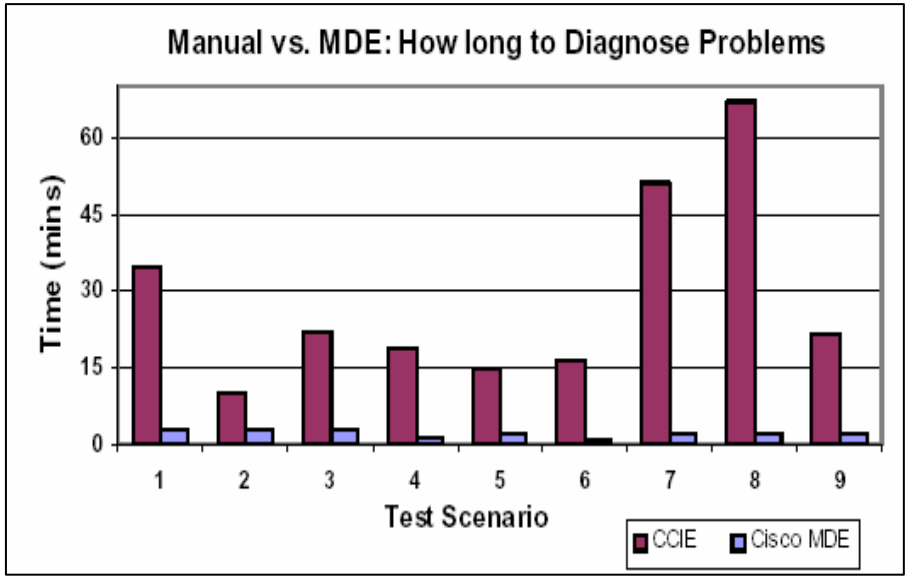
Miercom Independent MDE Test Manual Troubleshooting Vs MDE

Independent comparison test run by www.Miercom.com

- "Time & Motion" Study – comparison of Manual troubleshooting times vs MDE times
- Manual tester was MPLS CCIE
- MDE user was Miercom employee – no MPLS experience
- Small network: 7 P, 8 PE, 1 RR, 10 CE

Key findings and conclusions:

- The Cisco MPLS Diagnostics Expert (MDE) diagnose network problems more than 10 times faster than a veteran, MPLS-proficient CCIE
- Frees up CCIEs for more advanced tasks
- The Cisco MDE can be run by staff with minimal MPLS network-troubleshooting expertise



Troubleshooting times: Manual Expert vs MDE

Source: Miercom report #051120

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Bidirectional Forwarding Detection

BFD Feature Capabilities

Technology Components

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Drivers for BFD

- Link-layer detection misses some types of outages
  - e.g. Control Plane failure
- Control Plane failure detection is very conservative
  - 15-40 seconds in default configurations
- Link-layer failure detection is not consistent across media types
  - Less than 50ms on APS- protected SONET
  - A few seconds on Ethernet
  - Several seconds or more on WAN links
- Most current failure detection mechanisms are an order of magnitude too long for time-sensitive applications

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What is BFD?



Bi-directional Forwarding Detection:

- Extremely lightweight hello protocol
  - IPv4, IPv6, MPLS, P2MP
- 10s of milliseconds (technically, microsecond resolution) forwarding plane failure detection mechanism.
- Single mechanism, common and standardized
  - Multiple modes: Async (echo/non-echo), Demand
- Independent of Routing Protocols
- Levels of security, to match conditions and needs
- Facilitates close alignment with hardware

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Advantages Over Existing Alternatives

- Low overhead, Lightweight Hello Protocol.
  - Test data plane
  - Support for IPv4, IPv6, MPLS, PW
- Protocol-Independent:
  - Standard mechanism to detect data plane failure between 2 nodes: adjacent (single hop) or across multiple-hops
- Fast failure detection:
  - Fast forwarding plane failure detection (technically, microsecond resolution) mechanism Independent of Media/Data Link type/Interface, and routing protocols
- Media independent:
  - Provide a single data plane detection failures method for layer 2, 3 protocols.

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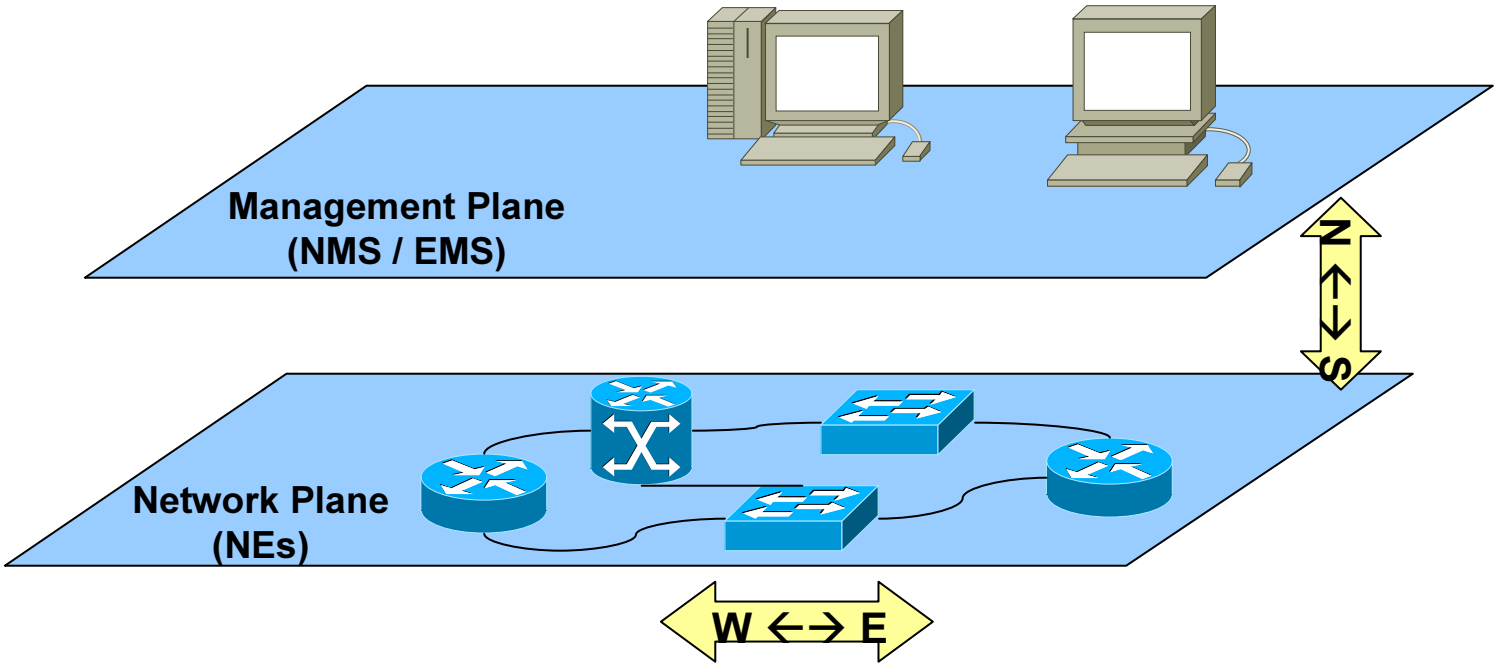


Ethernet OAM

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OAM &P: The Concept

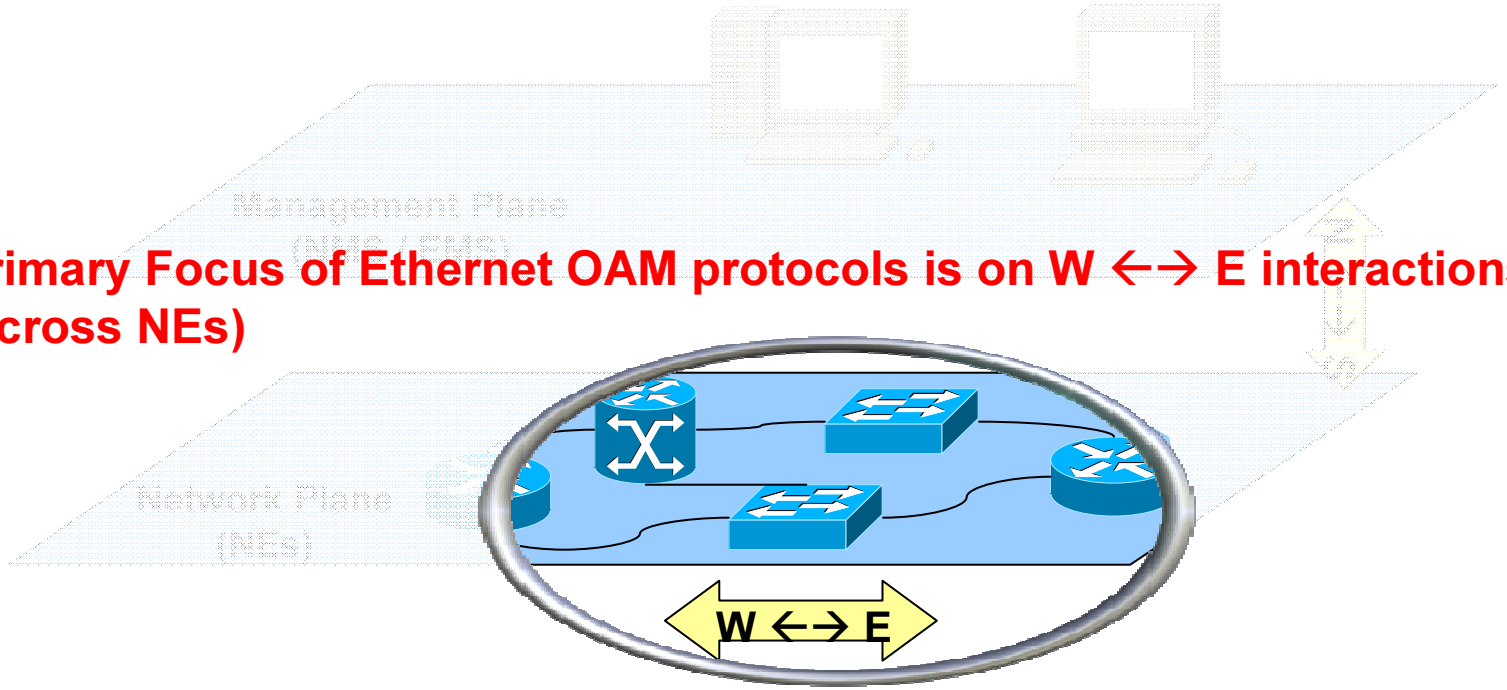
- Operations, Administration, Maintenance & Provisioning:
  - fault indication
  - performance monitoring
  - security management
  - diagnostic functions
  - configuration & service provisioning
- OAM covers both N←→ S and W←→E interfaces



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OAM &P: The Concept

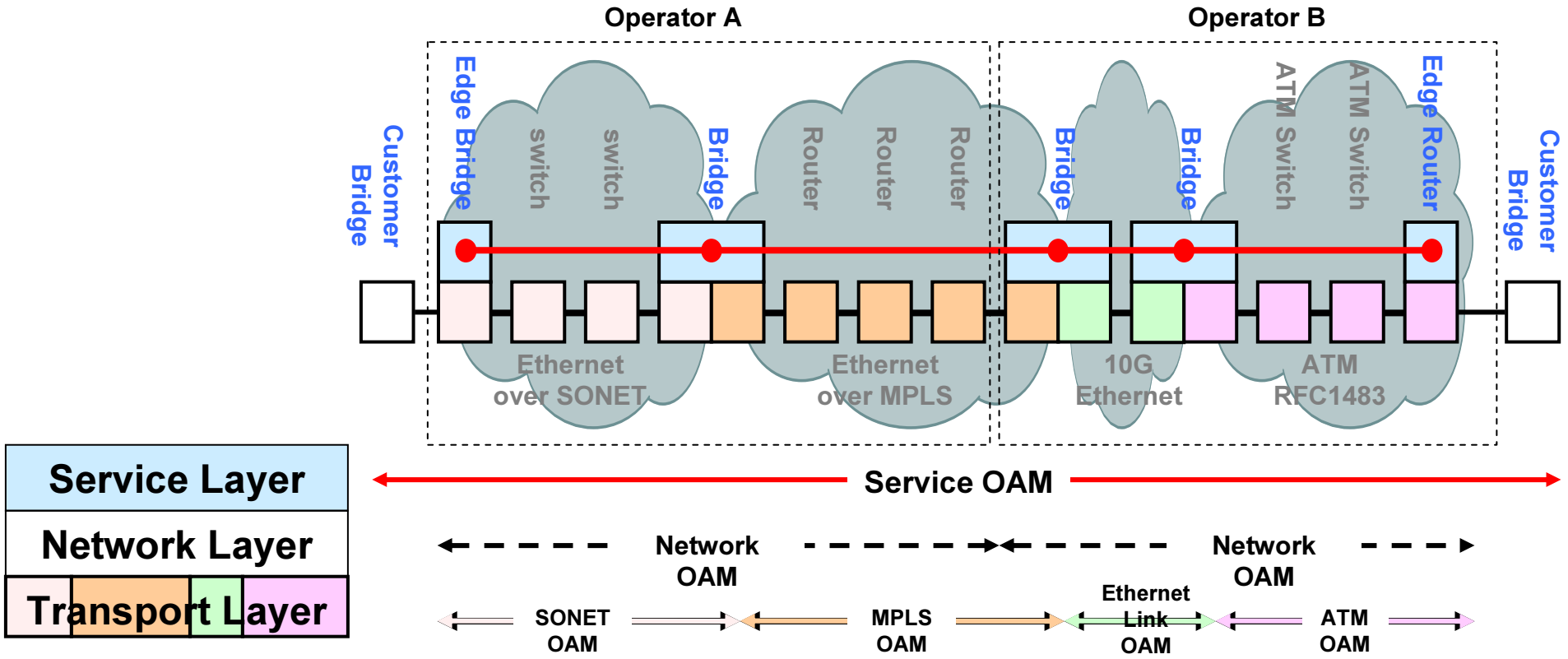
- Operations, Administration, Maintenance & Provisioning:
  - fault indication
  - performance monitoring
  - security management
  - diagnostic functions
  - configuration & service provisioning
- OAM covers both N←→ S and W←→E interfaces



Primary Focus of Ethernet OAM protocols is on W ←→ E interactions (across NEs)

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OAM Layering



- Model is iterative (layer stacking) & relative (service layer for Operator is transport layer for SP)
- Each Layer supports its own OAM mechanisms
- Inter-working across and within OAM layers is possible

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Overview of Ethernet OAM



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Drivers for Ethernet OAM

- Management Complexity
  - Large Span Networks
  - Multiple constituent networks belong to disparate organizations/companies
  - IP overlay or plug-n-play no longer valid options!
- Operational Efficiency
  - Reduce Opex, avoid truck-roll
  - Downtime cost
- OAM benchmarks set by TDM and existing WAN technologies

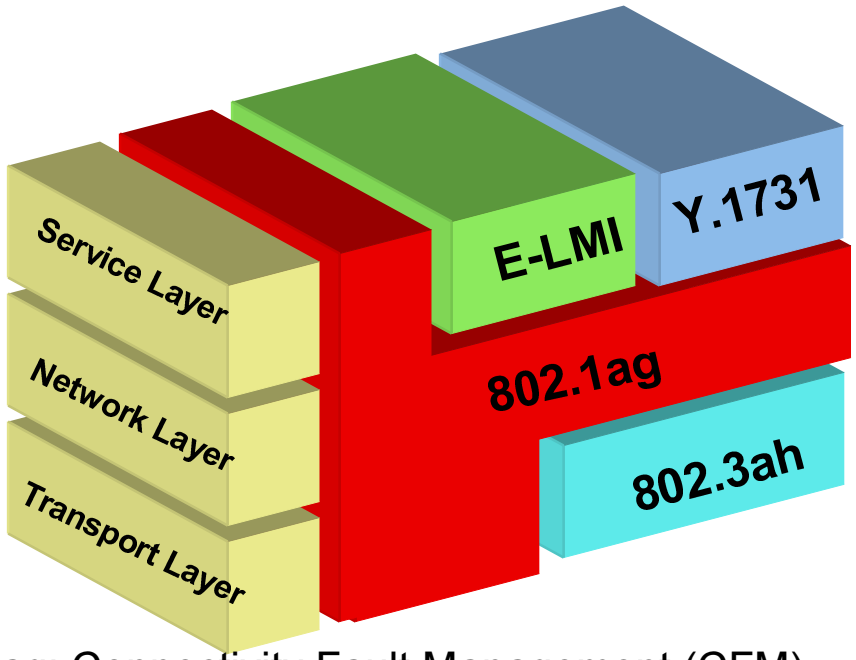
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The Promise of Ethernet OAM

- Simplify deployment of Ethernet WAN services
  - particularly important as Ethernet moves to a volume market with less technically sophisticated customers
- Enable monitoring & verification of SLAs
  - Not all customers and all services are equal.
  - SLAs will be a key differentiator for carriers and service providers.
- Provide End-to-End OAM
  - Not only within the franchise of a single carrier
  - Eliminate finger-pointing between carriers
- Facilitate deployment of Ethernet services over multiple access technologies
  - Service independence from access and transport
- Deliver SP's customers packaged plug and play solutions

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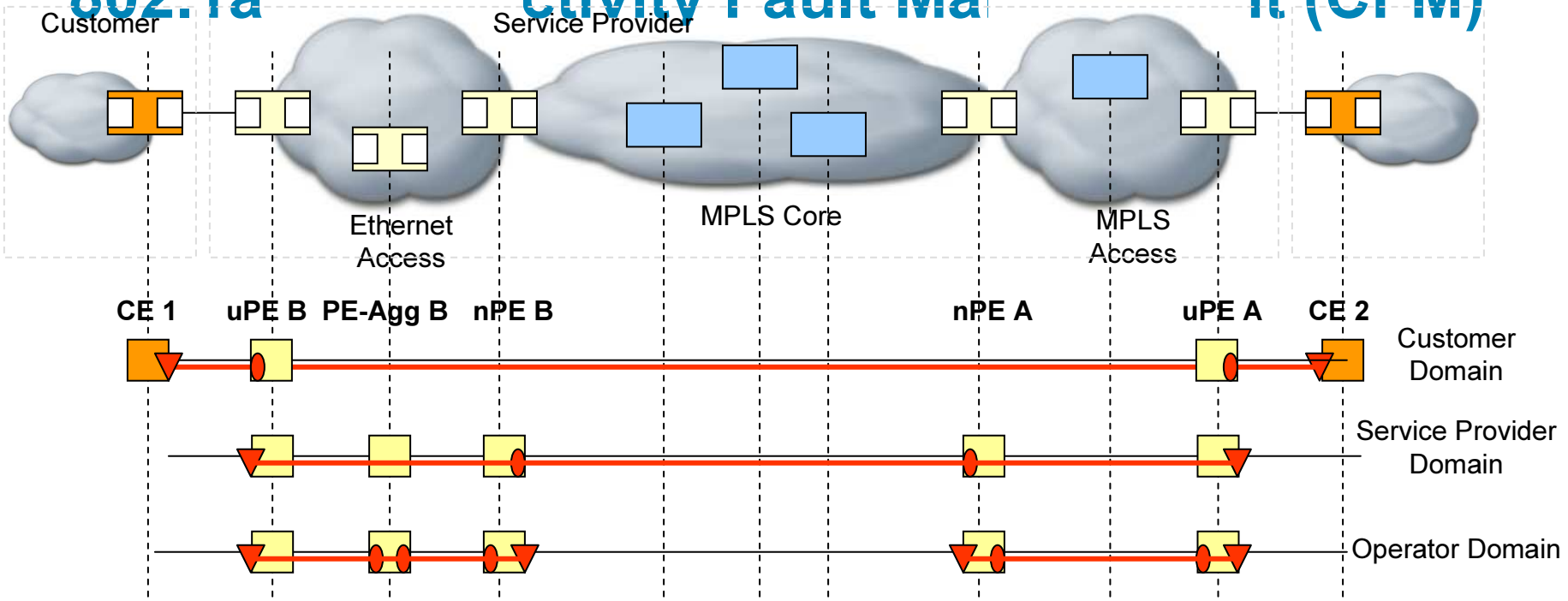
Ethernet OAM: the Protocol Building Blocks



- IEEE 802.1ag: Connectivity Fault Management (CFM)
- ITU-T Y.1731: OAM functions and mechanisms for Ethernet based networks
- IEEE 802.3ah: Ethernet Link OAM (EFM OAM)
- MEF E-LMI: Ethernet Local Management Interface

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802.1ag Connectivity Fault Management (CFM)



- End to End per EVC OAM
- Hierarchical Maintenance Domains
  - MEPs/MIPs
- Standard Ethernet Frames (in-band)
  - Continuity Check
  - Loopback
  - Link Trace

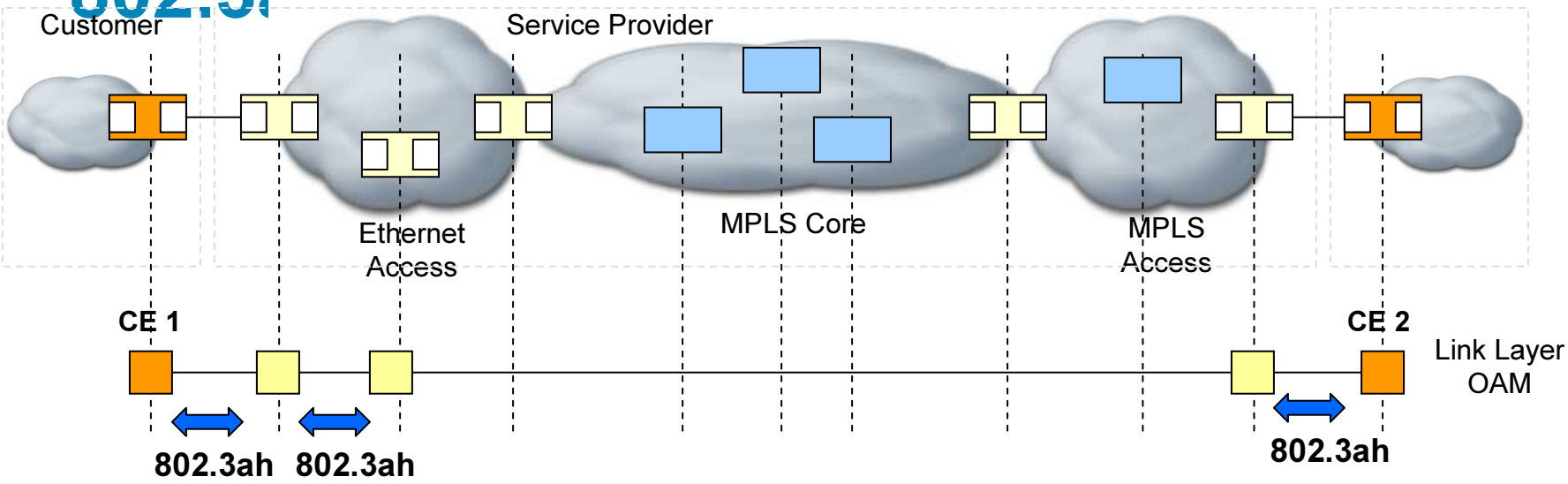
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Y.1731

- CFM plus…
  - ETH-LCK (out of service diagnostics)
  - Multicast Loopback
  - AIS
  - TEST
  - Maintenance Communication Channel
  - Experimental OAM
  - Performance Measurement (Delay, Packet loss …)

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802.3ah



- Link Level OAM
- Operates on point-to-point link, not propagated beyond a single hop.
- Slow Protocol (Max rate of 10 frames per second)
- Functions:

  OAM discovery – Discover OAM capabilities on peer device

  Link monitoring – Event notification when error thresholds exceeded

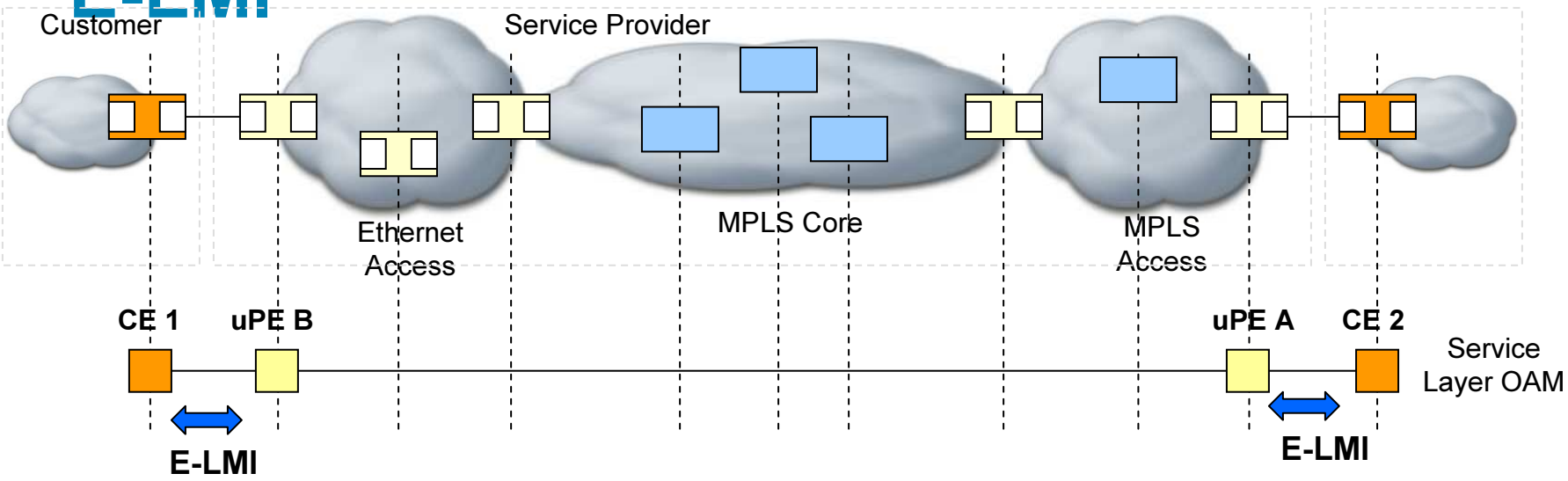
  Remote MIB Variable Retrieval – Polling and response (but not writing) of 802.3ah MIB

  Remote Failure indication – Inform peer that receive path is down.

  Remote Loopback – Puts peer in (near-end) intrusive loopback state. Statistics can be collected while testing link.

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E-LMI



- Asymmetric protocol, applicable on UNI only (uPE to CE)
- Specifies procedures & message formats exchanged and NOT how uPE collects OAM data – relies on Service/Network OAM running uPE to uPE
- Allows uPE to communicate to CE:
  - EVC Status
  - Remote UNI Status
  - CE-VLAN to EVC Map
  - BW Profiles

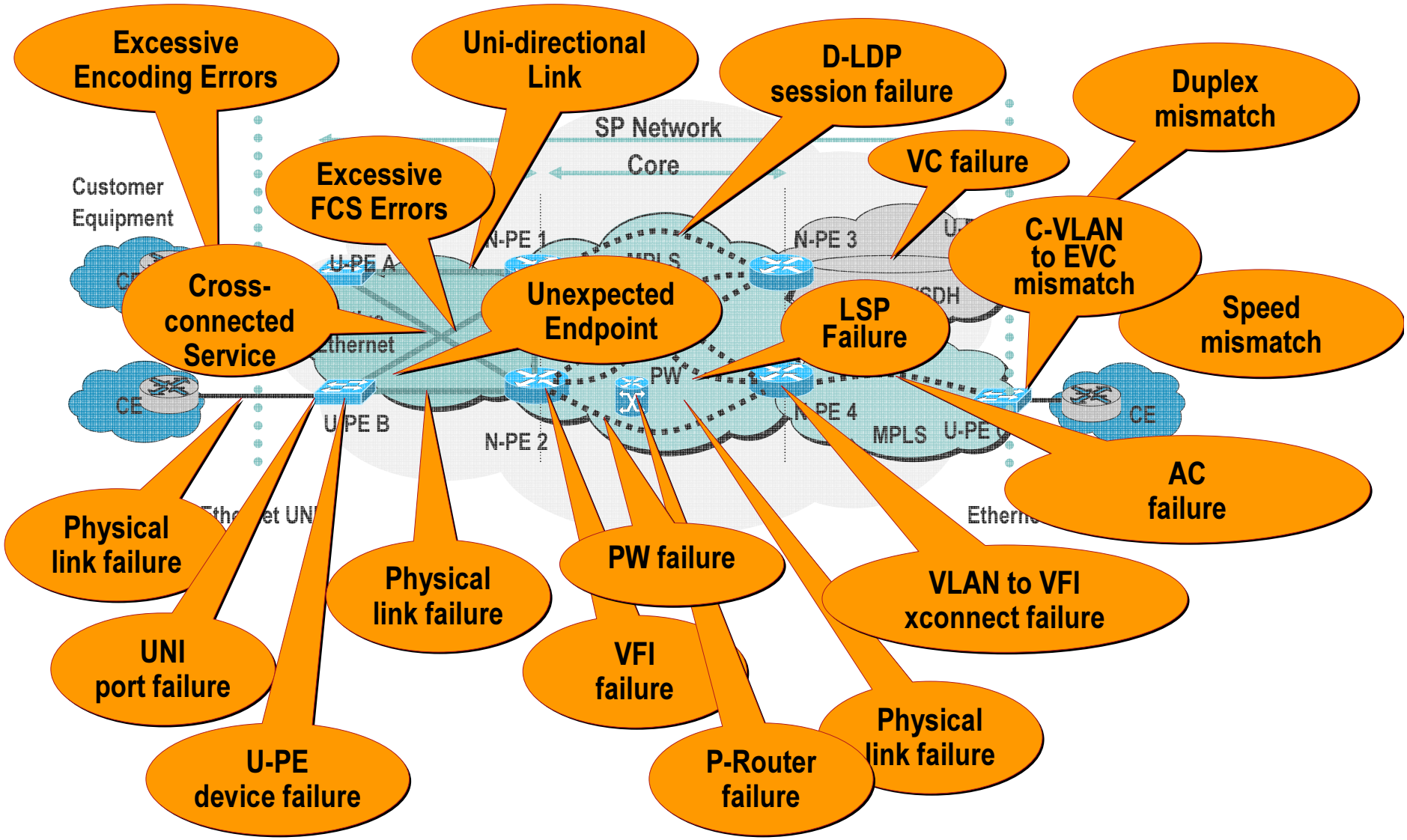
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In the Network: Using Ethernet OAM



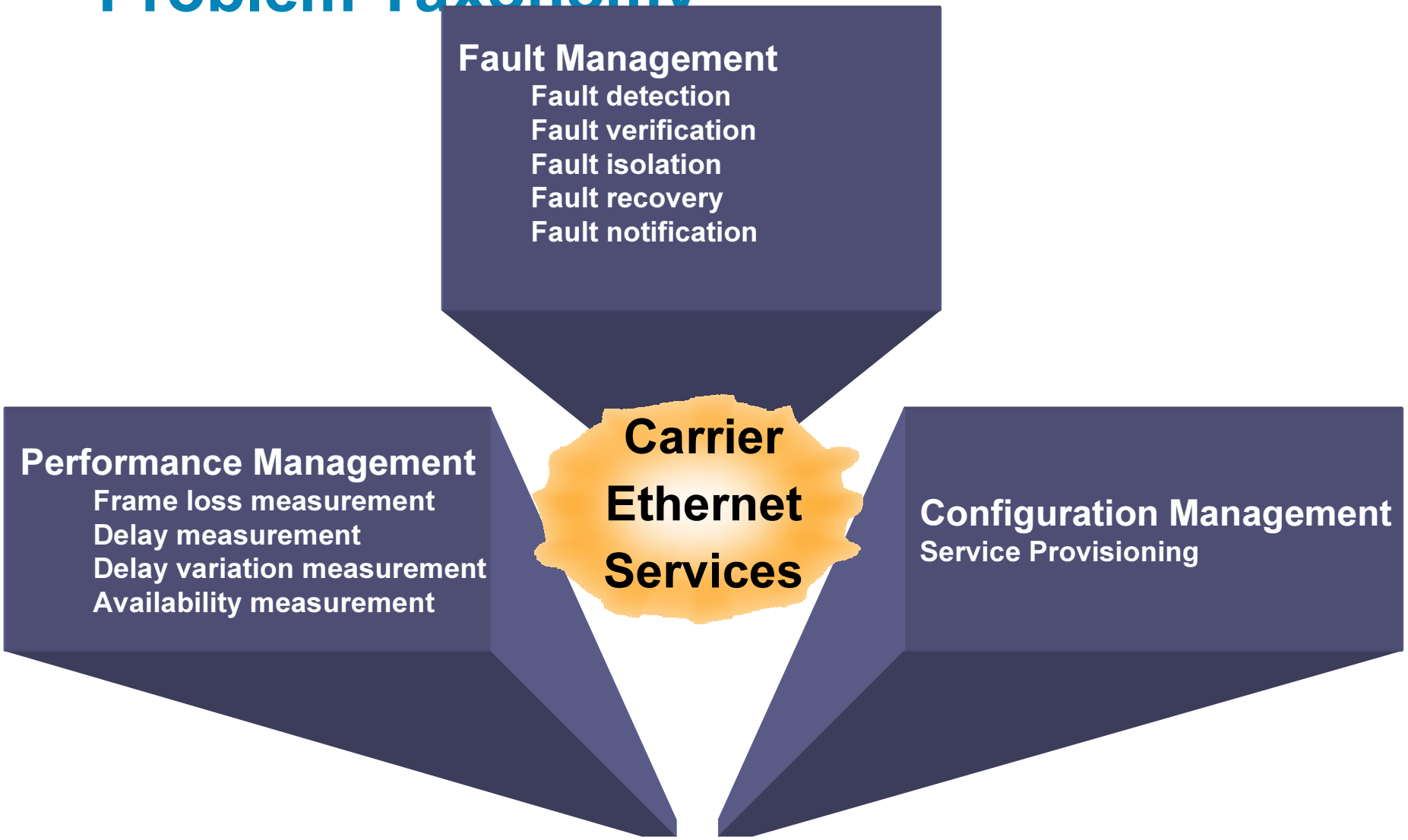
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Problem Scope
A few possible scenarios (incomprehensive)



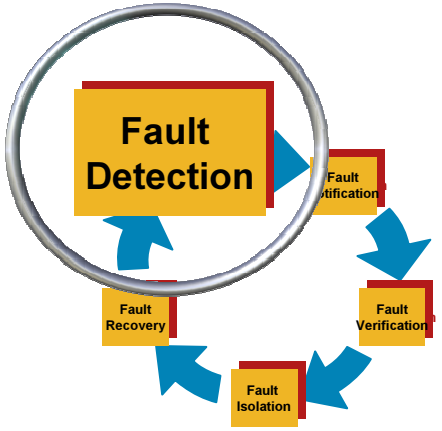
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Problem Taxonomy



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E-OAM Toolkit for Fault Management (1 of 5)



Fault Detection

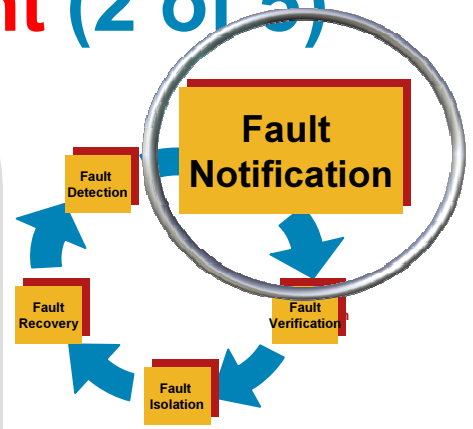
	Mechanism	Detectable Faults
802.1ag (CFM)	Continuity Check (CC)	• Unintended connectivity/service leaks • Unexpected sites • Loss of connectivity to a site • Link Connectivity failure • Device failure (soft & hard) • Forwarding plane loops • CFM Configuration Errors
802.3ah	Link Monitoring	• Unidirectional Link • Slowly deteriorating link quality (Frame/Symbol Errors)

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E-OAM Toolkit for Fault Management (2 of 5)

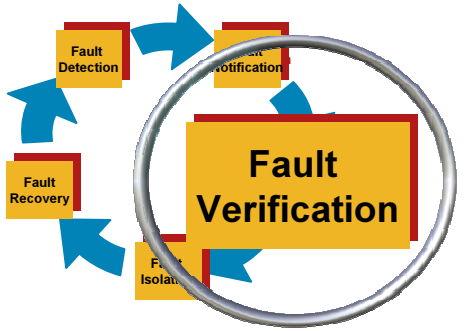
Fault Notification

	Mechanism	Triggers
Y.1731	Alarm Indication Signal (AIS)	• Loss of CFM Continuity Check • Indication from Server Layer OAM • Indication from lower ME Level CFM Domain
802.1ag (CFM)	Remote Defect Indication (RDI)	• Unidirectional service Connectivity (p2p) • Partial service connectivity (mp)
802.3ah	Remote Failure Indication	• Link Fault (receive path) • Dying Gasp • Critical Event
	Event Notification	• Error thresholds exceeded (frames/symbols per interval)
E-LMI	Status Message	• EVC Status Change • Remote UNI(s) Status Change



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E-OAM Toolkit for Fault Management (3 of 5)

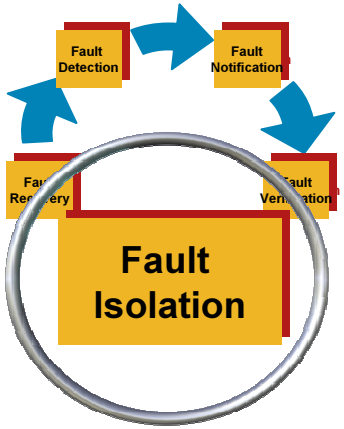


Fault Verification

	Mechanism	Capability
802.1ag (CFM)	Loopback	• Per EVC MAC Ping (source to single destination) • Verify bidirectional connectivity between two CFM Maintenance Points (for varied frame sizes)
Y.1731	Multicast Loopback	• Per EVC MAC Ping (source to all destinations) • Verify bidirectional connectivity between one CFM Maintenance End Point and all other End Points of a service (for varied frame sizes)

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E-OAM Toolkit for Fault Management (4 of 5)

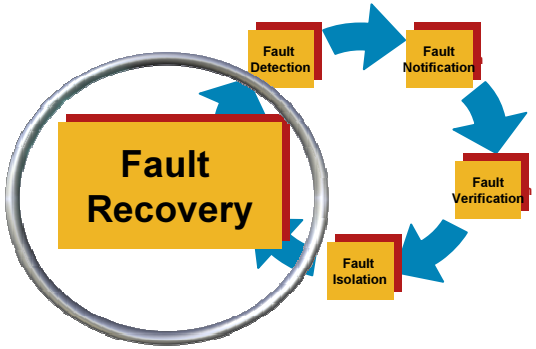


Fault Isolation

	Mechanism	Capability
802.1ag (CFM)	Link Trace	• Per EVC MAC Traceroute • Discover Maintenance Intermediate Points on path from source End Point to destination End Point • Report Ingress Action, Relay Action, Egress Action hop by hop. • Report encountered ACLs or STP-blocked ports

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E-OAM Toolkit for Fault Management (5 of 5)



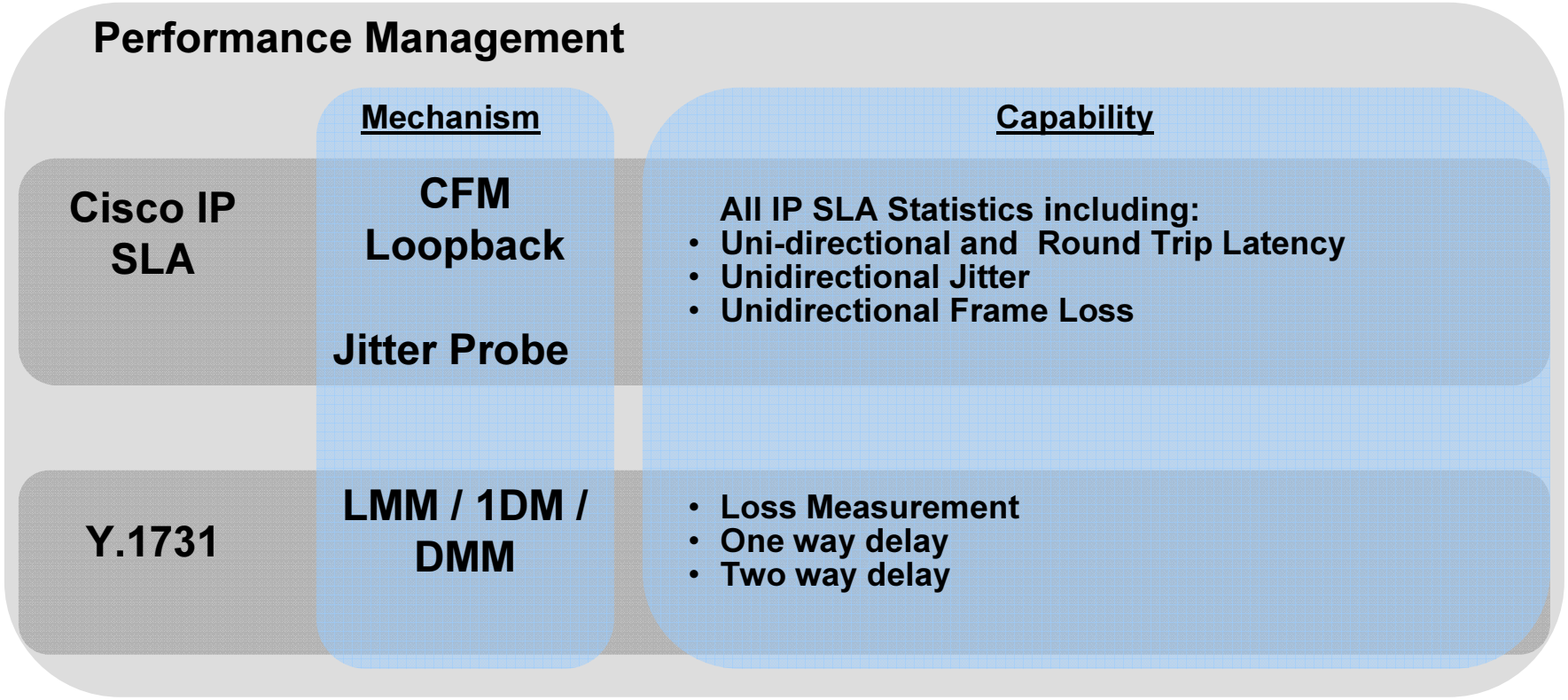
Fault Recovery

	Mechanism	Capability
G.8031	ETH-APS	• Use CFM mechanisms for monitoring redundant paths (order of msec)
Non-OAM Mechanisms	xSTP	

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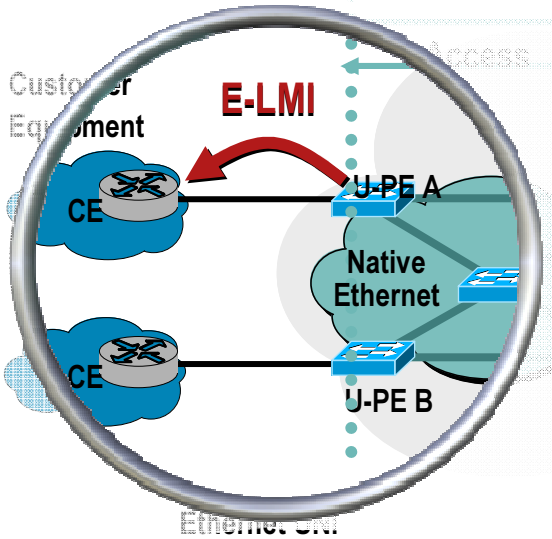
E-OAM Toolkit for Performance Management

- Frame loss measurement
- Delay measurement
- Delay variation (Jitter) measurement
- Availability measurement



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E-OAM Toolkit for Configuration Management

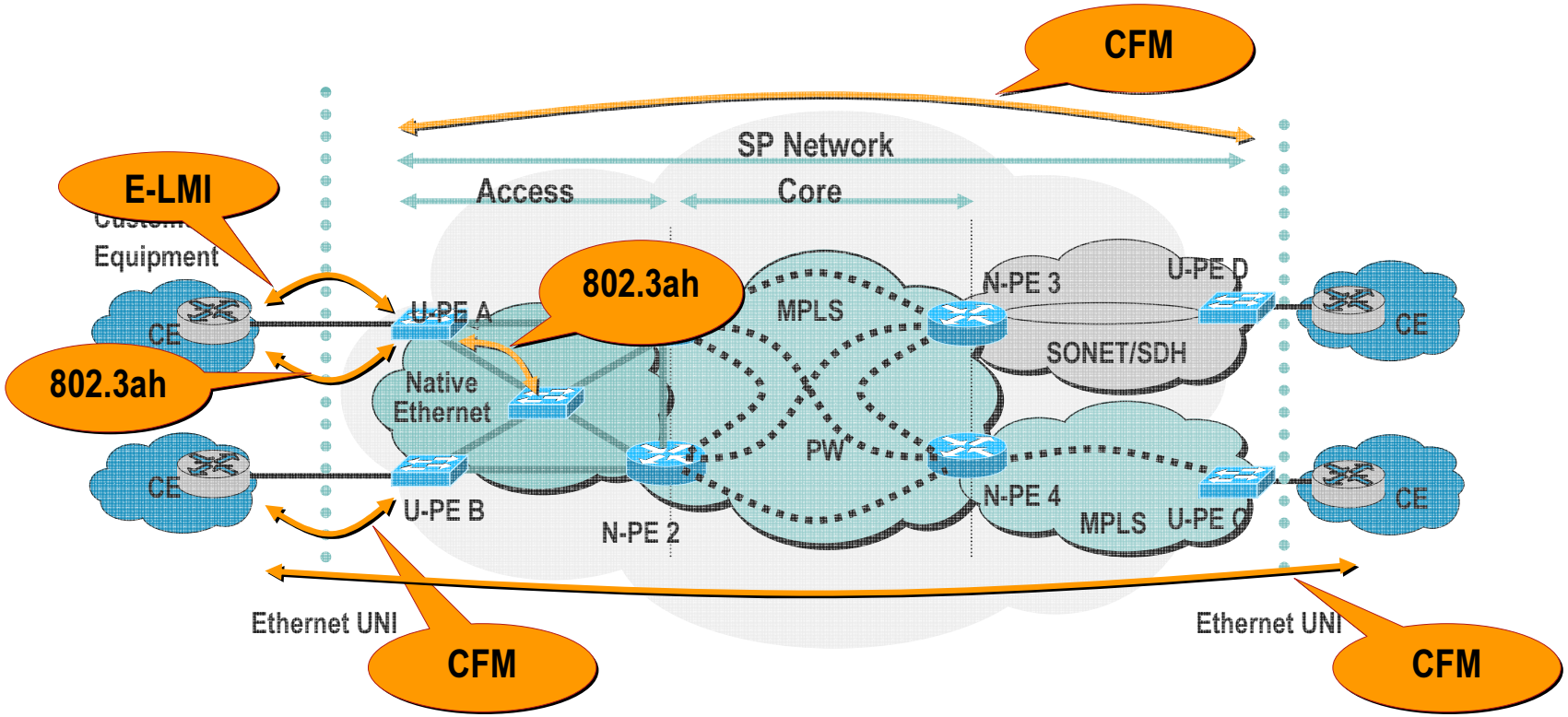


Configuration Management

	Mechanism	Capability
E-LMI	CE-VLAN / EVC Map	• Reduce Customer configuration errors • Ease deployment for Managed Services / Low-Touch Turn-key CE
	BW Profile	• Improve service performance by egress CE Shaping • Reduce Policing configurations on SP gear

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Which Protocol to Use & Where?



- Choice of protocols, their locality & span of operation varies with network topology & services – many variations possible
- Restrictions (inherent to protocol definitions):
  - E-LMI on UNI (uPE to CE)
  - 802.3ah on pt-to-pt (emulated ok) 802.3 PHY
  - CFM/Y.1731 require Ethernet MAC Layer awareness

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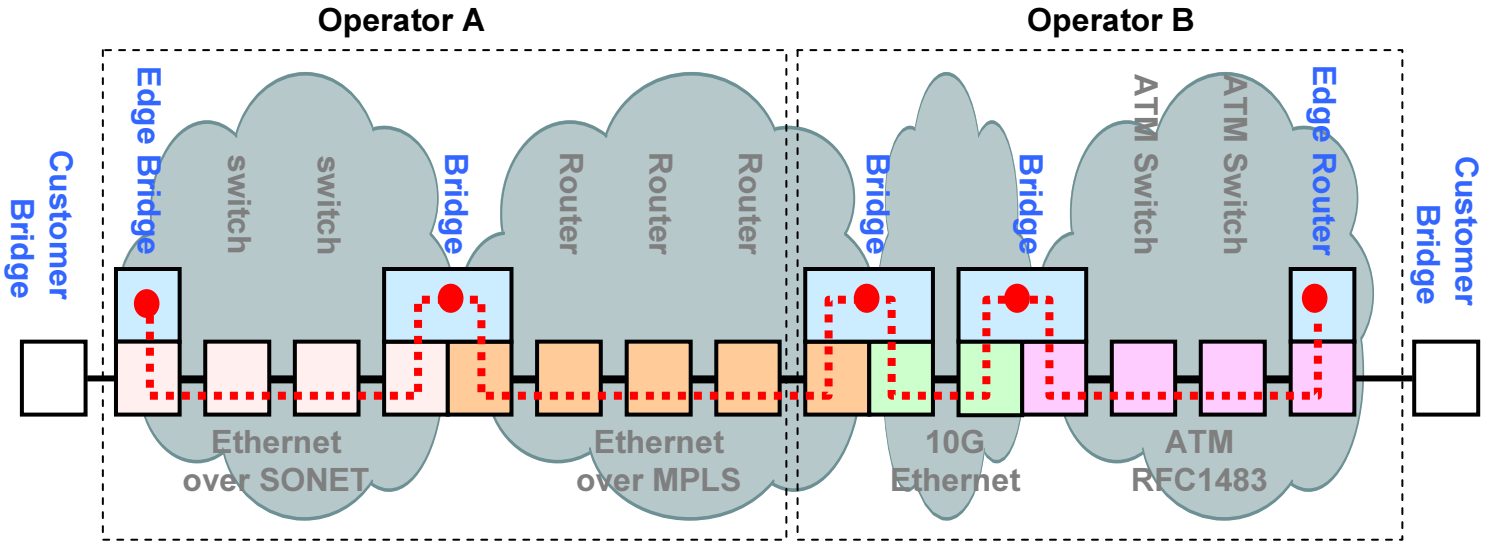
OAM Inter-working



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What is OAM Inter-working?

This is NOT OAM inter-working!

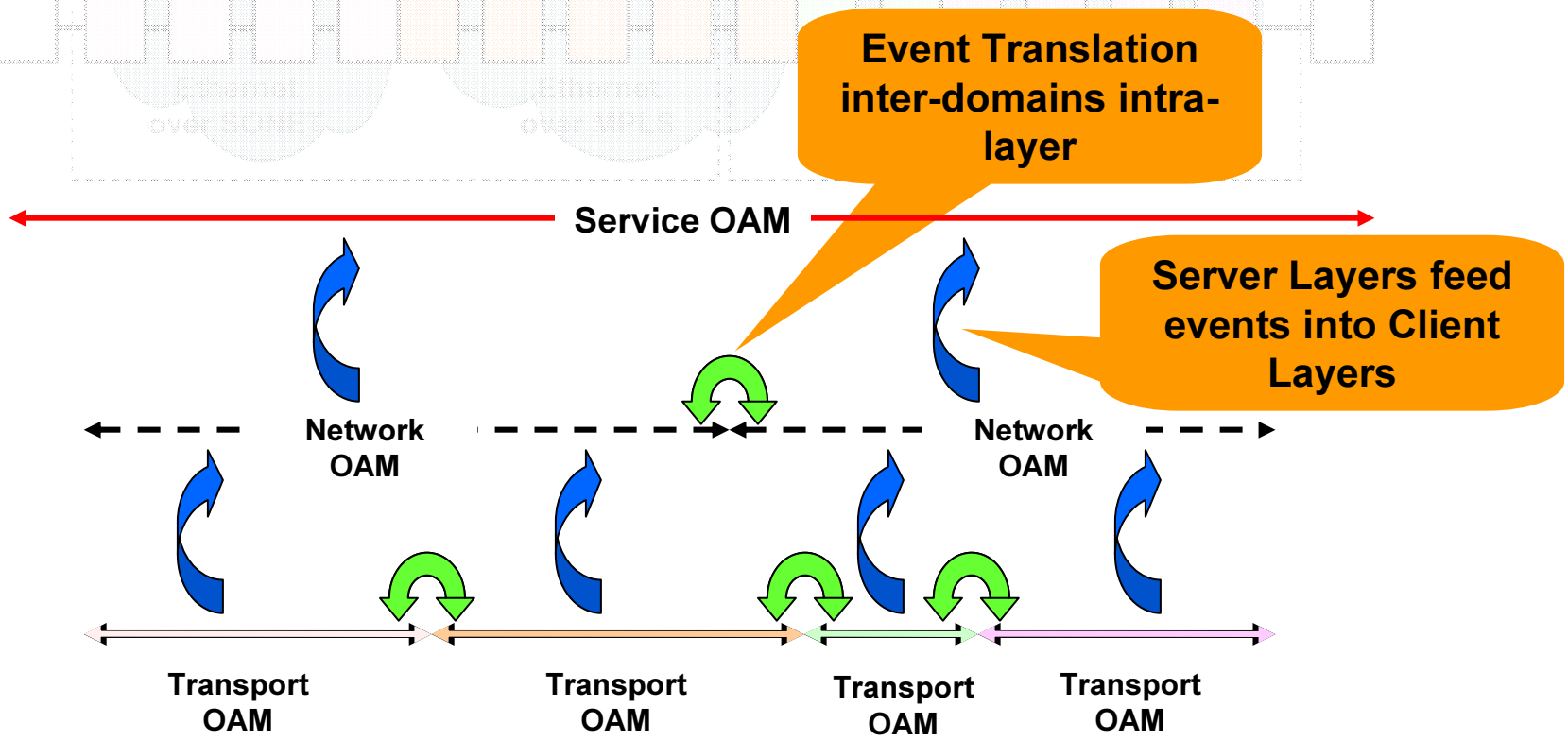


- Transport OAM message with embedded MAC address carried from bridge to bridge, visible to ETH layer (when present), and translated to new transport's OAM format when crossing physical media boundaries.
- Creates dependency on Physical layer and inter-operability issues.

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What is OAM Inter-working?

- Strict OAM layering should be honored: messages should not cross layers
- OAM Messages should not leak outside domain boundaries within a layer
- Inter-working is event translations & not necessarily 1:1 message mapping
- Inter-working may be inter-layer and intra-layer



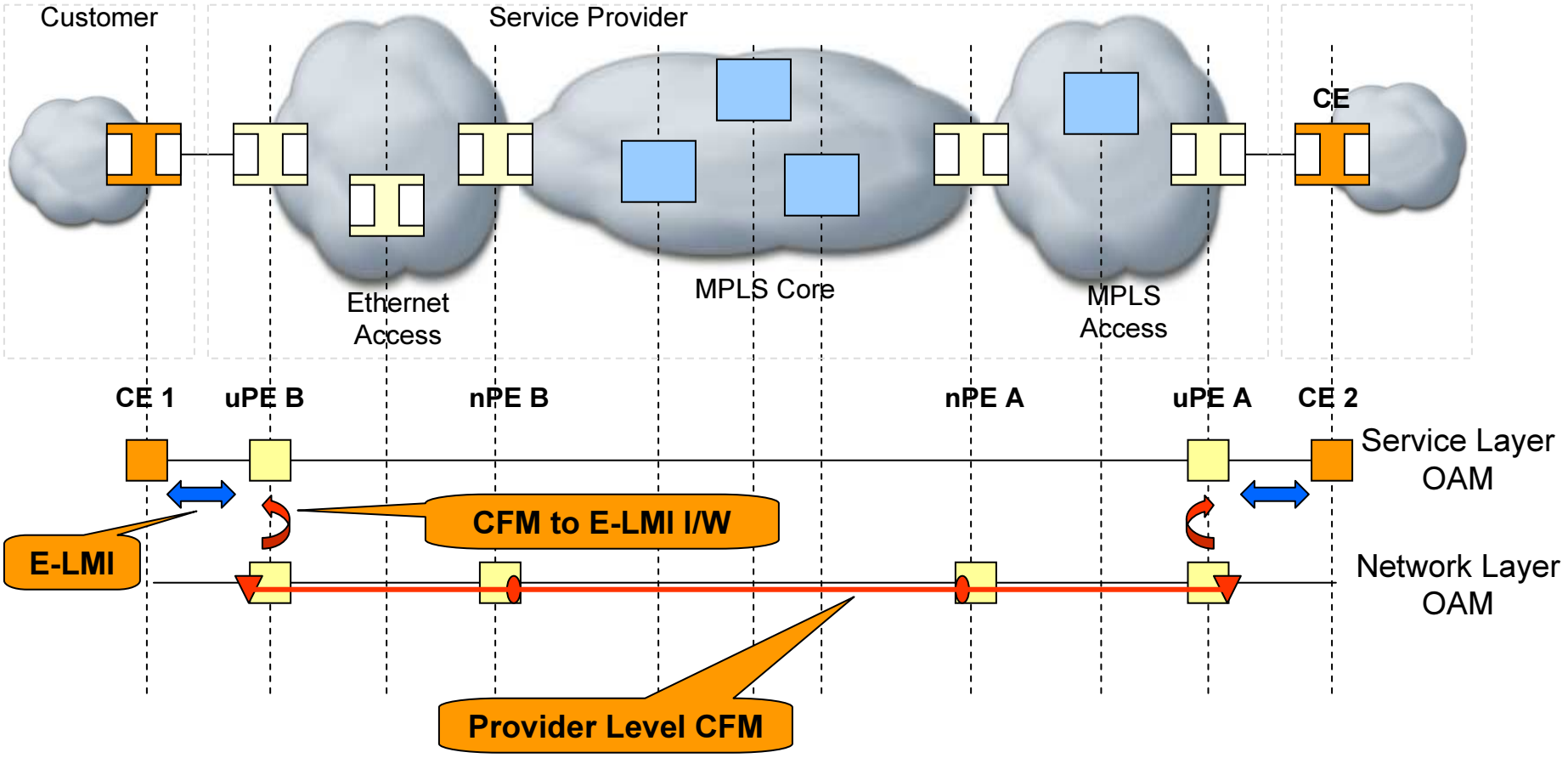
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Why OAM Inter-working?

- Enable management of ubiquitous Ethernet service offerings on:
  - mixed media
  - heterogeneous networks/transports
  - legacy equipment
- Leverage the benefits of server layer OAM mechanisms when applicable
  - Potentially faster failure detection times
  - Redundant Alarm Suppression (daisy chain effect)

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Inter-working Scenarios: CFM to E-LMI

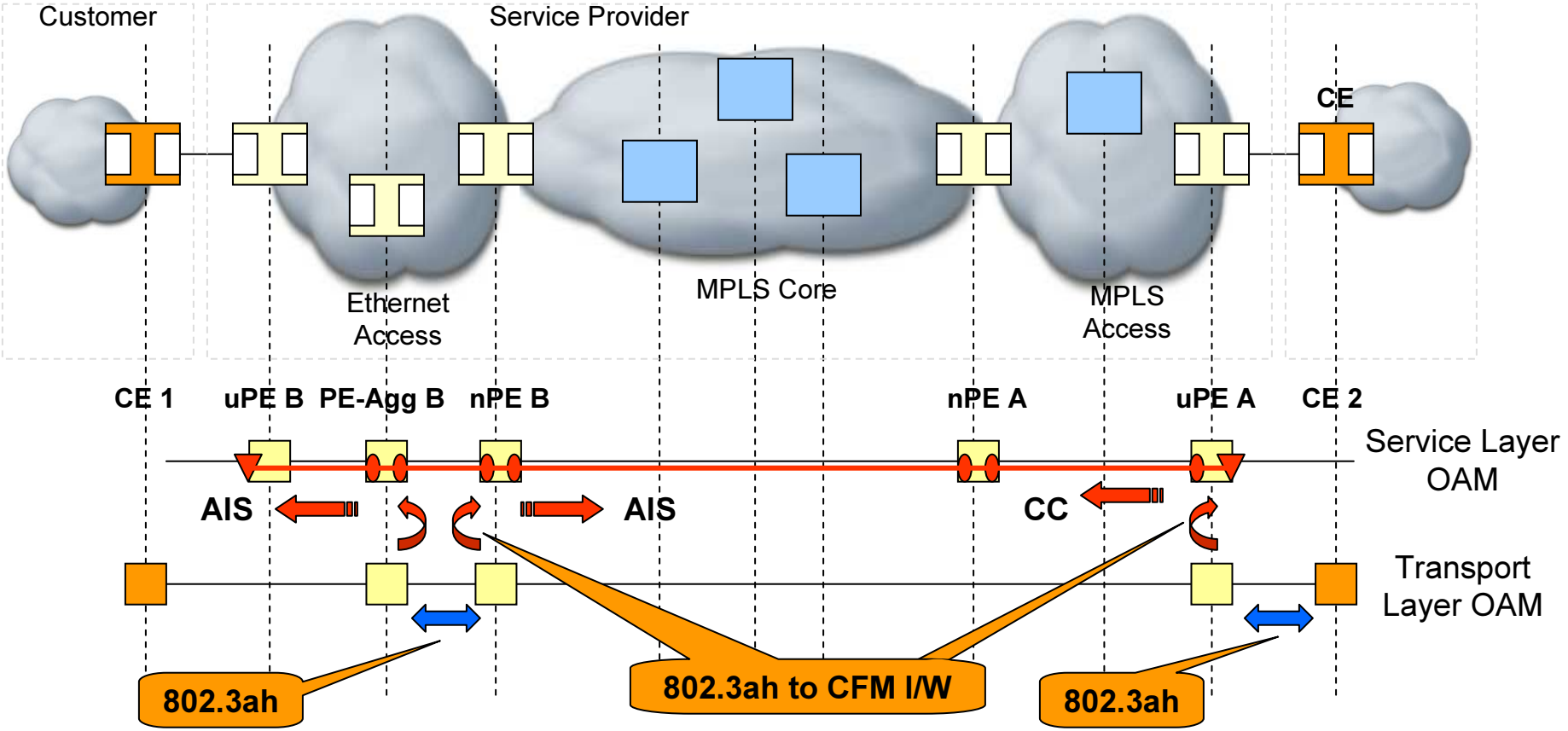


CFM @ Provider Level acts as MEN OAM: provides EVC Status and Remote UNI Status/Name to E-LMI

- Port State TLV of CC Messages carry remote UNI status
- MEP Name TLV of CC Messages carry remote UNI name
- Status of remote MEP in CCDB indicates EVC State

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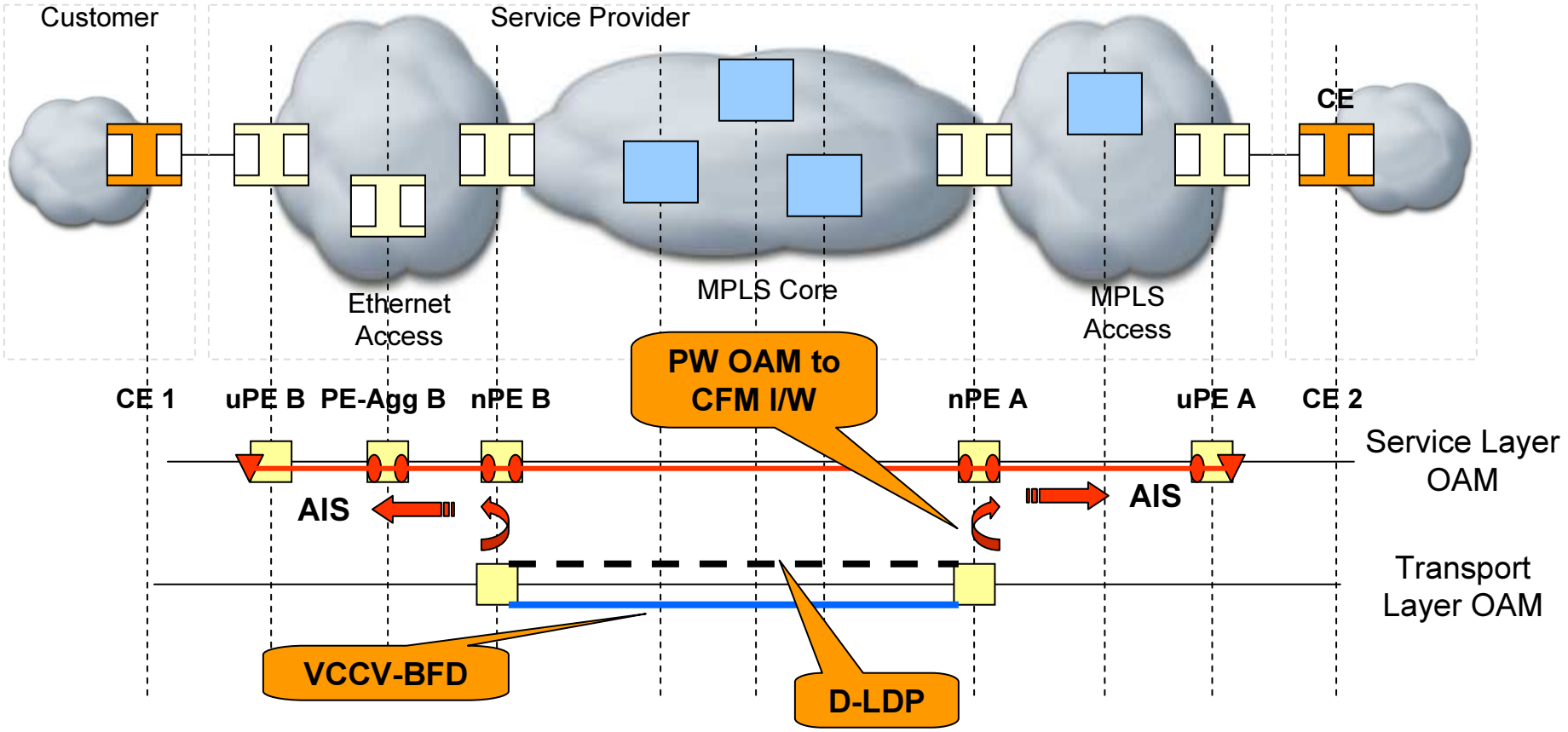
Inter-working Scenarios: 802.3ah to CFM



- Link Layer Defects detected by 802.3ah, relayed to CFM on same device.
- CFM notifies remote devices of localized fault.
- Two variants:
  - CC based (802.3ah on edge of domain)
  - AIS based (802.3ah within domain)

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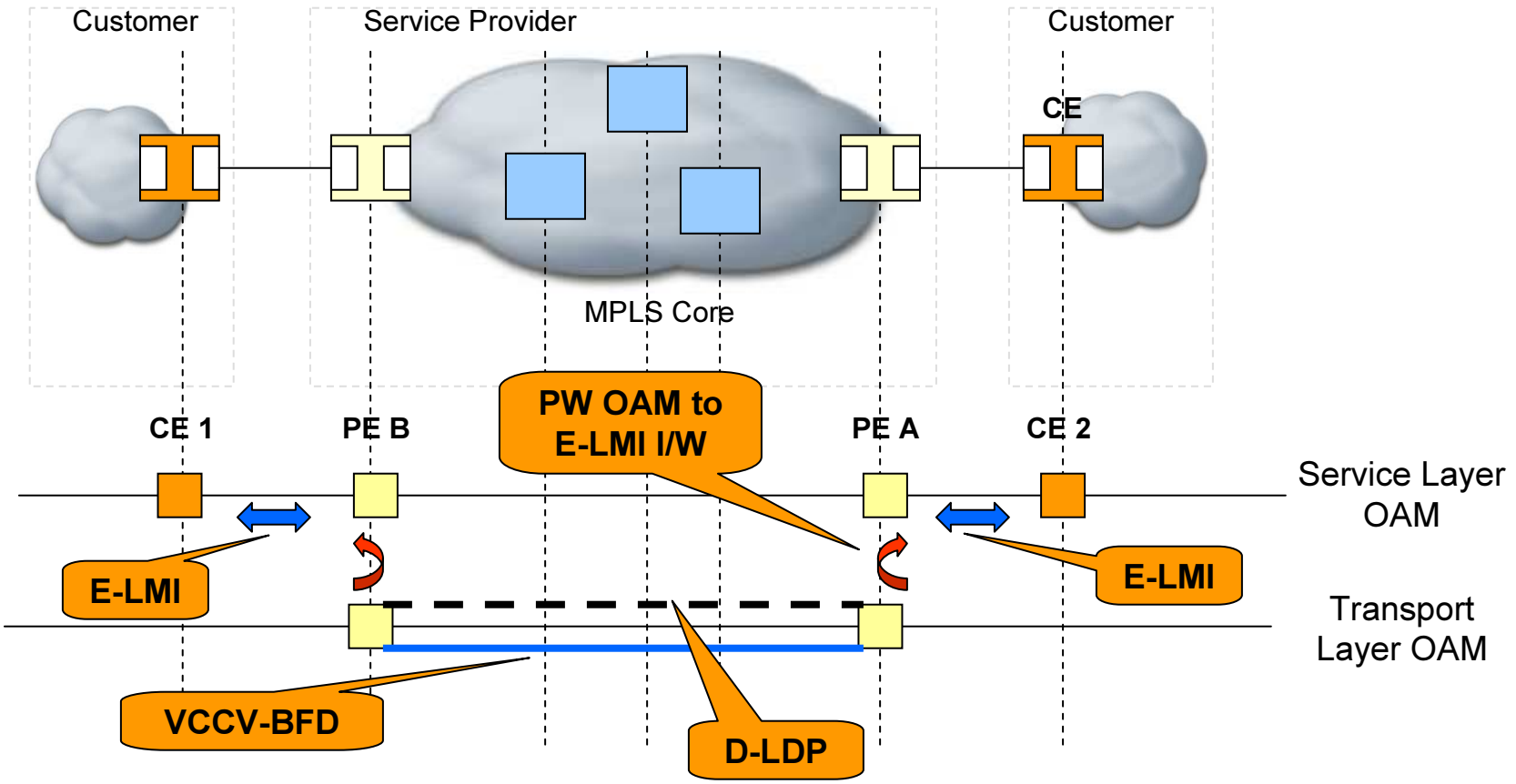
Inter-working Scenarios: MPLS PW OAM to CFM



- Directed-LDP & VCCV (BFD mode) running between nPEs.
- D-LDP for defect notification, VCCV for defect detection
- 3-way I/W function at nPE
- Requires CFM AIS/RDI

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Inter-working Scenarios: MPLS PW OAM to E-LMI



- Directed-LDP & VCCV (BFD mode) running between PEs.
- D-LDP for defect notification, VCCV for defect detection
- Defects detected/communicated by PW OAM are relayed to E-LMI via I/W function on PE.

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Ethernet OAM & Network Management



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Evolution of Ethernet Service Management

- Status Quo
  - – Rely on SNMP MIBs & CLI - Periodic device polling
  - – Correlation of failures to service/customer impact is not straight-forward
- With E-OAM Toolkit integration with NMS
  - – In-band E-OAM mechanisms (CC, AIS, RDI, RFI…) indicate failures to service endpoints
  - – Endpoints signal failures asynchronously to SP Management Device via traps
  - – SP may optionally inform Customer of failure via E-LMI (if Customer does not run CFM)
  - – SP utilizes E-OAM troubleshooting tools (Loopback, Linktrace) to verify and isolate problem.
  - – SP fixes problem and verifies service restoration through E-OAM tools (CC, Loopback)

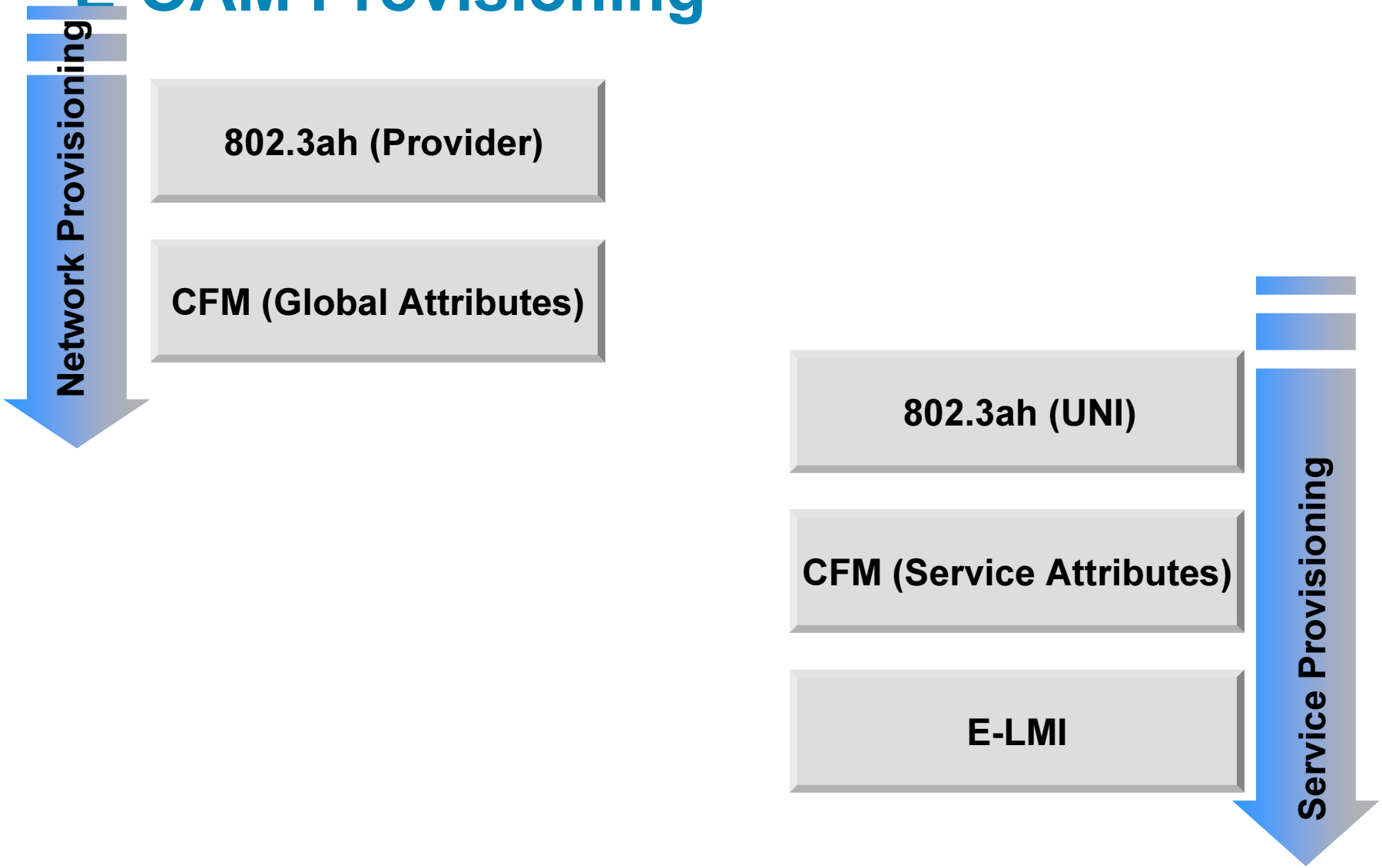
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NMS Complementing E-OAM Toolkit

- Tight integration with NMS to compliment device OAM instrumentation
- Value add:
  - Facilitate E-OAM Provisioning (Maintenance Domains, MIPs, MEPs, etc…)
  - Provide Post-Provisioning Service Connectivity Verification via crosscheck function
  - Proactive Monitoring, Redundant Alarm Suppression & Fault Correlation
  - Automate Proactive/Reactive Troubleshooting Mechanisms
  - Repair Verification

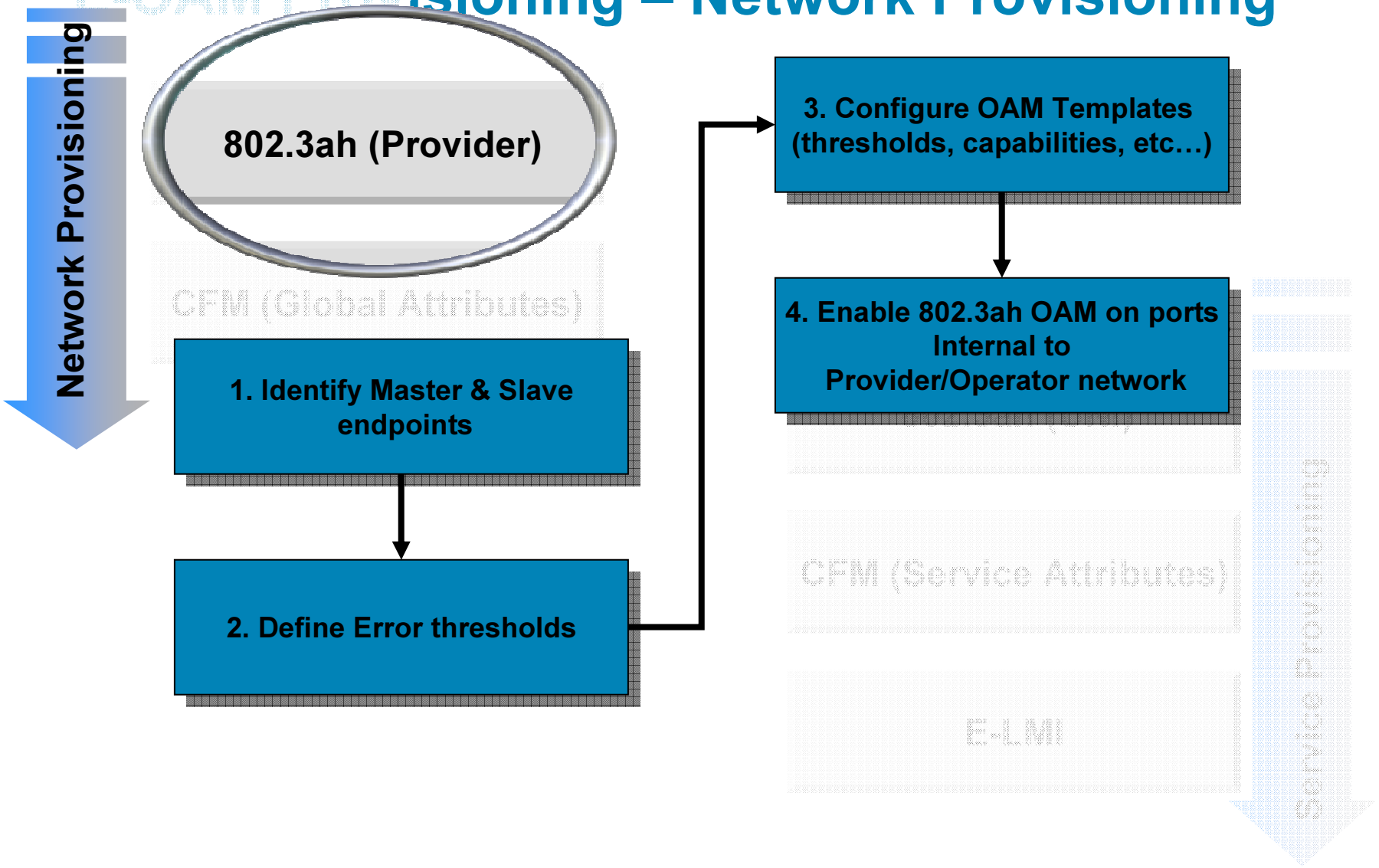
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E-OAM Provisioning



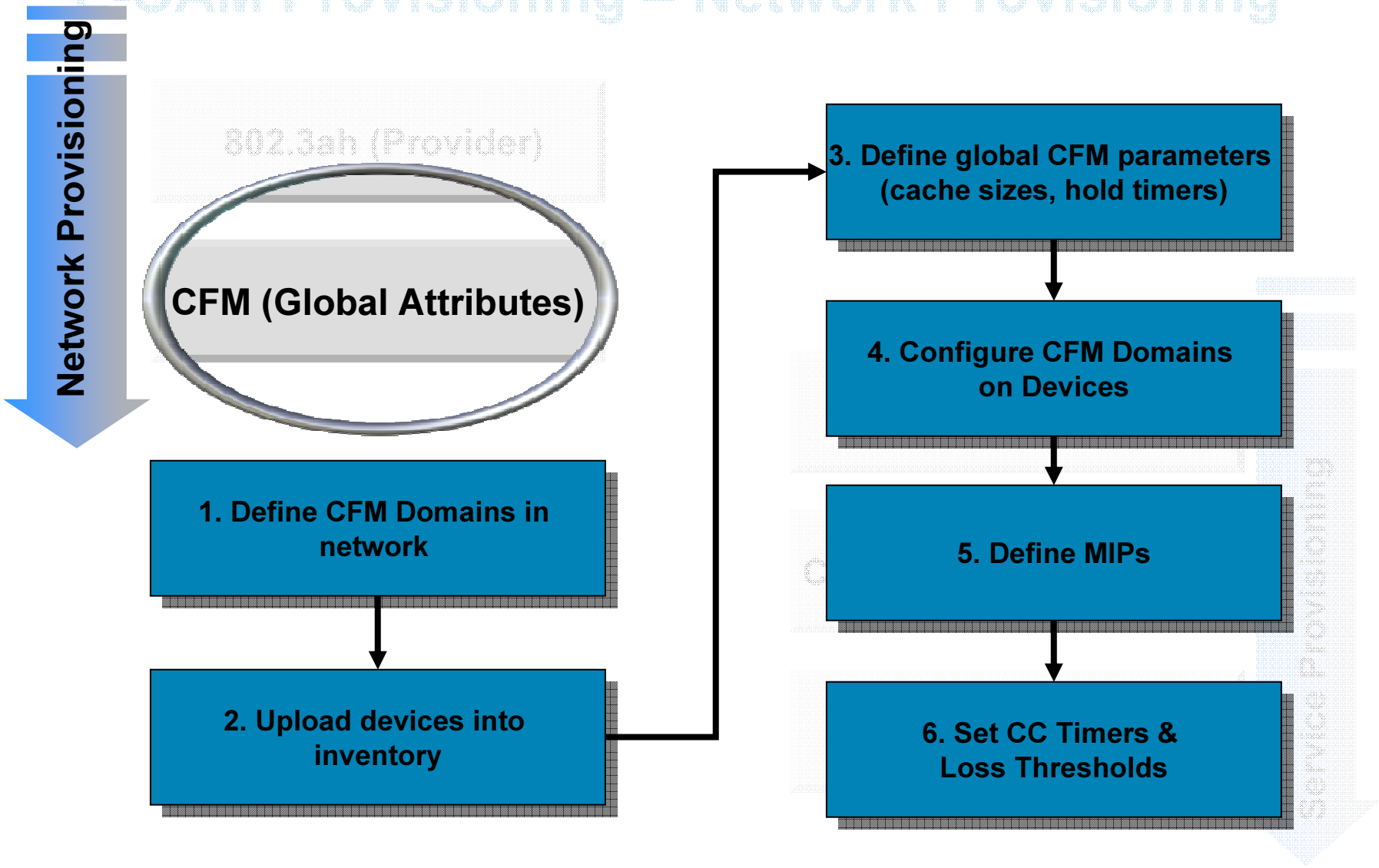
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E-OAM Provisioning – Network Provisioning



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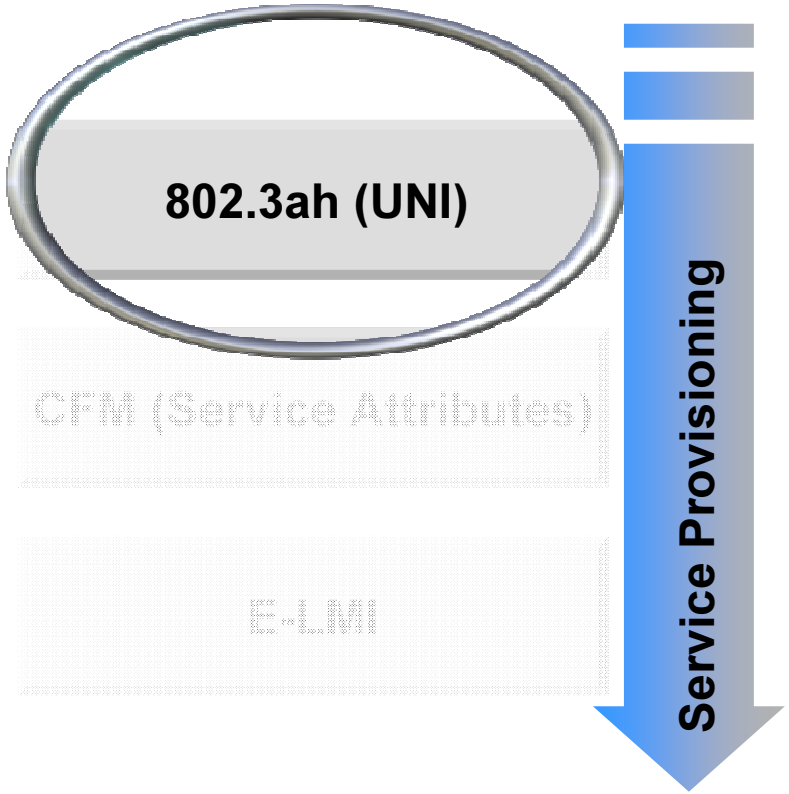
E-OAM Provisioning – Network Provisioning



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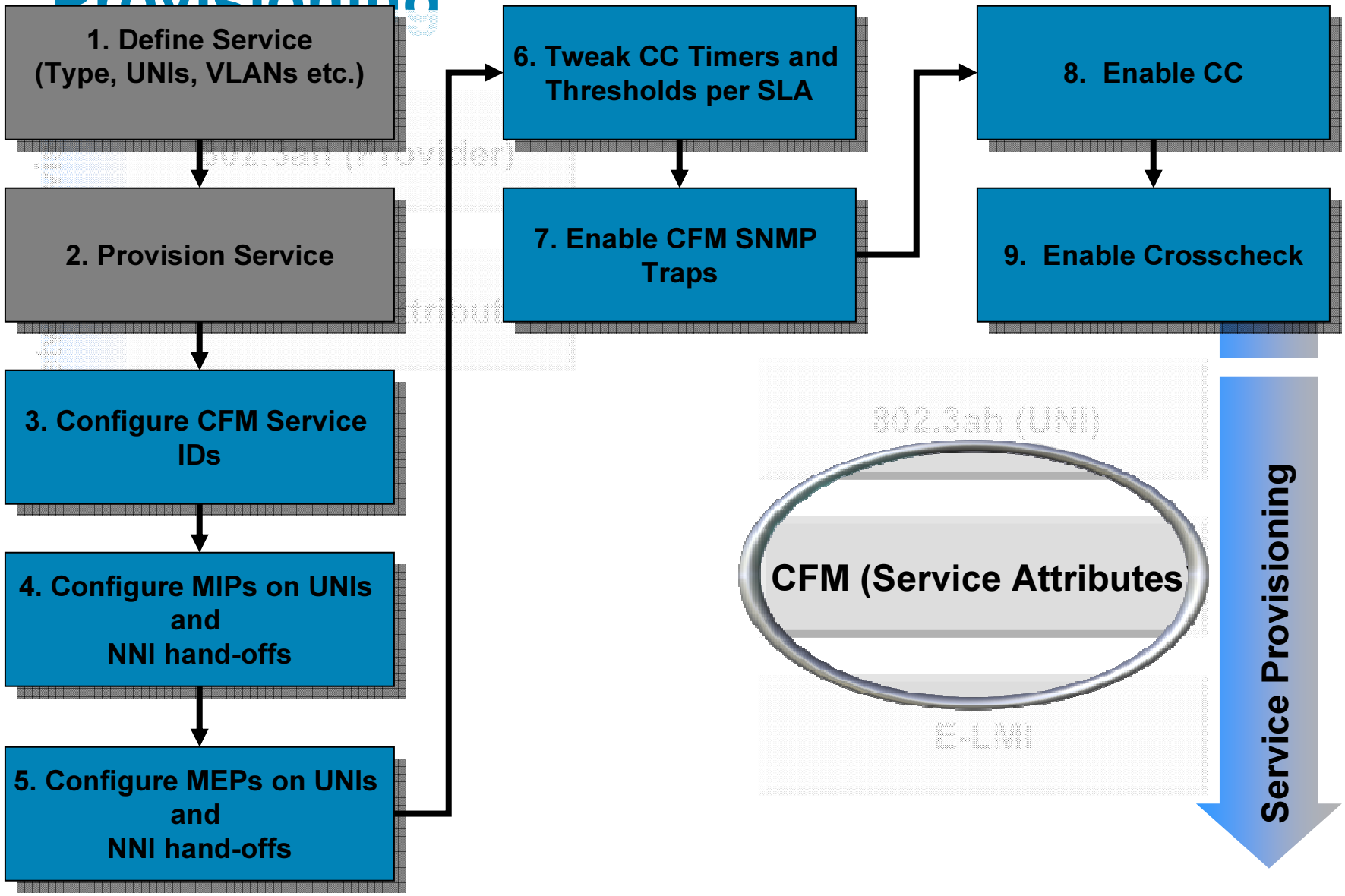
E-OAM Provisioning – Service Provisioning

1. Identify Master & Slave endpoints
2. Define Error thresholds
3. Configure OAM Templates (thresholds, capabilities, etc…)
4. Enable 802.3ah OAM on UNI ports



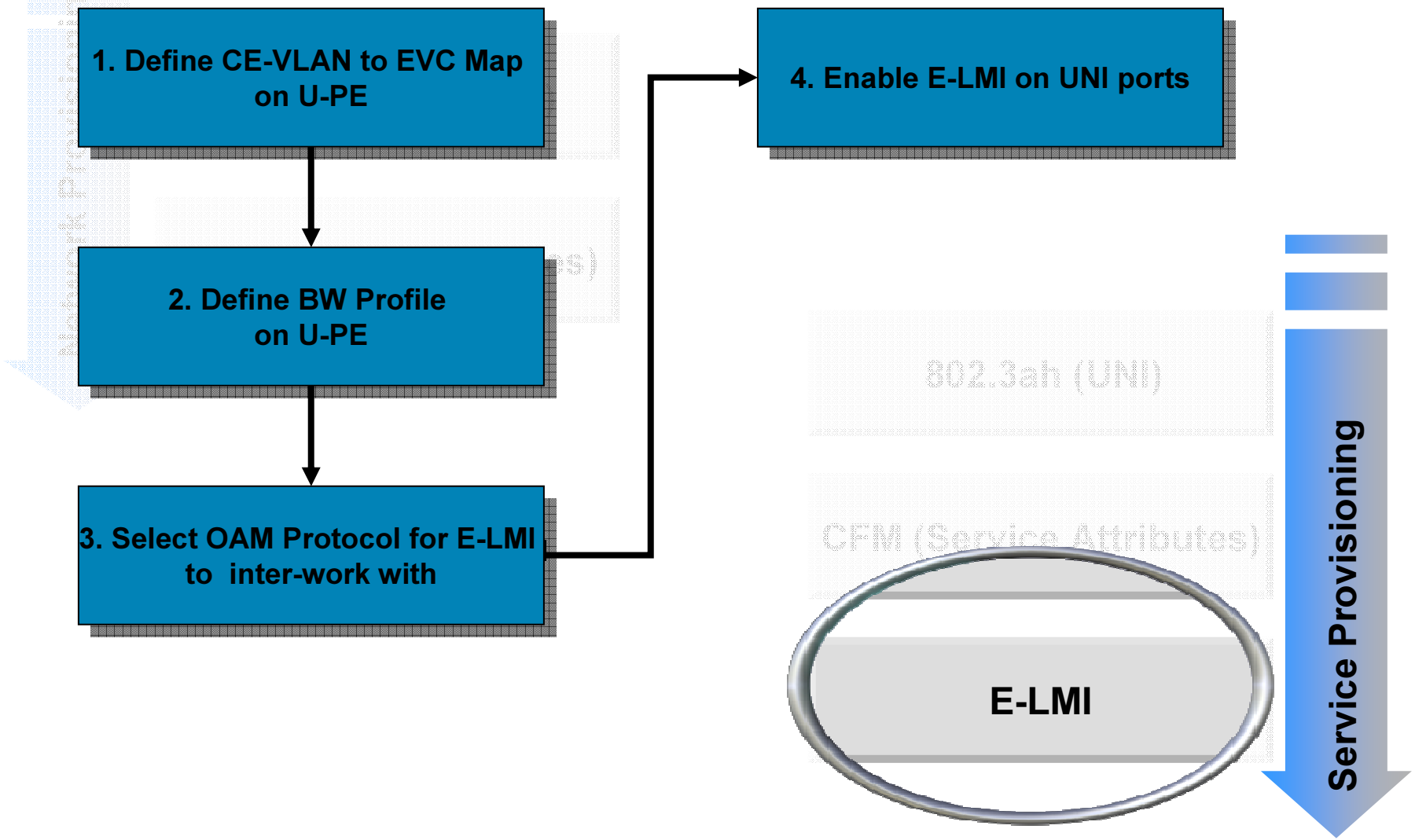
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E-OAM Provisioning – Service Provisioning



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E-OAM Provisioning – Service Provisioning

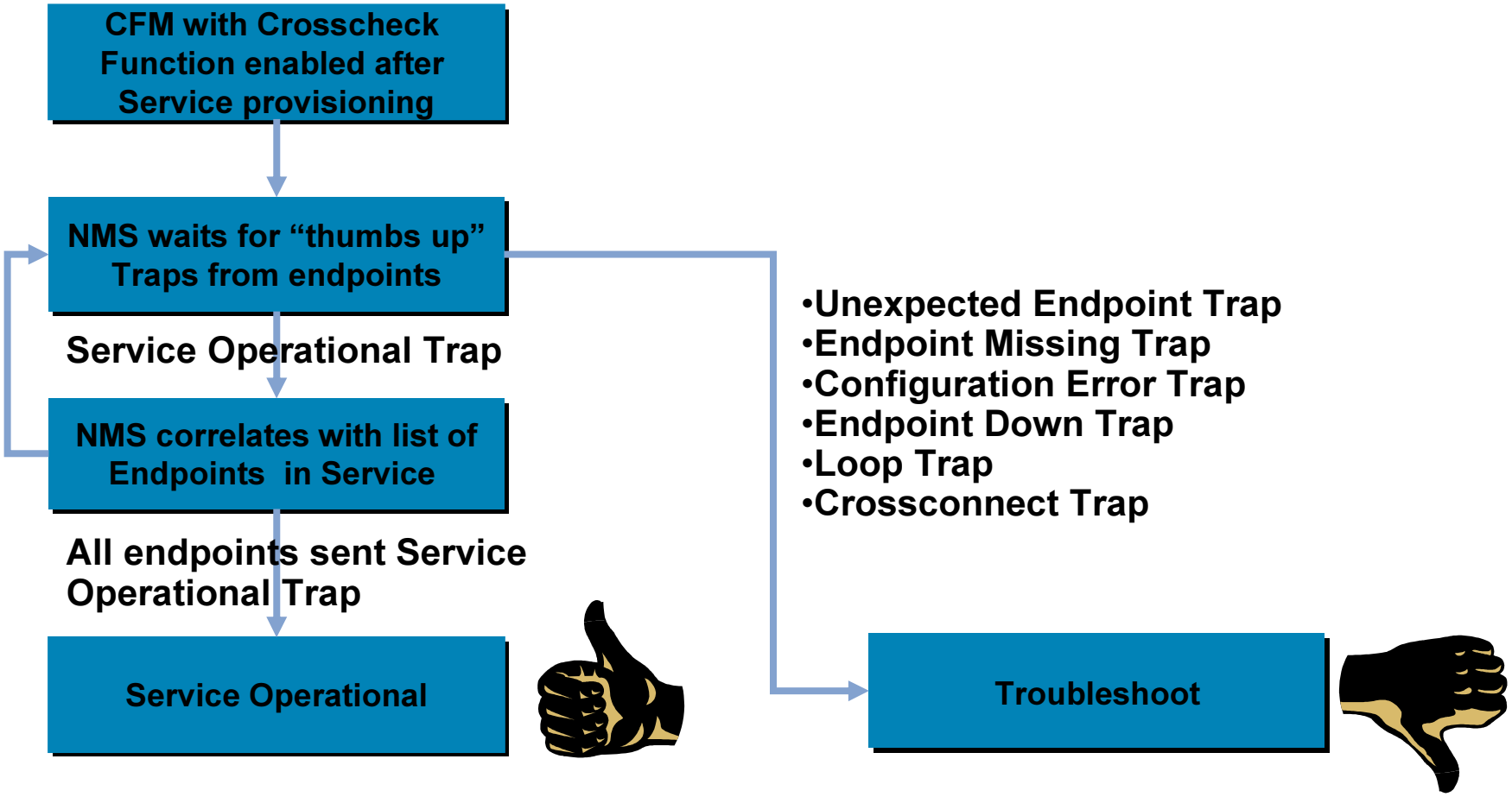


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Post-Provisioning Verification

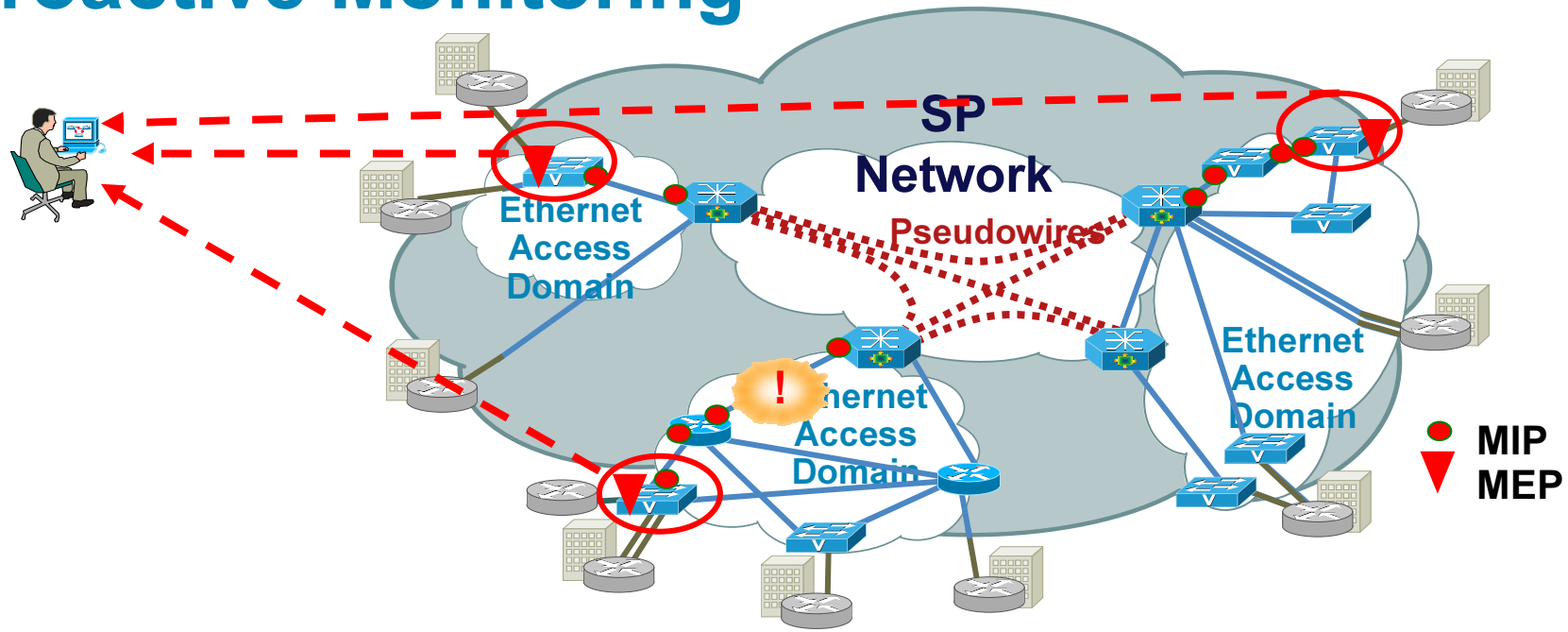
Problem: **How do we know that the service provisioning operation has been successful?**

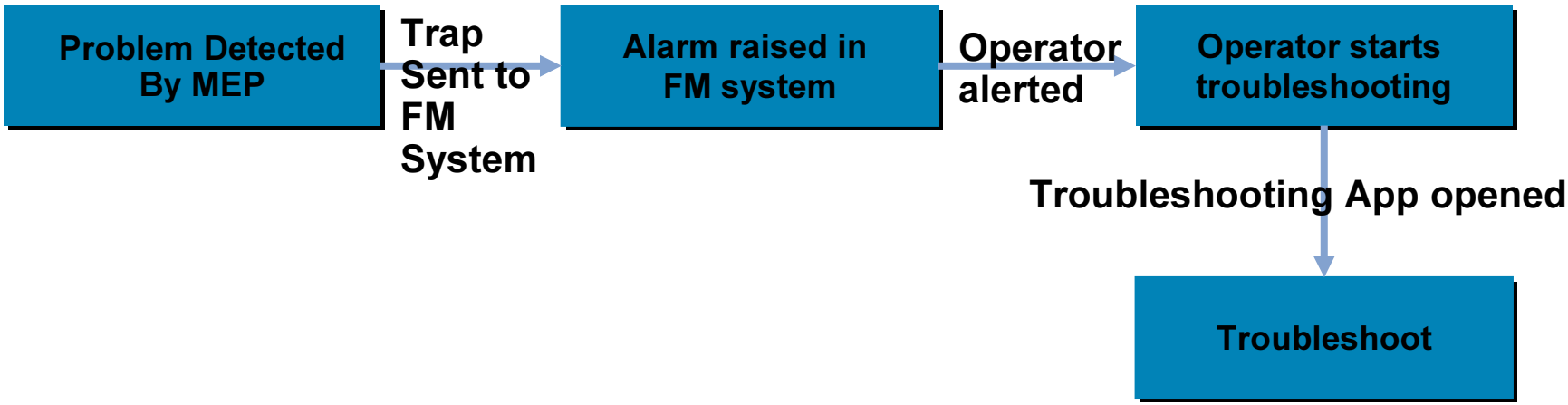
Solution: **Get the devices (NEs) to tell us via CFM Crosscheck Function**



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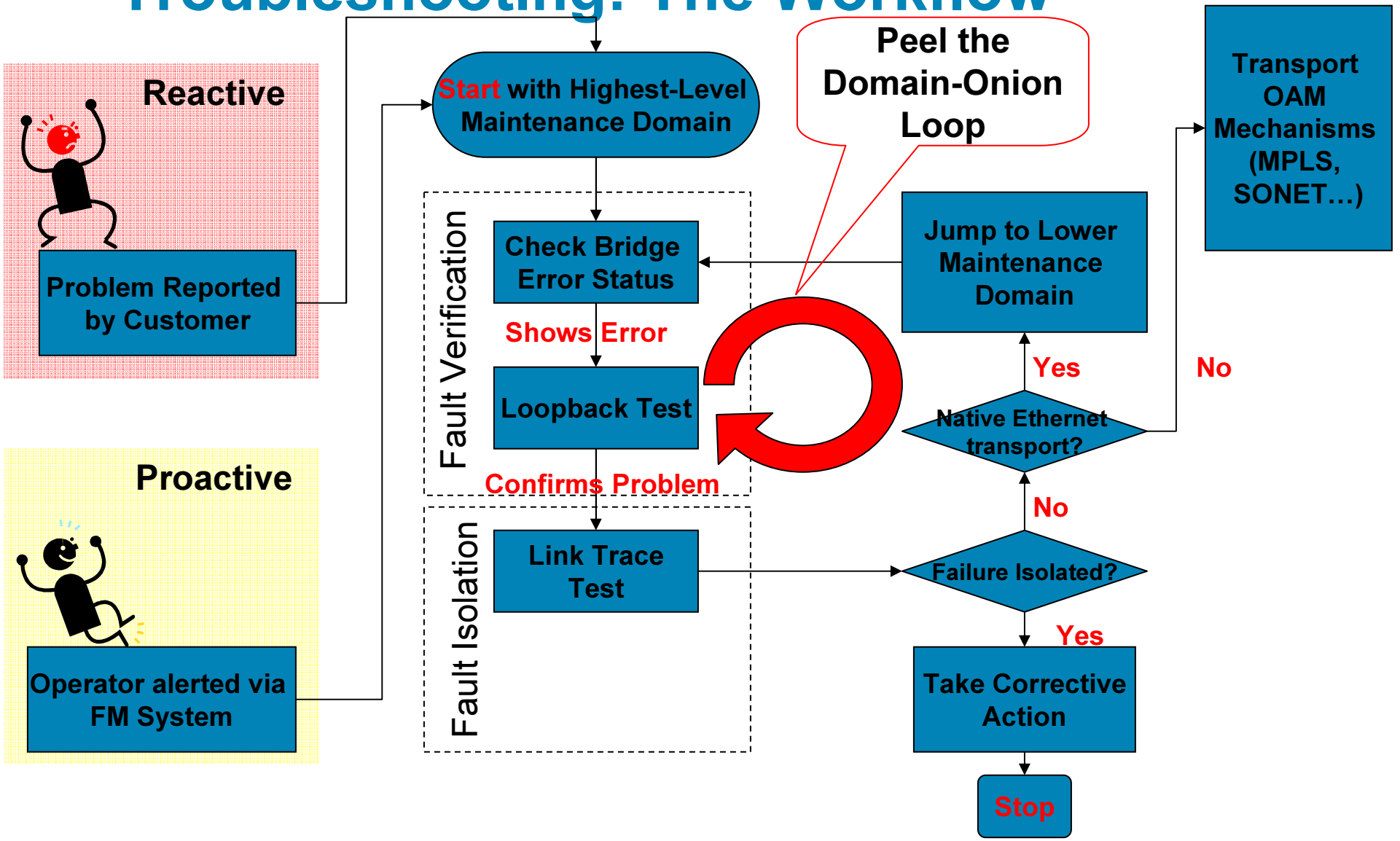
Proactive Monitoring





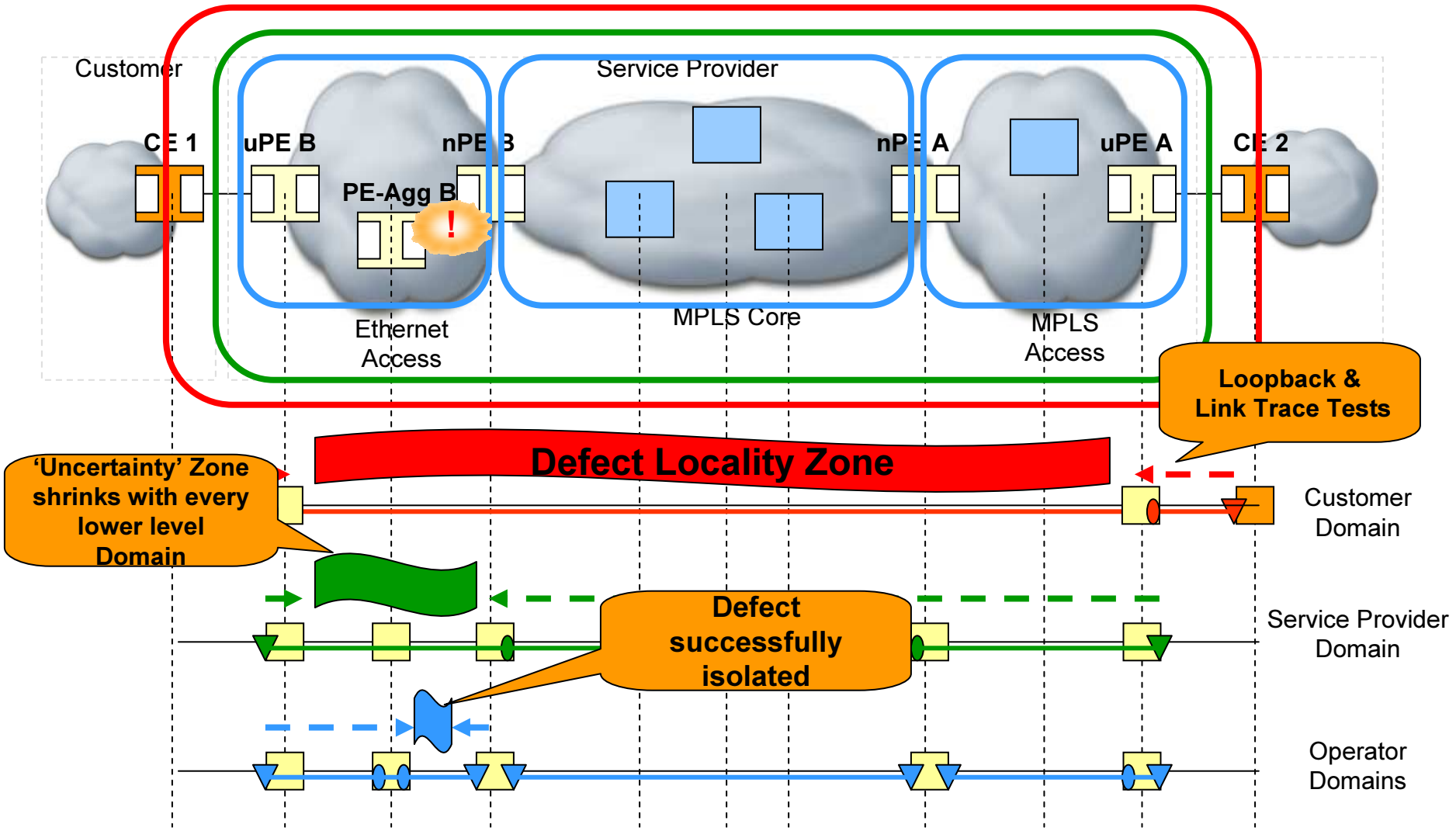
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Troubleshooting: The Workflow



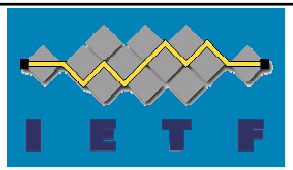
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Troubleshooting: Example



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Standards Landscape

Standard Organization	Ethernet OAM [W←→E, N← →S]
	Link OAM [E<>W] IEEE 802.3ah Fault Mgmt [E<>W] IEEE 802.1ag MIBS IEEE 802.3 + 802.1
	EMS/NMS [N<>S] EMS-NMS Info Model Service OAM [E<>W] Performance Mgmt
	Ethernet OAM [E<>W] Y.1731 Ethernet Protection G.8031 FCAPS/FAB SG4
	L2VPN OAM Req. & Framework [E<>W] L2VPN WG

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Summary



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